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Enhancing urban resilience capacity through multifunctional urban food gardens: Food-Water-Soil nexus approach

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ENHANCING URBAN RESILIENCE CAPACITY THROUGH
MULTIFUNCTIONAL URBAN FOOD GARDENS:
FOOD-WATER-SOIL NEXUS APPROACH

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PREFACE

I, Phạm Trung Kiên, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

All the work described in this thesis was carried out in the Water and Environmental Laboratory, Department of Geotechnics, Environment, Natural Risks and Earth Sciences, under the supervision of Dr. Béatrice Bechet. This dissertation is my own work and contains nothing that is the outcome of work done in collaboration with others except as specified in the text.

This dissertation is not substantially the same as any that I have submitted, or is being concurrently submitted, for a degree or diploma or other qualification at Gustave Eiffel University or any other University or similar institution except as declared in the Preface and specified in the text.

ACKNOWLEDGEMENTS

This will be written when I have some peace to myself before sending it out to the jury

for my inner child

“If at first you don’t succeed, have some cake”

By Charlie Mackesy in *The Boy, the Mole, the Fox and the Horse*

ABSTRACT

Urban Food Gardens (UFGs) are increasingly recognized as vital Nature-Based Solutions for enhancing urban resilience and sustainability. However, their long-term viability is challenged by a lack of assessment tools that can holistically capture their complex socio-ecological dynamics. Dominant paradigms often prove inadequate for these low-energy, soil-based systems, as they tend to marginalize the foundational role of soil, treating it as a static substrate rather than a dynamic capital asset.

This dissertation addresses this critical gap by formulating, developing, and validating a novel Food-Water-Soil (FWS) nexus framework, a paradigm shift that positions soil as the central mediating component in urban agroecosystems. The research employs a multi-stage, mixed-methods approach, beginning with a systematic review that critiques existing assessment methods and establishes the theoretical necessity of a soil-centric paradigm. Using the dual lenses of Ecosystem Services (ES) and Ecosystem Disservices (EDS) as an analytical tool, the study adapts the participatory MESMIS methodology to construct an operational FWS Sustainability Index.

This framework is empirically tested through an in-depth case study of 54 community gardens in Nantes Nord, France. The application of the index

generates a quantitative, multi-dimensional sustainability profile of the gardens, revealing critical strengths, weaknesses, and trade-offs between the food, water, and soil dimensions. Ultimately, this thesis contributes both a new theoretical framework and a validated, practical tool for the integrated assessment of UFGs. The FWS Sustainability Index provides a robust model for adaptive management by gardeners and evidence-based governance by urban planners, supporting the long-term resilience and strategic integration of urban green infrastructure.

IMPACT STATEMENT

Within the academic community, this research offers two primary contributions. First, it advances sustainability science by proposing a new conceptual model—the FWS nexus. By demonstrating the theoretical and practical necessity of a soil-centric paradigm, it challenges researchers to move beyond infrastructure-biased models and more accurately reflect the ecological realities of systems like UFGs. Second, it provides a robust, mixed-methods methodological blueprint that integrates quantitative performance metrics with qualitative social perceptions. This approach offers a replicable model for future socio-ecological research, showing how to bridge the divide between biophysical data and the lived experiences of community stakeholders.

Beyond academia, the FWS Sustainability Index is designed as a transformative tool for urban governance. For city planners and policymakers, it provides a means for evidence-based decision-making, allowing them to move from anecdotal accounts to a quantitative understanding of their green infrastructure's performance. The index can diagnose specific weaknesses across a network of gardens, such as systemic soil degradation or inefficient water use, enabling municipalities to deploy targeted support, such as compost delivery programs or water conservation workshops, where they are most needed. By making the interdependencies of food, water, and soil management explicit, the

framework encourages the breakdown of administrative silos and fosters integrated policy development across departments of parks, water, and waste management. For gardeners and the non-profit organizations that support them, this work offers a tool for empowerment and self-assessment. The FWS index can be adapted for use at the community level, allowing gardeners to collaboratively identify their strengths and weaknesses, set priorities, and track their progress over time. This fosters a cycle of adaptive management and collective learning, aligning with the participatory ethos of the MESMIS methodology from which the index was derived. Furthermore, the tangible, data-driven results generated by the index can serve as powerful evidence in advocacy efforts, strengthening the case for continued public support, land tenure security, and new funding opportunities.

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ABBREVIATIONS

ABM: Agent-Based Modelling

AHP: Analytic Hierarchy Process

ES: Ecosystem Services

EDS: Ecosystem Disservices

EFA: Exploratory factor analysis

FWS: Food-Water-Soil

GIS: Geographic Information System

KMO: Kaiser-Meyer-Olkin

MESMIS: Indicator-based Framework for Evaluation of Natural Resource
Management Systems

LCA: Life Cycle Assessment

LCC: Life Cycle Costing

MCDA: Multi-Criteria Decision Analysis

NbS: Nature-based Solutions

SDM: System Dynamics Modelling

UFG: Urban Food Gardens

WEF: Water-Energy-Food

1 INTRODUCTION

1.1 URBAN GARDENS AS CATALYSTS FOR URBAN RENEWAL

Urban gardens, in their various forms, generate a wide spectrum of benefits that extend far beyond the simple provision of food. Often viewed as multifunctional green infrastructure by urban planners and political actors, urban gardens deliver ecosystem services, strengthen social connections, and contribute to local economies (Delshad, 2022). Our understanding of these benefits enables various pathways to measure the outcomes of urban garden initiatives, moving beyond kilograms of produce to units of social cohesion, public health improvement, and ecological education. We should acknowledge that these “products” are no less vital to urban life than economic profitability or food production volume. This understanding is critical for justifying public investment, which should be perceived not as an agricultural subsidy but as a strategic expenditure on public health, community development, and green infrastructure that can be supported by multiple municipal budgets (Huq and Deacon, 2025).

Perhaps the most important argument for integrating urban gardens into broader solutions for city challenges is based on the vital ecosystem services they provide. Gardens play a direct role in climate actions, as plants' evapotranspiration process helps mitigate the urban heat island effect (Tomatis et al., 2023), absorb CO₂ through carbon sequestration in plants and soil (Mishra et al., 2024), manage stormwater runoff (Gittleman et al., 2017), reduce erosion, and boost biodiversity (Cabral et al., 2017).

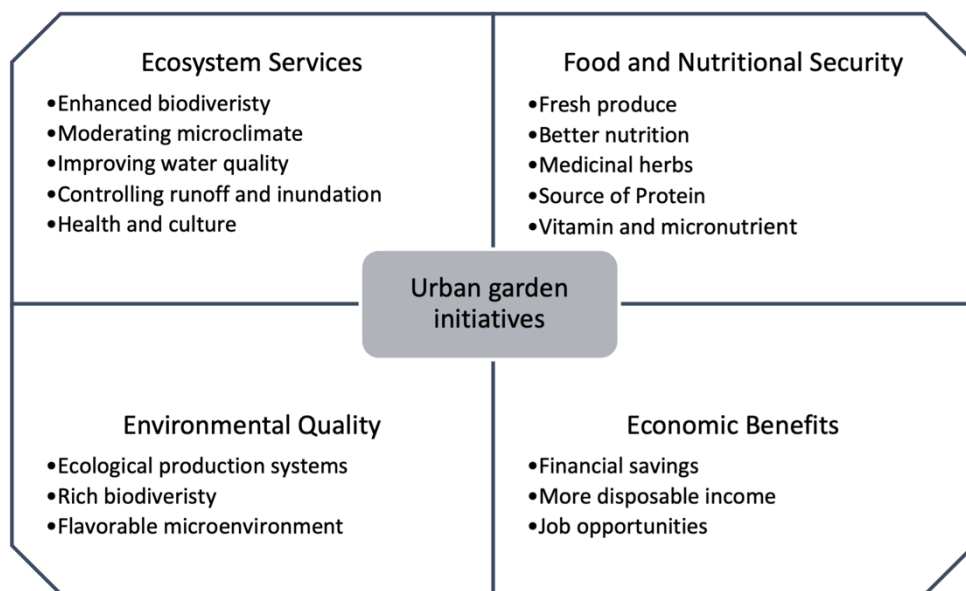


Figure 1.1. Example benefits of urban food gardens (adapted from Rattan, 2020)

In addition to the benefits of natural interactions among plants, water, and soil, garden spaces serve as a powerful tool for building stronger, healthier, and more connected communities. These functions, often referred to as cultural ecosystem services, are well supported by academic research, reports, and books. Gardens are consistently shown to be effective for fostering social cohesion, as they offer a neutral space where people from diverse backgrounds can gather, build trust,

and develop social networks (Kirby et al., 2021). There's no doubt that participation in gardening is closely linked to both physical and mental health benefits (Garcia et al., 2018), as it increases time spent outdoors and offers cognitive stimulation (Suto et al., 2021), thereby reducing stress and fostering a sense of belonging. Long-term gardening also improves access to fresh produce, leading to healthier eating habits with higher consumption of seasonal fruits and vegetables. Gardeners often value the learning opportunities about ecology, nutrition, and sustainability that come from working with plants and soil while aiming to protect biodiversity (Reyers et al., 2012).

During the global pandemic, COVID-19, home and community gardens became a crucial tool for reducing food insecurity in cities (Rattan, 2020; Schanbacher and Cavendish, 2023; Schoen et al., 2021), especially for vulnerable households. In post-industrial cities like Detroit, urban farming on vacant lots has become a key part of community efforts to rebuild the local food system and combat systemic food insecurity (Siegnier et al., 2018). Citizens in the global south, such as those in Hanoi, have been developing urban rooftop gardens as an essential way to ensure access to safe and healthy food (Turner et al., 2024). Although often not seen as their primary purpose, these examples show that urban gardens can offer tangible economic benefits and help create more resilient local food systems

1.2 KEY CONCEPTS IN URBAN GARDENS RESEARCH

1.2.1 Urban resilience and sustainability in the 21st century

By 2050, the Earth is expected to be home to approximately 9 billion people, with around 68% projected to reside in urban areas (United Nations, 2019). Cities worldwide face tremendous pressure to provide the necessary resources for their citizens while addressing social and environmental issues that arise from their impressive economic outcomes (Atta-ur-Rahman et al., 2016).

Although occupying a small fraction of Earth's surface, at around 3% (Liu et al., 2014), urban areas around the world are responsible for 60 to 80% of the energy used and 75% of the total carbon emissions (United Nations Environment Programme, 2020), significantly contributing to global warming. However, cities are vulnerable to abrupt changes in hydrometeorological phenomena induced by global warming, including flood hazards, storm surges, drought, and cyclonic storms (McCarthy et al., 2010; Puppim de Oliveira et al., 2022). These disasters threaten cities' food and water security. As cities grow and expand around the globe alongside technological advancements, more natural resources are required and consumed to meet our development needs. Consequently, the urbanisation process is often associated with excessive consumption and contamination of water, massive generation of greenhouse gases and air pollution, alteration of

local and regional climate through urban heat island effects, and the decline of biodiversity and ecosystems, to name a few (Grimm et al., 2008).

As a result of facing continuing challenges, cities and urban areas have been actively seeking more sustainable development pathways. Due to their capacity to serve as innovation and knowledge creation hubs, cities have been increasingly recognised for their role as the frontline in tackling climate change challenges and building a more resilient society (Puppim de Oliveira et al., 2022). Cities not only represent places of enhanced and clustered vulnerability to climate change and societal problems, but also places where climate actions and social changes can be leveraged through co-creation and co-benefits (*World Cities Report 2024*, 2024). In response to these complex and unpredictable threats, *urban resilience* and *sustainability* have become central goals for cities worldwide (Kong et al., 2022). This view illustrates the shift away from merely protecting against known hazards to building long-term capacity to withstand disruption and change.

The concepts of *resilience* and *sustainability*, while often driven by a strong concern about reducing disruptions and preventing unstable future development, differ considerably. In the urban context, *resilience* refers to a city or urban system's ability to adapt, withstand, recover, and transform quickly in response to environmental and social disruptions (Klein et al., 2003; Meerow et al., 2016), such as natural disasters, climate change, and other unexpected events. By integrating various aspects of adaptability, learning, and transformation, *resilience* is

positioned within the social-ecological system as a "bouncing forward" concept, which recognizes the process of disturbances and reorganization rather than the traditional "bouncing back" view (Comfort et al., 2010). The notion of *resilience* lies in the acceptance of disruption and radical surprises (Evans, 2011), as they are inevitable, and demands anticipation and preparation for changes to occur. *Urban resilience* nowadays is defined by several key characteristics, which are robustness, redundancy, flexibility, resourcefulness, reflectiveness, inclusiveness, and integration (Lowe et al., 2024; Meerow et al., 2016). In this view, *resilience* is portrayed as a precondition for *urban sustainability* (Romero-Lankao et al., 2016)

Meanwhile, the concept of *sustainability* extends beyond traditional ideas of environmental protection and conservation; it involves a long-term focus on conserving natural resources, meeting social needs, and maintaining economic resources. Originating as a normative political idea, the definition and understanding of *sustainability* have been quickly adopted by various scientific disciplines, often without explicitly acknowledging its political origins. After the 1992 United Nations Summit on Environment and Development in Rio de Janeiro, *sustainability* was subsequently implemented at the national and local levels of countries committed to sustainable development (United Nations, 1992). *Sustainability* is often expressed through the three pillars or Venn diagrams of economy, environment, and social equity, all of which are equally important and interconnected. At the core of sustainability lies the principle of equitable access

and distribution of natural resources to citizens, along with growth that minimizes environmental impact (Keivani, 2009; LaGro, 2019). Cities' resource and service needs extend beyond their borders, primarily focusing on essentials such as food, water, medical care, housing, sanitation, and ecosystem services to support citizens' well-being. In this context, the *urban sustainability* concept underpins collective efforts in urban planning and legislation to improve self-sufficient systems in cities, prevent excessive external resource consumption, and protect citizens from climate hazards

1.2.2 Urban nexus assessment

The *urban nexus* is a conceptual framework focusing on policy and planning approaches designed to understand and manage the intricate interdependencies between critical resource systems within an urban context. The idea is less focused on quantifying the input, transformation, and outflows of materials, which are linked to *urban metabolism*, and is more concerned with how resource flows are managed and distributed under the influence of social, political, and economic factors (Chen and Lu, 2015). *Urban metabolism* is a biological analogy, metaphorically describing the city as an organism that consumes resources and generates waste to sustain itself. Methodologies such as material flow analysis and life cycle assessment enable stakeholders to quantify the inputs, outputs, and internal transformations of particular materials and energy (Wang et al., 2023).

Urban nexus, on the other hand, is more concerned with the dynamic relationships among social, political, and economic factors and how they influence the movement of materials and energy. It shifts the analytical focus from resources alone to the interdependencies, trade-offs, and synergies across multiple sectors simultaneously (Chen and Chen, 2015). In other words, while *urban metabolism* offers an accounting framework for what flows through a city, the *urban nexus* provides a systems framework to understand why and how those flows are governed. **Table 1.1** outlines the main differences between these two concepts.

Table 1.1 Key distinctions between urban metabolism and urban nexus

Feature	Urban metabolism	Urban nexus	Reference
Core metaphor	City as an organism	City as a web of relationships	
Unit of analysis	Flows of a single resource (e.g., water, energy)	Interdependencies and trade-offs between multiple sectors	<i>Chen and Chen,</i>
Treatment of social factors	Often implicit or external drivers	Central to analysis (“latent nexus” of social relations)	<i>2015; Chen and Lu,</i>
Common methodologies	Material flow analysis, life cycle analysis, input-output analysis, ecological network analysis	Life cycle assessment, systems dynamic, indicator systems, system, network analysis	<i>2015; Gondhalekar and Erlbeck, 2021; IWRM Action Hub,” 2023.; Wang et al., 2023</i>
Primary Goal	Quantifying resource efficiency and environmental impact	Optimizing synergies and improving cross-sectoral governance	

The Water-Energy-Food (WEF) nexus sits at the heart of the urban nexus. It was formally recognized at the Bonn 2011 Nexus Conference (Martin-Nagle et al., 2012), emphasizing that these three critical sectors are deeply interconnected.

Actions or decisions in one sector often trigger cascading effects and create trade-offs or synergies in the others. For example, water is vital for almost all energy production, energy is needed for water treatment and transportation, and both are key inputs for food production. The WEF framework promotes integrated

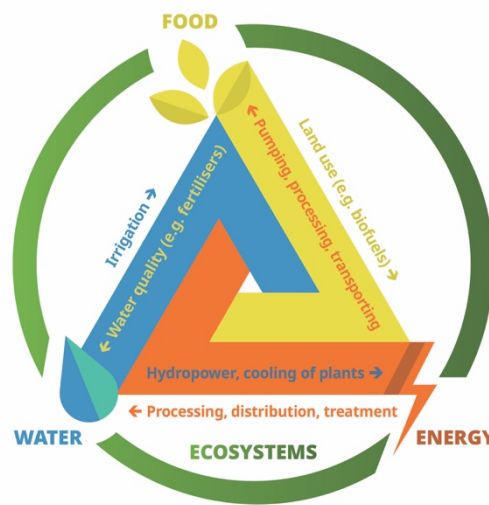


Figure 1.2 Water-Energy-Food-Ecosystem Nexus from (GWP-Med, 2019)

resource management, aiming to find cross-sector solutions, improve security and efficiency, and minimize ecological impacts (Arthur et al., 2019). It also underscores the importance of reconciling potentially conflicting interests among these sectors, while identifying opportunities and exploring emerging ones.

While originating with the WEF triad, the scope of urban nexus has expanded to encompass other critical urban systems. This evolution reflects a maturing understanding that the relationships between WEF are mediated by, and have profound impacts on, other domains. Urban nexus dialogue frequently includes

land use, waste management, climate resilience, ecosystem health, and broader socio-economic factors like public health and social equity (GWP-Med, 2019). This expansion is not an arbitrary addition of variables but a necessary progression of the paradigm from a resource management tool to a comprehensive framework for urban sustainability. Ecosystems, for example, are not just another sector but the fundamental platform upon which the nexus operates, while climate change is the primary external stressor altering the relationships within it (Mabhaudhi et al., 2024).

The core of the Nexus concept is to promote an integrated approach across sectors to address institutional fragmentation, often called "silo thinking." Urban governance typically operates along rigid sector-specific divisions, with separate departments handling water, energy, transport, and land use, often in isolation. This setup is more than just an administrative inconvenience; it reflects a deeply ingrained institutional culture, reinforced by sectoral budgets, specialized professional training, and unique legislative mandates, which can sometimes conflict (Artioli et al., 2017; Monstadt and Coutard, 2019). For example, a transport department might pave over a green space to expand a road, solving a traffic problem but inadvertently increasing stormwater runoff and heat island effects, thus creating new problems for the water and environment departments. Overcoming this institutional inertia requires a deliberate strategy

of multi-scalar integration, connecting actors both horizontally and vertically across levels of government (Bierwirth et al., 2024).

1.2.3 Urban Nature-based solution

Cities are at the forefront of global efforts to address the world's most pressing sustainability challenges, such as climate change and biodiversity loss. However, the solutions available are just as complex as the problems themselves, encompassing technological, social, and ecological approaches. Nonetheless, individual solutions alone do not seem to be a cure-all for the systemic issues, which is why the nexus approach we discussed earlier has gained importance. This has led to the rise of nature-based solutions (NbS)—an ecocentric approach offering multiple benefits for the environment, economy, and society in urban areas (Frantzeskaki, 2019; Laforzezza et al., 2018; Nesshöver et al., 2017). Despite its growing prominence, NbS is often considered as an “umbrella concept”. It is not a new invention but rather a synthesis that covers a wide range of pre-existing, nature-focused approaches. These include ecosystem-based adaptation, ecosystem-based mitigation, eco-disaster risk reduction, green and blue infrastructure, and ecological engineering (Cohen-Shacham et al., 2016).

The International Union for Conservation of Nature (IUCN) has the most widely cited and influential definition of NbS that shapes the global discourse, in which NbS is defined as:

“Actions to protect, sustainably manage, and restore natural or modified ecosystems, which address societal challenges, effectively and adaptively, providing human well-being and biodiversity benefits”

The IUCN definition of NbS emphasizes ‘actions’ from a human-focused perspective, integrating societal factors into its core framework to ensure solutions for human issues do not harm nature. Eggermont et al. (2015) shaped the NbS term by proposing a typology of NbS, categorizing them into three types based on the number of stakeholder groups, delivery services, and the level of engineering involved (see **Figure 1.3**). These NbS types and their hybrid versions should be seen as complementary, with no single type being better than the other, but space and time-appropriate.

In recent years, the European Commission (EU) has been a primary driver of NbS research and implementation, having its own definition of NbS that reflects a strong policy and innovation focus (EC, 2019). While aligned with the IUCN's core idea, the EC's definition frames NbS in a broader term – “solutions inspired and supported by nature”, and puts cost-effectiveness and economic benefits as core components, alongside environmental and social ones. The definition also explicitly mentions building resilience towards urban applications.

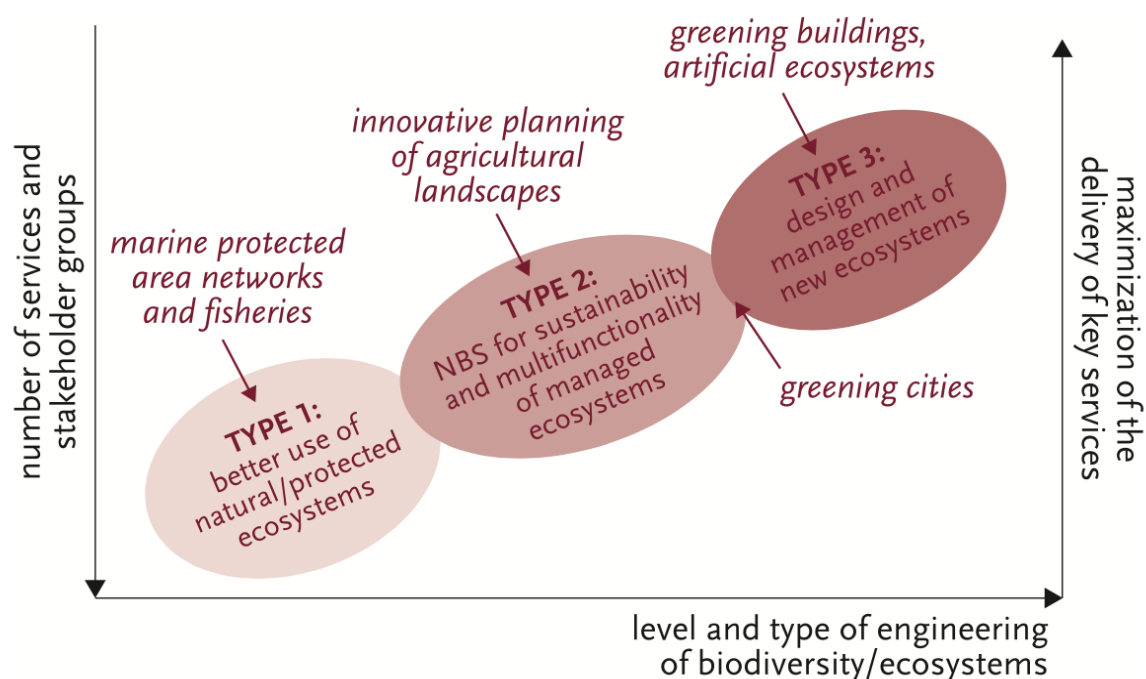


Figure 1.3 Typology of NbS from Eggermont et al., 2015

The application of NbS in cities is diverse, ranging from the protection of remaining natural areas to the creation of highly engineered green technologies integrated into the built environment. As seen in Figure 1.4, stakeholders can classify these

interventions by the degree of human intervention and ecosystem modification, from minimal intervention to extensive management or creation of new ecosystems. The following are some key challenges targeted by urban NbS:

- *Climate resilience and adaptation:* Interventions aim to mitigate the urban heat island effect by providing cooling effects through evapotranspiration and shading (Ramakreshnan and Aghamohammadi, 2024), manage stormwater to reduce flood risk from extreme precipitation events (Kõiv-Vainik et al., 2022), and enhance the city's capacity to adapt to a more volatile climate (Hobbie and Grimm, 2020).
- *Environmental quality improvement:* Vegetation can improve air quality by filtering pollutants and particulate matter (Nowak et al., 2006), reduce noise pollution (Zhao et al., 2021), improve water quality through the natural filtration process (Muerdter et al., 2018), and sequester atmospheric carbon (Thölix et al., 2025); all of these functions contribute significantly to a healthier urban environment.
- *Health and well-being:* The presence of green and blue spaces in cities is strongly linked to improved human health. These areas provide spaces for physical activity and recreation (Garcia et al., 2018), offer opportunities for stress relief and mental restoration (Sturm and Cohen, 2014; Vujcic et al., 2017), and can foster social cohesion by creating communal gathering places (Wilson et al., 2024). The

benefits are particularly important for vulnerable groups, e.g., children and the elderly.

- *Biodiversity conservation*: In an increasingly urbanized world, cities have to be part of the solution for conserving biodiversity. Urban NbS can create and connect habitats, supporting native flora and fauna within the urban matrix (Beaugeard et al., 2021; Threlfall et al., 2017).

- *Socio-economic benefits*: Beyond environmental and health benefits, urban NbS can deliver economic returns. For example, creating or restoring green spaces can increase tourism revenues (Mandić, 2019), raise property values (Costadone and Zhang, 2025), and create jobs in design, installation, and maintenance.

The multifunctionality of these interventions is a core advantage. A single NbS can address multiple challenges simultaneously, delivering a suite of co-benefits. **Table 1.2** provides an overview of common urban NbS types and their associated benefits, illustrating this principle.

While urban NbS show considerable promise, they are not a cure-all and have inherent limitations and potential trade-offs that need careful consideration during planning and implementation. First and foremost, the main concern of NbS is the uncertainty regarding their reliability and cost-effectiveness compared to traditional engineered solutions (Seddon et al., 2020), which calls for the

development of context-specific metrics to evaluate their effectiveness at various scales and associated trade-offs. Additionally, the concept has faced ongoing criticism for its vulnerability to “greenwashing”—where corporations and governments may misuse it to project an eco-friendly image without making meaningful changes to their core operations (Gałęcka-Drozda et al., 2021; Melanidis and Hagerman, 2022). Finally, a growing body of research and activism has brought the social justice dimensions of NbS to the center of the discourse, arguing that an NbS that is not socially just risks reinforcing or exacerbating social inequalities (Pineda-Pinto et al., 2021).

Table 1.2 Examples of NbS and associated benefits

NbS Typology	Primary Function	Key Co-benefits	Application Scale
<i>Green roofs</i>	Stormwater management	Building insulation (energy savings), urban heat island mitigation, habitat creation, aesthetic improvement.	Building
<i>Vertical Gardens & Green Walls</i>	Air Quality Improvement	Building cooling, noise reduction, aesthetic enhancement, biodiversity in dense areas	Building
<i>Rain gardens & swales</i>	Stormwater management	Groundwater recharge, water quality improvement (pollutant filtering), habitat creation, green amenity	Site, street, neighbourhood
<i>Street trees & urban forests</i>	Urban heat island mitigation	Air quality improvement, carbon sequestration, noise reduction, mental health benefits, aesthetic value.	Street, neighbourhood, city
<i>Urban parks & green spaces</i>	Recreation & Social Cohesion	Mental & physical health, biodiversity support, stormwater infiltration, air cooling, tourism revenue.	Neighbourhood, city
<i>Community gardens</i>	Food provision	Mental & physical health, biodiversity support, social cohesion, biodiversity support, recreational and educational opportunities.	Neighbourhood, city

<i>Constructed wetlands</i>	Water Purification / Wastewater Treatment	Flood attenuation, habitat for wildlife, carbon storage, recreational and educational opportunities.	Neighbourhood, city
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1.3 PROBLEM STATEMENT

Cities worldwide are adopting NbS, such as UFGs, as a key strategy to tackle future sustainability challenges. This global greening effort, driven by initiatives like the One Health moment (Bruno et al., 2024), has created an urgent need for effective assessment tools to measure the performance and long-term viability of these complex social-ecological systems. However, many NbS still lack frameworks and parameters for evaluating their effectiveness from a social-ecological perspective, especially across different scales and dimensions (Kato-Huerta and Geneletti, 2022).

One of the primary challenges in establishing such a framework lies in the heterogeneity of urban contexts across the globe, which are deeply influenced by varying cultural and institutional factors. Consequently, it is essential to develop context-specific indicators that are customised to local circumstances to enhance the usability, scalability, and transferability of NbS (Debele et al., 2023). This endeavour often requires a substantial allocation of both financial and human resources, which are generally not readily accessible to the majority of urban centres. Secondly, the majority of existing research focuses primarily on evaluating how modelling tools and parameters can be utilized to identify optimal locations

for NbS and UFGs, or to facilitate their design (Caneva et al., 2020; Wickenberg et al., 2021), often overlook the importance of long-term assessment of implemented NbS, including the potential disservices associated with them. An additional challenge pertains to the insufficient systematic understanding and documentation of the synergies and trade-offs among various impacts of NbS (Dumitru et al., 2020).

Given the significant role of UFGs in enhancing urban resilience, UFGs—especially gardens operated collectively with a non-profit structure—are widely recognized as nature-based solutions (NbS) in many different contexts (Laforteza and Sanesi, 2019; Moniz et al., 2022; Skar et al., 2020; van der Jagt et al., 2017). Despite increasing interest in UFGs, these initiatives often receive limited long-term support and struggle to maintain sustainability amid ongoing development (Lawson, 2004). Beyond the need for institutional backing, gardeners face financial and technical obstacles after establishing the gardens, mainly due to persistent resource demands such as soil, compost, fertilizers, and water (Becker and von der Wall, 2018). The use of these resources as inputs for UFGs is linked to the safety of the harvested products and greatly influences their sustainability, yet they are frequently evaluated separately. A fragmented approach to analyzing these components risks a disconnect between sectoral planning efforts and undermines the overall assessment's integrity. Therefore, it is essential to adopt an integrated

and holistic approach to these components to foster synergies and ensure the effective use of these vital resources (Gu et al., 2021; Skar et al., 2020).

1.4 RESEARCH AIM

This dissertation aims to contribute to the field of NbS research by developing a theoretical and technical framework to support the long-term assessment of UFGs. The framework examines the complex relationships among the key components of UFGs—such as natural capital and human capital—and provides recommendations to optimize synergies and mitigate trade-offs between these components. Ultimately, this approach seeks to promote the sustainability of UFGs and their contribution to urban resilience. The main research question of this doctoral research is as follows: **How to assess the sustainability of UFGs through a food-water-soil nexus framework – a more relevant and accurate paradigm for soil-based urban ecosystems?**

The dissertation is structured around sub-questions that systematically address the primary research objectives. It comprises a series of studies, beginning with an examination of the existing theoretical and methodological nexus framework for UFGs (Chapter 2). This is followed by the development of an index system designed to assess the sustainability of nexus systems (Chapter 3). A case study applying this framework is presented in Chapter 4, culminating in a comprehensive critical discussion of the key findings and limitations of the index system (Chapter 5). The sub-questions guiding this research are as follows:

- Sub Research Question 1: What is the current state of knowledge concerning the nexus assessment frameworks of UFGs, and what research gaps remain?
- Sub Research Question 2: How can a Food-Water-Soil (FWS) nexus be conceptualized as a more appropriate paradigm for these systems?
- Sub Research Question 3: How to incorporate perceived priorities, benefits, and risks associated with UFGs from gardeners into the assessment criteria?
- Sub Research Question 4: How do the sustainability profiles of UFGs provide insights regarding their key strengths, weaknesses, and trade-offs among food, water, and soil functions?
- Sub Research Question 5: How does the application of the framework provide a more robust and responsible model for the strategic management of urban green spaces?

1.5 METHODOLOGICAL FRAMEWORK

1.5.1 Research setting – the CLEAR-Doc project

This doctoral research is a part of CLEAR-Doc – a European Horizon 2020 Marie Skłodowska-Curie COFUND programme that focuses on all research areas related to “Cities”. The project employed a total of 30 doctoral students working on designing and developing resource-efficient urban spaces, understanding and managing “urban risk” for safe and resilient cities, and developing the digital city as a catalyst for social, environmental, and economic performance (“City Lab Education And Research Doctoral Programme,” 2022). Within six thematic fields of the programme, this dissertation identifies and covers two of them, which are:

- C2 Safety, security, and risk management within the city
- C5 Sustainable urban life and tourism

In this doctoral research work, the investigation was conducted in two significantly different urban areas - Nantes city (France) and Thu Dau Mot city (Vietnam). This research benefits from datasets collected through various previous projects conducted on urban garden systems by the Institute for Research and Urban Science and Technology (IRSTV - l’Institut de Recherches en Sciences et Techniques de la Ville) and the Water and Environmental Laboratory (Eau et Environnement Laboratory) at Gustave Eiffel University. The case studies for this doctoral research include soil data from Nantes, gathered through two research projects, URBiNAT and WISE Cities, between 2019 and 2021 (Doux et al., 2022; Moniz et al., 2022). An additional dataset on earthworm populations was obtained from the JASSURE

project (Le Guern et al., 2018). All studied gardens in Nantes are situated in deprived neighbourhoods with a significant need for improvement. More detailed socio-demographic profiles of the studied populations are provided in chapters 4 and 5.

1.5.2 Step 1: Forming a Theoretical Framework

A scoping literature review aims to provide an overview of the current applied nexus frameworks for UFGs. It examines the methods, variables, and indicators used to evaluate urban gardening initiatives and summarizes these approaches in Chapter 2. Besides highlighting the connections between nexus components, the chapter also reviews existing literature to identify gaps and limitations in these methods. The chapter concludes by suggesting a conceptual framework of the food-water-soil (FWS) nexus and alternative strategies for effectively integrating the soil dimension into research agendas for UFGs.

1.5.3 Step 2: Building a Food-Water-Soil Assessment Framework

Building on the scientific insights from the review, particularly those derived from the MESMIS nexus methodology, this chapter introduces our FWS assessment framework designed to evaluate the sustainability attributes of UFGs in Nantes' North. The framework adapts the existing MESMIS approach for the water-food-energy nexus, replacing the energy component with soil considerations, given that UFGs are predominantly soil-based systems with minimal energy consumption. The original MESMIS nexus was modified into two separate tiers – foundational viability and sustainability performance. Foundational Viability is built upon the argument that sustainability is meaningless if the garden system is fundamentally unsafe; thus, a list of non-compensatory indicators is established to ensure safe ecological and health boundaries. UFGs advancing past the first tier would then progress to tier 2, where their sustainability attributions would be evaluated. More details are discussed in Chapter 3.

1.5.4 Step 3: Data collection

This doctoral research employs three complementary data sources for data collection. Two of these sources were introduced in Section 1.5.1. The third dataset encompasses a comprehensive sample of urban gardeners in Nantes and Thu Dau Mot, capturing their perceived priorities, benefits, risks, and gardening practices. Data were collected via a detailed questionnaire administered to Nantes' North gardeners, municipal officials, and non-profit workers, yielding a

total of 54 responses collected between April and June 2025. The survey was subsequently replicated in Thu Dau Mot from May to July, resulting in an additional 32 responses from the city gardeners. The dataset provides personal context indicators, such as years of gardening, age, gender, and employment status, in addition to the self-perceived risks and benefits of UFGs. The complete questionnaire can be found in the appendix (A3.1).

1.5.5 Step 4: Coding, weighting indicators, and statistical analysis

Based on the collected data and the characteristics of the observed UFGs, gardens that passed the initial tier were moved to tier 2 for evaluating sustainability indicators. The data were categorized according to their respective indicators, and sustainability scores were assigned within each dimension—such as food, water, and soil—based on the weighted contribution of each indicator. Statistical analyses were conducted to assess the normality of the scores and to compare attributes across different groups. Detailed information about the indicators and their weights can be found in Chapter 3.

1.6 RELEVANCE AND IMPACT

This research advances efforts to protect and enhance soil conditions within urban environments, an urgent priority for the European Union in pursuit of healthy soil

(European Commission, 2023). It frames soil as interconnected with critical resources such as food and water in garden systems, enabling their co-evaluation through relevant drivers and pressures to promote long-term sustainability and resilience. Additionally, the research offers a framework and a set of indicators to assist urban planning practitioners and decision-makers in designing sustainable UFGs. This approach encourages integrated, cross-sectoral collaboration and moves beyond siloed perspectives, fostering more holistic urban development.

In the scientific context, the proposed research offers a solution to incorporate soil into the core of ecological nexuses (Hatfield et al., 2017; Lal et al., 2017). It develops a theoretical framework, establishes common research guidelines, and provides methodological support for the FWS nexus. This approach could be valuable for future studies aiming to apply the FWS nexus to other NbS systems, thereby enabling the monitoring and assessment of their sustainability performance.

1.6.1 Thesis structure

This thesis is organized into five chapters. Chapter 1 provides a general introduction to the background and research problem, followed by research aims and objectives.

Chapter 2 offers a literature review of the existing theoretical and methodological nexus framework for UFGs. In addition to an overview of the available methods, tools, and indicators for assessing UFGs, the chapter also presents arguments and suggestive pathways to incorporate soil into the current research paradigms.

Chapter 3 continues with the detailed examination of the food-water-soil nexus concept through ESs and EDSs lenses. modification and development of the two-tier FWS nexus assessment framework, its indicators, and the associated ranking system. It explains how these indicators can be distributed to form five sustainability attributes.

Chapter 4 expands upon the FWS framework to analyse and discuss the sustainability of UFGs within four allotment gardens located in Nantes. This chapter details the datasets and data collection methods employed, alongside the statistical analyses conducted. Additionally, it offers an interpretive discussion of the results, highlighting how prioritizing one sustainability dimension may influence the performance of others.

Chapter 5 synthesizes the overarching findings from the preceding chapters, presenting a cohesive integration of key insights, practical implications, policy recommendations, and contributions to existing knowledge. The chapter also addresses the study's limitations and suggests directions for future research. It concludes by highlighting the critical importance of developing context-sensitive,

participatory, and integrated tools to foster the advancement of sustainable urban food systems.

2 MAPPING THE LANDSCAPE OF URBAN FOOD GARDEN INTEGRATED ASSESSMENT: A CRITICAL REVIEW

DISCLAIMER

A part of this chapter has been submitted as an opinion paper. It is currently under review in the journal *Earth: Environmental Sustainability*, under the title “*Untangling the Food-Water-Soil nexus through the lens of Ecosystem Services and Disservices in urban gardens*”. The paper was authored by:

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Abstract

Urban food gardens (UFGs) are increasingly seen as key to urban sustainability, but the methods for evaluating their complex benefits and drawbacks are still disjointed. This systematic review charts and critically examines the various integrated assessment approaches for planning and evaluating UFGs. By adhering to PRISMA guidelines, 58 peer-reviewed articles were analyzed to map the dominant analytical methods. The analysis shows that the field is organized into five main methodological groups: (1) Quantitative Sustainability Assessment, (2) Spatial Planning & Optimization, (3) Systems & Nexus Modeling, (4) Socio-Ecological & Community Assessment, and (5) Empirical & Field-Based Studies. While material nexus (e.g., water-energy-food) is a popular high-level concept, most studies (60.3%) do not explicitly use it but other interconnections of non-material nexuses, revealing a gap between academic discourse and practical research. The discussion breaks down the inherent limitations of each group and identifies two main areas for future research: the need for better integration of different methodologies to connect quantitative and qualitative approaches, and the need to "re-ground" assessments in the dynamic dimension of soil—a vital yet consistently overlooked element of the urban food-energy-water nexus.

Keywords: *urban agriculture, nexus thinking, integrated assessment, social-ecological system.*

2.1 INTRODUCTION

The 21st century is marked by rapid urbanization, with most of the world's population now living in cities or semi-dense areas (Ehrlich et al., 2024). While this demographic shift drives economic growth, it also places unprecedented pressure on our urban systems, raising global concerns about food security, resource management, and climate resilience (Foley et al., 2005). In response, cities are actively pursuing more sustainable development pathways and are often seen as leaders in addressing climate change and fostering innovation for a resilient society. Among both traditional and modern approaches, urban edible green infrastructures (e.g., community gardens, allotments, and advanced rooftop farms) have become a vital strategy for promoting sustainable and resilient urban environments (Cabral et al., 2017; De la Sota et al., 2019). Although these forms of agriculture have been tied to human existence, enhanced understanding of social-ecological systems has highlighted urban agriculture's intricate connections with city life and its interactions with water, energy, land use, and socioeconomic systems (Barthel et al., 2010). This interconnectedness means that any intervention can trigger cascading effects, resulting in both beneficial synergies and challenging trade-offs (Depietri, 2022; B. Zhang & MacKenzie, 2024).

Amid efforts to create more resilient and sustainable urban food systems, a new approach—centered on interactive and place-based research—is emerging across the European Union: the urban living laboratory for co-designing nature-based

solutions (NbS) (Lupp et al., 2020). These environments are not laboratories in the traditional sense. Rather, they are real-world spaces woven into the fabric of the city, providing a vibrant setting where diverse stakeholders—researchers, local residents, municipal authorities, and private sector partners—can come together. Within these living laboratories, participants collaboratively design, test, and adapt innovations in urban agriculture, addressing challenges as they arise in urban contexts (Bulkeley et al., 2019; Ternell et al., 2024). What makes this paradigm particularly compelling is its departure from conventional, linear models of research and development. Instead, urban living laboratories foster a genuinely dynamic process of co-creation and social learning of various agriculture NbS, such as community gardens, green corridors, or rooftop gardens (Galdámez et al., 2025; Nunes et al., 2021). The European Union, through strategic initiatives under frameworks such as Horizon Europe, has actively championed the living lab model as a critical instrument for accelerating the agro-ecological transition. Programs like the "Mission 'A Soil Deal for Europe'" explicitly advocate for the establishment of urban living labs to address challenges of soil health and land use (Panagos et al., 2024), which are fundamental to the viability of urban farming. These initiatives recognize that the complexities of urban gardens—spanning spatial constraints, circular economy principles, and social equity—demand a multi-stakeholder and adaptive approach to innovation (Mumaw & Mata, 2022).

These complexities underscore a critical gap between its widespread promotion of urban agriculture and its effective, evidence-based implementation: the need for robust analytical tools and frameworks (Lehmann, 2018). Simple, ad-hoc, or single-issue approaches to planning invariably fall short of navigating the multifaceted dimensions of these systems. Suppose agroecological systems are to be successfully integrated into urban planning and policy; in that case, decision-makers should have access to systematic methods capable of assessing, modeling, and evaluating the intricate web of impacts stemming from various urban edible green infrastructure typologies and management strategies. Embracing such a comprehensive and analytical approach not only enables more strategic decision-making but also creates opportunities for meaningful progress toward sustainable urban food systems (Jahrl et al., 2021; Russo & Cirella, 2019).

While the 'nexus' concept has become a dominant paradigm in sustainability science (Ghodsvalli et al., 2019), its application to the hyper-local, socially-embedded context of urban gardens remains contested.. Meaningful application nexus thinking to edible urban gardens requires us to broaden our conceptual horizons. The complex reality of these spaces called for knowledge beyond the familiar water-food-energy (WEF) triad, embracing more nuanced frameworks such as the food-energy-waste-water (FEWW) nexus (Garcia et al., 2019) or soil biodiversity-climate change-society nexus (Eisenhauer et al., 2024). This broadening of scope is both necessary and challenging. The systems interacting

within a single community garden exemplify this complexity. These gardens are uniquely situated: they are intimately managed at the community level while simultaneously embedded within larger flows of food and materials. The factors that influence their success or failure are not confined to the biophysical, but encompass social, cultural, and political processes as well. As such, any effective nexus-based planning or assessment does not simply call for interdisciplinary insight; it demands profound inter- and transdisciplinary collaboration. Successful adaptation of such systems could significantly improve food security in urban environments, as evidenced by the Cuban urban agriculture system, which accounted for 60% of the country's demand for vegetables (Hanon, 2015).

The global interest in urban agriculture, particularly those initiatives linked to NbS, has seen a remarkable increase in recent years (Artmann & Sartison, 2018; Yan et al., 2022). Emerging themes related to cultivated urban areas often encompass the optimization of site selection, the assessment of environmental performance, and the evaluation of the benefits these systems provide. In summarizing the outcomes of urban agriculture, Rao et al., (2022) emphasizes the lack of an integrated assessment that captures the multiple benefits arising from these practices; this gap significantly hampers the mainstreaming and formalization of urban agriculture across various sectors. Similarly, Kamali et al., (2025), while reviewing methods for selecting appropriate sites for urban food production, advocates for a greater integration of social criteria within spatial models. This

approach is particularly important for addressing inequities faced by vulnerable populations. Such integration is crucial for ensuring that urban agriculture not only thrives but also serves as a tool for social equity and environmental sustainability.

Thus, this review investigates the extent to which nexus thinking has been adopted and critically assesses the broader landscape of integrated assessment tools used in its place. We presents a systematic review of the academic literature to identify, categorize, and synthesize the current landscape of assessment frameworks and tools designed to analyze the nexus of urban food gardens. In developing this review, we found academic dicussion surrounding urban food production suffers from a critical lack of terminological precision. Broad terms like 'urban agriculture' often conflate vastly different models, from industrial, profit-driven vertical farms to non-profit, community-managed plots. This ambiguity makes a systematic analysis of assessment tools difficult, as methodologies designed for one model may be inappropriate for the other. For the purposes of this review, we establish a precise analytical category Urban Food Garden (UFG), which is defined as ***“a public or private space within or on the fringe of urban boundaries, which is primarily used for growing vegetative plants for self-consumed food and other non-commercialized purposes”***. This definition is crucial because it deliberately excludes profit-driven models, allowing our analysis to focus squarely on practices aligned with NbS and ecosystem service (ES) benefits. Some commonly known forms of UFG are communal gardens, allotments, domestic gardens, urban forest

gardens, and guerrilla gardens. Unlike the broader term 'edible green infrastructure,' which focuses on food provisioning (Russo et al., 2017), our use of UFG specifically emphasizes soil-based, low-energy gardens that provide a diverse collection of cultural and regulating ES in addition to food.

By adopting PRISMA guidelines (Page et al., 2021), this review aims to answer the following questions: 1) What are the nexus analytical/modelling approaches that assist urban food gardens planning and evaluation? 2) What are the key characteristics and limitations of these frameworks? 3) What are the research gaps and future directions for developing integrated assessment tools? This literature review employs a quantitative and evidence-based approach, enabling the authors to identify nexus systems, their respective scales, methodological approaches, and the impact of these systems on planning and governance. Finally, this literature and insights could help craft an integrative research approach to the urban food garden system, proposing alternative pathways and dialogue to understand nexus systems dynamics and interactions.

2.2 METHODS

2.2.1 Literature search design

In this paper, the focus was on original research works that present an integrated assessment on UFGs, linking multiple input factors or outcomes (i.e., social, environmental, and economic) for planning and optimizing UFGs. We applied a systematic literature review to examine integrative research methodologies for UFG systems, as described in Snyder, (2019). While the UFG topic is not new, the application of multidimensional research approaches for such systems remains limited. Aiming at providing a synthesis lens and mapping out diverse methodological approaches, we defined three searching pillars that cover subject contexts, study methodological approach, and study scale. Multiple related synonyms and word variants were included to ensure a comprehensive search.

The three searching themes are:

- i) The integrative concept: studies with an explicit focus on the interconnectedness
- ii) The assessment concept: studies presenting a methodological approach for evaluation
- iii) The subject concept: specific typologies of UFGs, excluding studies on general urban food systems or rural agriculture. Since researchers could use

different terms for UFGs or conduct assessments across multiple UFG systems, the keywords were expanded to include urban agriculture and peri-urban Search procedure and selection criteria

2.2.2 Search procedure and selection criteria

A search for academic literature was performed on Web of Science core collection on 12 March 2025, and one more time in August 2025 to enhance coverage and relevance. We use WOS as it is widely recognized as one of the most reliable sources of bibliographic data in the scientific community. The search was limited to peer-reviewed articles published in the English language until August 2025 to capture the modern evolution of the field. The search string was constructed using three conceptual themes, which the operator combined “AND”. Terms within each thematic poctics were combined using the OR operator.

- Query 1: ("nexus" OR "interlinkage*" OR "interconnection*" OR "interdependen*" OR "trade-off*" OR "tradeoff*" OR "synerg*" OR "cross-sector*" OR "integrated approach" OR "co-benefit*")
- Query 2: ("assess*" OR "framework*" OR "evaluat*" OR "model*" OR "indicator*" OR "index" OR "indices" OR "tool*" OR "metric*" OR "analy*")
- Query 3 iii: ("urban agriculture" OR "urban farming" OR "allotment garden*" OR "community garden*" OR "edible green infrastructure" OR "backyard

garden*" OR "kitchen garden*" OR "edible landscape*" OR "edible garden*" OR
"communal garden*" OR "edible green roof*" OR "school garden*" OR "edible
botanic garden*" OR "edible green wall*" OR "edible forest garden*" OR
"neighborhood garden*" OR "residential garden*" OR "guerrilla garden*")

After excluding book chapters, conference proceedings, and meeting abstracts, the literature search yielded a total of 298 research publications. We performed an initial screening of the documents based on their title and abstracts, removing studies that are not related to the UFG concept or focus on a single issue or outcome. This resulted in 137 articles, all of which the authors were able to access the full text. A second round of screening based on reading of the full text reduced this number to 58 articles by removing conceptual papers that do not propose or apply a replicable assessment framework, model, or tool, purely descriptive studies (e.g., single-case ethnographies, historical analyses, policy commentaries) that do not present a replicable methodology for assessment purposes. Papers were selected if they satisfied the following criteria:

- Study subject: one or more specific typologies of UFGs.
- Study lens: adopt an integrated, nexus, or relational perspective, analyzing the trade-offs, synergies, co-benefits, or interconnections between multiple (e.g., social, environmental, economic) factors or outcomes.

- Study methodology: propose or apply a replicable assessment framework, model, or tool. This includes quantitative models (e.g., Life Cycle Assessment (LCA), Life Cycle Costing (LCC), Multi-Criteria Decision Analysis (MCDA), optimization models, statistical models), empirical experiments, and structured qualitative frameworks designed for systematic assessment (e.g., a formal

protocol for comparative case study analysis, a specific multi-criteria qualitative evaluation scheme).

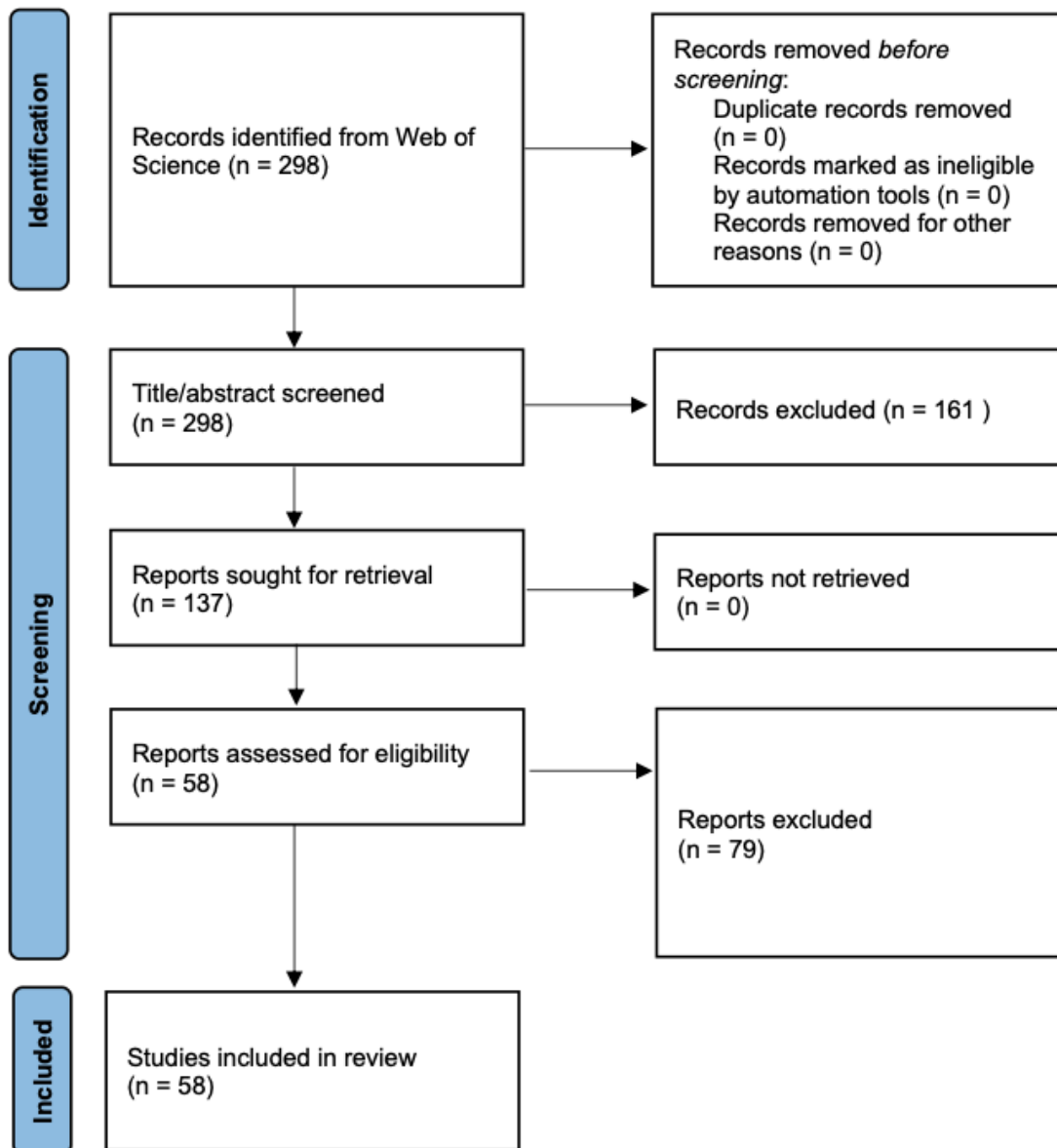


Figure 2.1. The PRISMA flow diagram for systematic review of UFGs

2.2.3 Data Analysis

We conducted a comprehensive bibliometric analysis of the final dataset, which comprised 58 documents selected after meticulous screening. This analysis aimed to systematically document the study subjects, scales, locations, methodological approaches, input parameters, and key performance indicators associated with these works. In the subsequent stage, we undertook a detailed full-text analysis to explore the thematic groupings evident within the database, examining their evolution and the relationships between various research methods and their primary subjects of inquiry. Through this analysis, we identified common patterns and indicators frequently associated with specific methodological clusters.

Moreover, this in-depth examination provided valuable insights into the connections between these research approaches, as well as the limitations acknowledged by the authors and the existing knowledge gaps in the literature. As our review reveals the intricate landscape of assessment tools for UFGs, it also facilitates a discussion of the potential synergies among these categorized methods and outlines promising directions for future research.

2.3 RESULTS AND ANALYSIS

2.3.1 Overview of analytical landscape and methodological flows

This review analysed 58 papers that met our specific criteria. Although research on city and regional-scale nexuses is still emerging, the earliest publication we found dates back to 2015. That pioneering study introduced a water footprint model for urban agriculture, incorporating spatial analysis, an evapotranspiration model, and potential crop yield assessment. As shown in Figure 2.2, the range of nexus approaches for UFGs has expanded significantly over time. Despite the relatively small number of publications overall, those that

Number of per-reviewed articles and total citation on systematic assessment of UFGs per year

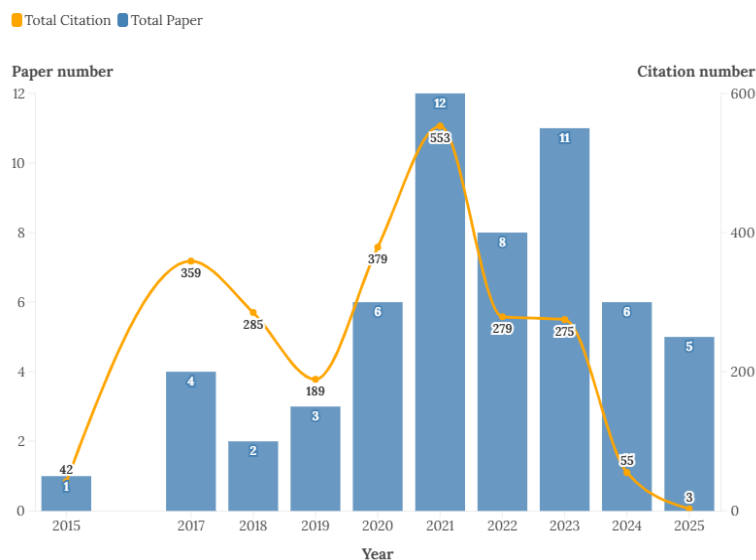


Figure 2.2. Academic publication overview on the integrated approaches for UFGs assessment

involve integrated assessments of UFGs have garnered a notably high number of citations, around 42 per paper, underscoring their substantial influence within the academic community.

Among the analysed papers, 23 (39.7%) directly engaged with the core components of the WEF nexus, although many underwent conceptual evolution over time. The WEF nexus framework was explicitly employed in 10 studies (17.7%), primarily focusing on resource efficiency and the mechanics of circularity. Interestingly, an additional five studies (8.6%) implicitly integrated all three WEF components without explicitly naming the nexus, indicating a broader recognition of the integrated approach in studying these interconnected systems. Furthermore, eight studies expanded beyond the core WEF framework to incorporate other dimensions. These additional dimensions often represented a conceptual shift from a purely technocratic model to a socio-ecological perspective (e.g., *people* and *governance* dimensions) or toward a more comprehensive circular economy model (e.g., *waste* dimension). Researchers also adapted the core nexus idea to address pressing, context-specific urban challenges that extend beyond biophysical resource flows, encompassing factors such as *climate*, *transportation*, *land use*, and *economic* dimensions. Beyond WEF-related studies, most research, about 60.3%, or 35 papers, does not utilize an explicit, multi-resource nexus framework (i.e., WEF). Instead, these studies focus on understanding specific interconnections: nexus between agricultural practices and

measurable outcomes (*practice-to-performance*), nexus between people and the gardens themselves (*people-to-place*), or the nexus between a garden's location and a particular urban problem (*site-to-city*). These studies tend to adopt a more methodologically focused approach, often with a narrower scope and specific techniques. Despite this, they provide essential foundational knowledge on yields, social benefits, and environmental impacts.

We identified five main but interconnected methodological groups. Each group has its own key research questions, preferred tools, and typical ways of working. These groups include: (1) Quantitative Sustainability Assessment, (2) Spatial Planning & Optimization, (3) Systems & Nexus Modeling, (4) Socio-Ecological Assessment, and (5) Empirical & Field-Based Studies. The structural flow between these groups connects, along with their methods and the scales they work at, to follow a clear pathway outlined in **Table 2.1** and discussed in the following sections.

Table 2.1. Overview of methodological clusters and research methods in UFGs

Methodological Cluster	Research Method	Count	Reference
Quantitative Sustainability Assessment	LCA	8	Caputo et al., 2021; Dorr et al., 2017;
			Imam et al., 2025; Meng et al., 2023;
			Sanjuan-Delmás et al., 2018; Stone et
			al., 2021; Toboso-Chavero et al., 2019; Zhen et al., 2020)

	LCC	2	Dorr et al., 2017; Zhen et al., 2020
	Environment Foot printing & Emergy	2	McDougall et al., 2019; Stone et al., 2021
Spatial Planning & Optimization			(Camacho-Caballero et al., 2024; Chen et al., 2023; Cumo et al., 2025; Fanfani et al., 2022; Kafle et al., 2023; Lin et al., 2025; Liu et al., 2022; Lowan-Trudeau et al., 2020; Lucertini & Di Giustino, 2021; M. Susana et al., 2023; Pedro et al., 2021; Small et al., 2023; Smith et al., 2021; Toboso-Chavero et al., 2023; Ustaoglu et al., 2021; Zambrano-Prado et al., 2021; Y. Zhang et al., 2022; Zheng et al., 2025)
	GIS & Remote Sensing	18	
	MCDA & AHP/Delphi	9	Babaei et al., 2024; Camacho-Caballero et al., 2024; Kong et al., 2025; M. Susana et al., 2023; Nie et al., 2024; Smith et al., 2021; Toboso-Chavero, Villalba, Gabarrell Durany, & Madrid-Lopez, 2021; Ustaoglu et al., 2021; Valencia, Qiu, et al., 2022
	Optimization Modelling	6	Kafle et al., 2023; Liu et al., 2022; M. Susana et al., 2023; Valencia, Zhang, et al., 2022; Y. Zhang et al., 2022; Zheng et al., 2025

Systems & Nexus Modelling	Systems Modelling (SDM, ABM, MFA)	11	Elkamel et al., 2023; Gondhalekar & Ramsauer, 2017; Goodarzi et al., 2022; Huang & Chang, 2021; Jans-Singh et al., 2021; Pueyo-Ros et al., 2024; Valencia, Hossain, et al., 2022; Valencia, Zhang, et al., 2022; Weidner & Yang, 2020
Socio-Ecological & Community Assessment	Survey, semi- structured interview, statistical analysis	10	Babaei et al., 2024; Caputo et al., 2021; Kirby et al., 2021; Kong et al., 2025; Lowan-Trudeau et al., 2020; Nie et al., 2024; Rodríguez González et al., 2023; Siegner et al., 2020; Toboso-Chavero, Madrid-López, Durany, & Villalba, 2021; Valencia, Qiu, et al., 2022
Empirical & Field- Based Studies	Field Measurement and ES Evaluation	9	Caputo et al., 2023; Dennis & James, 2017; Dorr et al., 2023; Grard et al., 2018; Holt et al., 2017; Jha et al., 2023; Mulier, van de Ven, et al., 2022; Taylor, 2020; Wright et al., 2021
<i>Note: Total label is higher than the number of analyzed paper because a single paper often employs multiple methods. Further analysis is at section 2.3.3</i>			

2.3.1.1 Quantitative Sustainability Assessment

This cluster encompasses studies that aim to develop objective, quantifiable metrics for assessing environmental and economic performance. It is based on the principle that sustainability can be measured, compared, and benchmarked.

Central to this approach is LCA, which evaluates environmental impacts across various concerns such as global warming potential, eutrophication, and fossil fuel depletion (Dorr et al., 2017; Meng et al., 2023; Sanjuan-Delmás et al., 2018). LCA is often complemented with LCC to provide a parallel economic assessment (Dorr et al., 2017; Zhen et al., 2020), thereby enabling an eco-efficiency analysis for decision-makers. Additional methods employed within this cluster include Environmental Footprinting (e.g., carbon, water) and Energy Analysis, which quantify the total embodied energy of a system (McDougall et al., 2019). Primary inputs were detailed inventories of material and energy flows, meticulously cataloged for LCA. The reviewed papers consistently draw upon inputs from infrastructure materials, substrates, fertilizers, and energy consumption, as the process aims to create a comprehensive ledger of all resources extracted from the economy and environment to support the production process of UFGs. **Figure 3** illustrates common input parameters across five methodological clusters.

Research within this cluster generally adheres to a highly structured and linear methodology. Workflows typically start with establishing the system boundaries and the definition of the functional unit (e.g., “1kg of tomatoes produced.”) This is followed by a comprehensive life cycle inventory phase, during which all material and energy inputs and outputs are meticulously documented. The final phase involves life cycle impact assessment, which translates the inventory data into environmental impact indicators. This systematic approach enables reliable

comparisons between different systems, such as an integrated rooftop greenhouse versus a conventional one (Sanjuan-Delmás et al., 2018). Therefore, the methodological framework is frequently applied at the plot and building scales, where the well-defined boundaries of a single system (e.g., a rooftop farm) facilitate detailed inventory analysis. It can also be employed at a conceptual or multi-scale level to compare entire food systems, for example, contrasting large-scale conventional agriculture with small-scale local production (Stone et al., 2021).

2.3.1.2 Spatial Planning & Optimization

The Spatial Planning & Optimization cluster focuses on determining optimal locations and configurations for UFG interventions to maximize their benefits. Its primary aim is to develop decision-support tools for urban planners, policymakers, and community organizers. The core tool utilized by the cluster is Geographic Information System (GIS), which facilitates mapping and analysis of spatial data alongside a broad spectrum of socio-economic information. GIS is typically combined with MCDA, a framework for assessing alternatives based on multiple, often conflicting, criteria such as social equity, environmental benefits, and economic costs. To operationalize MCDA, methods like the Analytic Hierarchy Process (AHP) are used to derive criteria weights from expert or stakeholder input (Ustaoglu et al., 2021). For more complex applications, optimization algorithms are integrated to identify the most efficient spatial configurations that address

multiple objectives simultaneously—such as reducing food deserts and urban heat islands (Y. Zhang et al., 2022). The most frequent inputs for this cluster are spatial representations: Land Use / Land Cover (appearing in 18 papers), Building Data (14 papers), and Infrastructure Networks (11 papers). These are frequently overlaid with socio-economic and demographic data, such as population density, income levels, and health vulnerability indicators.

Spatial analysis typically involves selecting specific criteria and gathering corresponding spatial data layers, such as income levels, vacant land inventories, and land surface temperatures. A key step in the process is assigning weights to these criteria, often through participatory expert rankings. The weighted layers are then overlaid within a GIS environment to generate a final suitability or priority map, helping to identify optimal locations for new UFG projects. Since the methods are designed for large-area analysis, this cluster operates mainly at community, neighborhood, city, and regional scales. Studies have employed this approach for strategic planning, such as identifying priority zones for community gardens across metropolitan areas (Smith et al., 2021) or planning for renewable energy communities (Cummo et al., 2025).

2.3.1.3 Systems & Nexus Modelling

This cluster seeks to understand and simulate the complex, dynamic, and non-linear interactions within urban resource systems. The literature clearly

demonstrates a conceptual maturation of the nexus framework within the cluster; it moves beyond static assessments to explore feedback loops, trade-offs, and systemic behaviour over time. While early applications focused on the core WEF triad, the nexus expanded to other resource nexuses to address key urban challenges. The inclusion of "Waste" to form a Food-Energy-Water-Waste nexus is a significant pattern, seen in studies such as Valencia et al., (2022) and Weidner & Yang (2020), which explicitly model waste streams (e.g., organic waste, wastewater) as resource inputs for UFGs, thus embedding the principle of a circular economy. A further evolution is the inclusion of the "People" dimension, as articulated by Caputo et al., (2021). This is a critical theoretical shift, arguing that social dynamics, behaviours, and well-being are not just outcomes but integral components of the nexus that influence resource flows. Adding the "People" dimension adds complexity, often requiring the integration of social survey data or agent-based models (Elkamel et al., 2023) to capture human decision-making. To reflect the behaviours of systems over time, this cluster often employs rates, stocks, and feedback parameters. The most common inputs are resource demand and supply data for water, energy, and food (appearing in 10-11 papers each), often as time series. These are supplemented by technology specifications (e.g., PV efficiency) and external drivers like climate change projections.

The primary methods within this cluster are simulation-based approaches. System Dynamics Modelling (SDM) is employed to capture feedback loops and the stock-and-flow dynamics, while Agent-Based Modelling (ABM) simulates the emergent, system-level consequences stemming from individual actors' behaviours. The process is inherently iterative, beginning with a conceptual mapping of the system's components and their interrelationships. This conceptual framework is subsequently translated into a quantitative simulation model, which is calibrated and validated using real-world data. The core of the analysis involves executing the model under various "what-if" scenarios (e.g., climate change projections or policy interventions) to explore potential systemic responses and trade-offs. Besides SDM and ABM, specialized models have been developed for more specific objectives and criteria. These include a co-simulation model developed by Jans-Singh et al., (2021) for detailed technical simulations of energy and growth dynamics of a building-integrated greenhouse. Meanwhile, Pueyo-Ros et al., (2024) developed a stochastic modelling of to estimate UFGs contribution to sustainability across a Mediterranean city. At the city and regional level, these methods were utilized to model urban metabolism and the interactions among extensive infrastructure systems (Valencia, Zhang, et al., 2022). These simulation models could also integrate spatial data from GIS to provide context and inputs (Gondhalekar & Ramsauer, 2017), or LCA to assess the environmental impacts of different resource flow scenarios modelled within the system (Valencia, Hossain, et al., 2022)

2.3.1.4 Socio-Ecological & Community Assessment

The Socio-Ecological & Community cluster systematically explores the human dimensions of UFGs, emphasizing the social, cultural, and political factors that influence its practices and outcomes. Research within this domain often prioritizes understanding participant experiences and broader community impacts through a rigorous mixed-method approach, combining qualitative and quantitative techniques. Commonly employed tools include surveys, semi-structured interviews, and case studies to collect quantitative data on motivations and perceived impacts (Kirby et al., 2021; Siegner et al., 2020). Frameworks such as ES and sustainable livelihoods frequently guide data collection and interpretation (Rodríguez González et al., 2023). Studies uniquely focus on human perceptions, values, and well-being when applying social science methodologies to measure concepts like general well-being, food security, and social cohesion. MDCA and AHP were also employed to determine indicators and their respective weights for the sustainability assessment of UFGs (Babaei et al., 2024; Nie et al., 2024). In this paradigm, an "indicator" is not just a material resource but can also be the social trust or traditional knowledge that enables a garden to function. Therefore, their main inputs are socio-economic demographics of participants (10 papers), participant motivations, stakeholder perceptions (often gathered via AHP/Delphi), and governance structures.

Researchers engaged in this cluster typically collaborate directly with community members and decision-makers to facilitate data collection. Data analysis involves thematic coding of qualitative data (Siegner et al., 2020) or statistical analysis of surveys, with findings interpreted through established social or ecological theories. Therefore, this cluster predominantly operates at the community and neighbourhood levels, or is extended to conceptual or multi-scale frameworks, as the methods effectively investigate detailed social dynamics within specific communities or develop broader understandings of sustainability that incorporate social values.

2.3.1.5 Empirical & Field-based studies

Empirical and field-based studies are rooted in the scientific method, emphasizing direct observation and measurement. These studies frequently aim to generate primary, empirical evidence concerning the performance and impacts of UFGs under both real-world and controlled experimental conditions. For example, Dennis & James, (2016) conducted on-site measurements of variables such as crop yield, water usage, soil quality, and biodiversity within operational gardens. Similarly, Taylor compared different management practices in controlled environments. Researchers also utilize citizen science, employing standardized protocols to collect data. The final outcomes typically involve measurements of on-the-ground processes, acting as proxies for ES. Instances include quantifying rainwater runoff reduction as a percentage metric for stormwater management

(Wright et al., 2021), or measuring aphid removal rates as an indicator of pest control efficiency by beneficial insects (Jha et al., 2023). These key performance indicators are distinctive because they represent direct, empirical measurements rather than modelled or perceived outcomes typical of other methodological approaches. A notable synergy in this approach is the application of GIS to analyse how surrounding landscape contexts influence the measured variables, thereby linking local processes to broader spatial patterns (Jha et al., 2023). The analysis depends heavily on statistical methods to assess the significance of observed results. This approach provides foundational data, such as crop yields and resource use efficiencies, that are often used to parameterize models in other research clusters. Its inputs are defined by what can be directly and physically measured on-site. The most common inputs are site and cultivated area, measured water consumption, and soil amendment application rates. Crucially, this is the only cluster where soil quality metrics appear as a significant input parameter (Taylor, 2020). This reflects a bottom-up approach to knowledge generation, where the analysis is built upon a foundation of primary, ground-truthed data. The inputs are not proxies for a larger system but are direct measurements of the system itself, albeit at a very localized scale.

The research process adheres to a traditional empirical workflow: forming hypotheses, designing experiments or data collection protocols, collecting and analysing primary data through statistical methods, and drawing evidence-based

conclusions. This methodology is versatile, applicable across various scales, from detailed, plot- or building-level studies (Grard et al., 2018) to multi-site analyses, such as evaluating the resource nexus among multiple farms within an urban area (Caputo et al., 2023; Mulier et al., 2022).

INPUT INDICATORS FOR SYSTEMATIC ASSESSMENT OF URBAN FOOD GARDEN

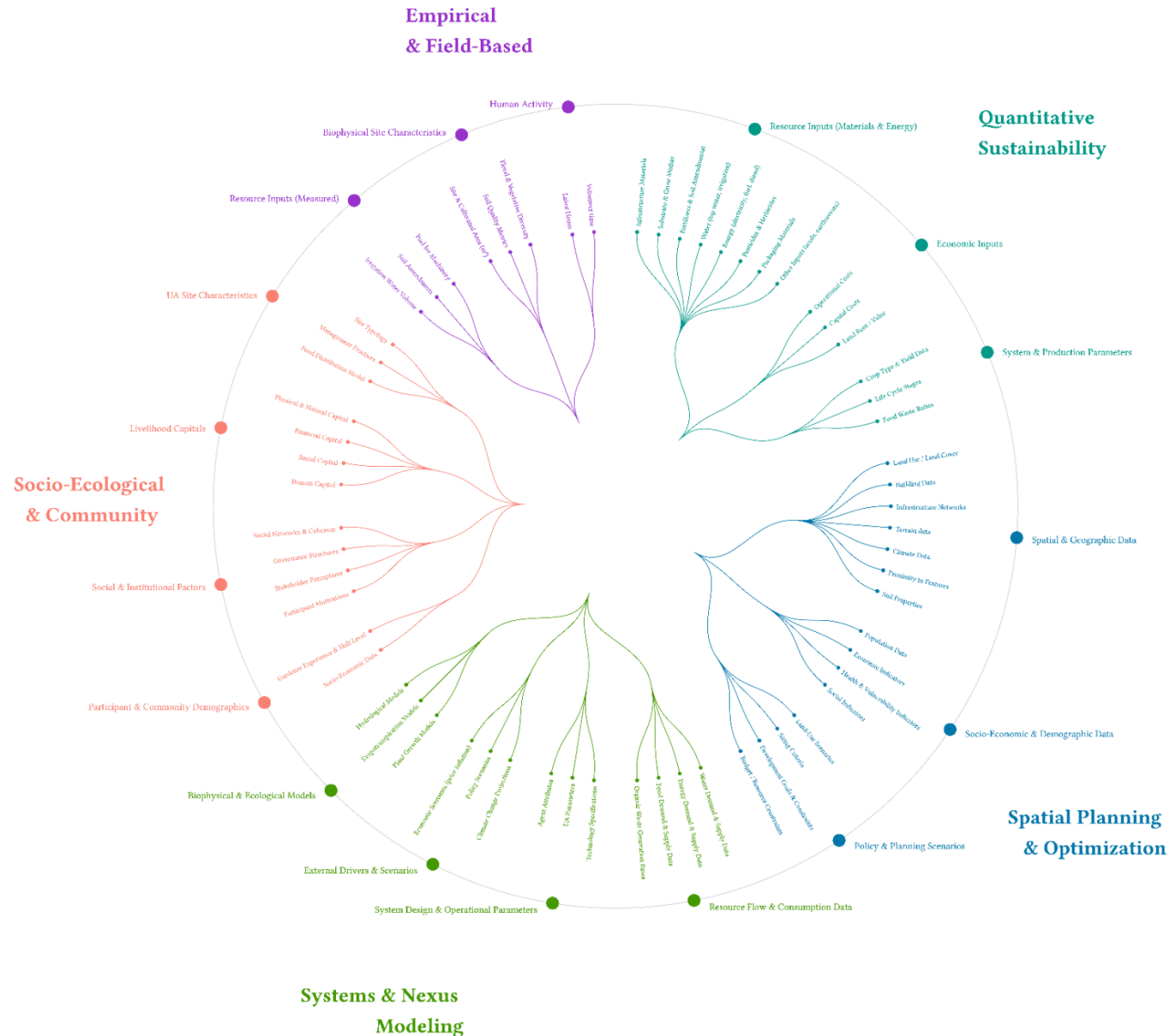


Figure 2.3. Indicators used in UFGs assessment (an interactive version can be assessed at <https://kien-pham.com/UFGs>)

2.3.2 Overlap in methodological cluster

2.3.2.1 Fundamental overlapping between clusters

While categorizing research into distinct methodological clusters offers a useful perspective, some individual papers in our database employed more than one scientific approach to develop a more holistic and problem-oriented understanding. **Table 2.2** highlights seven studies out of the 58 analysed that primarily utilize a methodology from one cluster while extensively integrating key methods or frameworks from another. Notably, spatial assessments using GIS often incorporate a variety of input data, as demonstrated in Table 2.3, where GIS plays an essential role across different methodological clusters, serving as either a primary or secondary approach.

Table 2.2. Overview of studies that employed core methods from a secondary cluster

Primary Cluster	Secondary Cluster	Reference	Rationale for Overlap
Spatial Planning & Optimization	Quantitative Sustainability Assessment	Liu et al., 2022	Uses LCA to quantify the environmental impacts of the spatially optimized WEF practices.

Spatial Planning & Optimization	Systems & Nexus Modeling	Camacho-Caballero et al., 2024	Integrates ecosystem service and urban metabolism (a nexus concept) analysis into its spatial MCDA framework.
Spatial Planning & Optimization	Socio-Ecological Assessment	Lowan-Trudeau et al., 2020	Combines GIS mapping with statistical analysis of census data to explore the socio-demographic context of gardens.
Systems & Nexus Modeling	Quantitative Sustainability Assessment	Valencia, Hossain, et al., 2022	Integrates LCA principles into its system dynamics model to quantify decarbonization pathways.
Systems & Nexus Modeling	Spatial Planning & Optimization	Gondhalekar & Ramsauer, 2017	Uses detailed GIS and spatial statistics as the foundational input for its city-scale WEF nexus model.
Socio-Ecological Assessment	Quantitative Sustainability Assessment	Caputo et al., 2021	Combines a participatory social assessment with formal LCA and Material Flow Analysis in its "FEWP" nexus framework.
Empirical & Field-Based Studies	Spatial Planning & Optimization	Jha et al., 2023	Uses GIS to analyze how the surrounding landscape context (a spatial variable)

mediates the empirically
measured ecosystem services.

2.3.2.2 Scenario applications across methodological clusters

Scenario methods were employed in 22 papers, representing 37.9% across various methodological clusters. Though widely used, these methods differ in logic and form but often share characteristics within clusters (see **Table 2.3**). They were mainly employed in the System & Nexus Modelling and Spatial Planning & Optimization clusters.

The most common scenario type is the comparative scenario. For instance, in the Quantitative Sustainability Assessment cluster, Sanjuan-Delmás et al., (2018) compare an integrated rooftop greenhouse with a conventional one. Similarly, Empirical & Field-Based Studies use experimental treatments as micro-scenarios (e.g., Taylor, (2020)'s soil management systems). In Spatial Planning & Optimization, scenarios are often normative, depicting desirable futures to guide policy, such as Fanfani et al., (2022)'s comparison of current land use with an agroecological vision. The Systems & Nexus Modelling cluster employs exploratory scenarios to navigate uncertainty, testing models against external forces like climate projections (e.g., RCP scenarios in Goodarzi et al., (2022) or socio-economic pathways (Elkamel et al., 2023). The goal is to find strategies robust

across multiple futures rather than predict specific outcomes. Finally, the Socio-Ecological cluster features participatory scenarios co-created with stakeholders, reflecting diverse values. For example, Kong et al., (2025) assess urban farms under different governance scenarios focused on social and political futures.

Table 2.3. Type of scenario analysis deployed in different methodological clusters

Methodological Cluster	Papers Using Scenarios	Frequency within Cluster	Primary Scenario Type
Systems & Nexus Modelling	9	81.80%	Exploratory Scenarios: Used to test the resilience of a system against external, uncontrollable forces (e.g., climate change projections, policy shocks, price inflation).
Spatial Planning & Optimization	7	38.90%	Normative Scenarios: Used to compare alternative, designed land-use plans or policy pathways to achieve a desirable future state.
Quantitative Sustainability Assessment	3	33.30%	Comparative Scenarios: Used to compare the performance of different technologies, management practices, or system configurations (e.g., integrated vs. conventional greenhouse).

Socio-Ecological & Community Assessment	2	20.00%	Participatory Scenarios: Used to represent different stakeholder-defined visions for the future or alternative governance structures.
Empirical & Field-Based Studies	1	11.10%	Treatment Scenarios: The comparison of different experimental treatments (e.g., soil management systems) can be understood as a form of micro-scenario analysis.

2.3.3 Scale of Analysis

The analysed papers were categorized by scale, from individual plots and buildings to city regions, as shown in Table 4. Methodologies such as Spatial Planning & Optimization primarily targeted mid to large scales, focusing on Community & Neighbourhood (45%) and City & Regional (40%) levels. In contrast, System & Nexus Modelling and Quantitative Sustainability Assessment centred on micro-levels, with 42% and 33%, respectively.

The Plot and Building scale includes gardens, rooftop farms, or integrated building systems. Research here mainly involves controlled experiments aimed at understanding biophysical processes under specific conditions. This approach isolates variables to establish clear causal relationships. Empirical & Field Studies and Systems & Nexus Modelling dominate this scale, with models like the co-

simulation of greenhouse and building energy balances (Jans-Singh et al., 2021), and LCA-based assessments such as Sanjuan-Delmás et al., (2018).

At the Community & Neighbourhood level, research shifts focus from biophysical processes to social dynamics, spatial access, and lived experiences. Social science methods like surveys and case studies are prevalent, exemplified by the Socio-Ecological & Community Assessment cluster (3 papers). Spatial Planning & Optimization (5 papers) examines the distribution of neighbourhood gardens, optimizing locations to meet community needs (e.g., Y. Zhang et al., 2022). Multi-site empirical studies (2 papers) analyse interactions between social and ecological variables across different gardens (Caputo et al., 2023; Jha et al., 2023)

The city and regional scale rely on models and large datasets to represent complex urban systems, as direct observation is impractical. This scale predominantly employs Spatial Planning & Optimization (8 papers) and Systems & Nexus Modelling (6 papers). GIS-based analyses are central here, as seen in city-wide suitability assessments (Ustaoglu et al., 2021) and ecosystem service mapping (M. Susana et al., 2023). System models simulate urban metabolism, exemplified by WEF nexus studies (Elkamel et al., 2023; Gondhalekar & Ramsauer, 2017). Other clusters are less frequent, often in multi-site studies, generalizing findings across cities.

Finally, the Conceptual/Multi-scale category is analytical rather than physical. It emphasizes developing and testing general frameworks, comparing systems across contexts, or synthesizing multi-scale findings. The Socio-Ecological & Community Assessment cluster (5 papers) here develops broad indicators (Babaei et al., 2024) or compares social impacts internationally (Kirby et al., 2021). The Quantitative Sustainability Assessment (5 papers) compares entire food systems using methods like LCA (e.g., Stone et al., 2021). Spatial Planning & Optimization (3 papers) and Empirical studies (1 paper) contribute primarily when creating generalizable frameworks.

Table 2.4. Overview of the methodological clusters with the scale of analysis

Scale of Analysis	Quantitative Sustainability Assessment	Spatial Planning & Optimization	Systems & Nexus Modelling	Socio-Ecological & Community Assessment	Empirical & Field-Based Studies
Plot & Building	4 (33%)	-	5 (42%)	-	3 (25%)
Community & Neighbourhood	-	5 (45%)	1 (9%)	3 (27%)	2 (18%)
City & Regional	-	8 (40%)	6 (30%)	2 (10%)	4 (20%)
Conceptual / Multi-Scale	5 (36%)	3 (21%)	-	5 (36%)	1 (7%)

2.4 DISCUSSION

The results demonstrate a clear and logical alignment between the scale of inquiry and the chosen methodology, with quantitative, model-driven approaches dominating city- and regional-scale analyses, while qualitative and empirical methods provide deep, contextualized knowledge at the plot and community levels. This descriptive mapping, however, naturally gives rise to a more critical set of questions. Having established what methods are used and where, it is now necessary to explore what this methodological architecture implies for the production of knowledge in the field. The following discussion deconstructs the inherent limitations of each paradigm, explores the unstated assumptions and epistemological blind spots that shape their findings. By examining the tensions between quantification and justice, optimization and participation, and modelling and lived experience, this section seeks to illuminate not only the strengths of the current research but also the critical frontiers where a more integrated and reflexive science is urgently needed.

2.4.1 Critical reflection on nexus approach in UFGs

While we framed this review through a nexus lens to capture the interconnectedness of UFGs initially, a majority of the studies themselves (60.3%) do not explicitly adopt a multi-component nexus framework like WEF. This finding reveals a mismatch when applying the conventional nexus framework for the

unique context of UFGs. The WEF nexus framework was originally conceptualized to address resource security challenges at large geopolitical scales, such as transboundary river basins or national resource strategies (Hoff, 2011; Palazzo et al., 2024). In contrast, a community garden operates at a hyper-local scale, often characterized by decentralized, informal, and socially mediated resource management. Resource flows from various sources are significantly challenged to capture in traditional nexus models (e.g., irrigation water may come from municipal tap, or rainwater harvesting barrel, or a neighbour's hose). Therefore, the direct application of a large-scale framework to a neighbourhood-scale UFG can be an inappropriate, analytical fit.

A significant challenge arises from the conventional epistemological orientation of the nexus framework, which is often perceived as technocratic and resource-focused approach as it emphasizes quantifiable biophysical flows and technical efficiency (Allouche, 2024). However, Our research indicates that much of the UFG literature centres around the Socio-Ecological & Community Assessment cluster, emphasizing non-material benefits, often termed "incommensurable" by ecological economists (Satz et al., 2013), resist easy measurement or comparison with a cubic meter of water or a kilowatt-hour of energy. The evolution of the nexus concept within our reviewed studies underscores this tension. The introduction of a "People" dimension to form a FEWP (Food-Energy-Water-People) nexus by Caputo et al., (2021) aims to address this technocratic focus by

recognizing social dynamics as essential components of the system. This marks a necessary shift from a purely technical analysis of resource flows towards a more socio-ecological understanding.

We would argue that future research should not abandon the nexus concept but rather applying more critically and reflexively by moving beyond the WEF triad and embracing more context-specific framework, such as the FWEP nexus mentioned above or WEFS (Water-Energy-Food-Soil) nexus that grounds the system in its material foundation. Besides, nexus is one tool among many. For questions centred on power, equity, or social change, it must be supplemented with other analytical frameworks to bridge the gap between the biophysical and social, ensuring that UFGs assessments are socially just and meaningful.

2.4.2 Critical perspectives in integrated assessments of UFGs

In the Quantitative Sustainability Assessment cluster, LCA is a powerful tool aimed at creating an objective, comprehensive ledger of sustainability. While papers in our database often acknowledge limitations such as system boundary constraints and reliance on generic life cycle inventory databases, they rarely address the challenges posed by the unique nature of UFGs. A major issue is applying an industrial assessment tool to these non-industrial, complex systems. LCA was designed for standardized, linear production chains, but urban gardens are often

polycultural and highly biodiverse, complicating the estimation of nutrient flows, water use, and yields for multiple intercropped species. Existing life cycle inventory databases lack the detailed data needed for such diverse systems, leading researchers to make simplifying assumptions that may not be accurate. Another significant limitation is LCA's inability to capture the social dimensions of community gardens. While it can quantify the carbon footprint of producing a kilogram of tomatoes, it cannot reflect social trust, intergenerational knowledge transfer, or cultural and aesthetic values. Emerging fields like Social Life Cycle Assessment attempt to address this but face philosophical challenges in creating a single impact unit for social and environmental values. As ecological economists Martinez-Alier (2002) argued, reducing incommensurable social benefits to a single ledger is a political and ethical act, not merely a technical one. Focusing only on measurable factors risks undervaluing the social benefits that often define urban agriculture's most cherished outcomes.

Meanwhile, the Spatial Planning & Optimization cluster offers valuable tools to bring order and strategic intent to urban agriculture development, but its pursuit of an "optimal" solution has notable limitations often understated in technical presentations. While studies like Smith et al., (2021) recognize the subjectivity in criteria weighting, a larger issue is that these solutions are optimal only within simplified models, not reflecting real-world complexity. Consequently, maps of "food deserts" or "high priority" areas are analytical constructs, not inherent

features. More fundamentally, these models risk perpetuating inequality. Urban political ecology highlights that greening efforts can become gentrification catalysts, displacing communities they aim to help (Wolch et al., 2014). Current models cannot account for these socio-economic feedbacks, and their focus on efficiency may conflict with community-driven decisions, potentially disempowering local participation and fostering technocratic reasoning (read more in Kamali et al., 2025).

The Systems & Nexus Modelling cluster is defined by its holistic ambition to capture the complex, dynamic, and interconnected nature of urban systems. Its models provide a powerful "God's-eye view" of the city's metabolism. The most frequently acknowledged limitation in these papers is data availability and uncertainty. However, the deeper limitation is the risk of creating a "black box." As the models become more sophisticated, they can become opaquer, making it difficult for stakeholders to understand or question their assumptions. This can create a false sense of predictive certainty about complex adaptive systems that are inherently unpredictable. As Meadows & Wright (2015) cautioned, we should not fall in love with our models; they are always simplifications. Besides, reliance on complex nexus models can alienate public discourse. Furthermore, the very act of defining the model's boundaries is an act of interpretation that shapes what is seen as important. By focusing on quantifiable resource flows in the WEF nexus, as many of the papers included in this review do, we risk overlooking other critical

flows of knowledge, of social capital, of political power, those may be equally important in shaping the sustainability of urban systems.

In contrast to the lack of social dimension in LCA, Socio-Ecological and Community assessments often correct the technocratic biases of other paradigms by emphasizing the human aspects of UFGs. A common critique is that these findings lack statistical generalizability due to small, non-random samples (Cox, 2015). Another often unstated limitation is the tendency to overlook broader structural constraints. Focusing on community gardeners' resilience and empowerment can unintentionally shift the burden of fixing a broken food system onto under-resourced communities, while masking the influence of larger forces like land markets, national policies, and systemic racism. This raises a fundamental sociological question: how much can grassroots action challenge powerful political and economic structures?

And last but not least, the Empirical & Field-Based Studies provide the primary, ground-truthed data upon which much of the other research relies. It is grounded in the scientific method of direct observation and measurement, seeking to produce objective, verifiable knowledge about the performance of urban agroecosystems. The primary and most frequently acknowledged limitation in these papers is the question of external validity. Findings from a neatly managed, replicated experimental plot, as in the work of Taylor, (2020) on soil management, may not accurately reflect the messy, complex, and socially embedded reality of a

real community garden, which is subject to variable management skills, fluctuating participation, and site-specific constraints. This tension becomes even more acute when the cluster turns its attention to assessing ES. While providing invaluable quantitative data, the empirical approach to ES in studies like Dennis & James, (2017) and Jha et al., (2023) carries a series of unstated limitations that reflect a broader academic debate about the ES framework itself. Some of them include the simplification of the proxy and ecological reductionism, the utilitarian and managerial worldview of the results in terms of "synergies and trade-offs" (e.g., Sylla et al., 2020) and the systemic neglect of ecosystem disservices (Lyytimäki & Sipilä, 2009), and the struggles to capture the social co-production of ecosystem services.

2.4.3 Beyond the boundaries of methodological clusters

The previous analysis has broken down the landscape of UFGs research into five distinct methodological clusters, each characterized by its own internal logic, preferred tools, and inherent limitations. This categorization serves as a useful heuristic, while also highlighting that numerous studies have attempted to bridge the gap between these approaches by employing multiple core methodologies across different clusters. Building upon the reviewed literature and broader academic discussions, we consider the potential synergies, challenges, and

philosophical implications of integrating these methodological clusters to advance a more holistic and influential science of UFGs.

A particularly promising and necessary integration involves combining the "*where*" aspect of Spatial Planning & Optimization with the "*for whom*" and "*how*" aspects of Socio-Ecological & Community Assessment. The former offers robust tools for land allocation, while the latter provides a nuanced understanding of community needs and values. The process could start with qualitative methods from the socio-ecological cluster—such as interviews, focus groups, and participatory mapping—to collaboratively identify community-defined criteria for successful and desirable garden locations. This approach departs from the expert-led criteria selection seen in studies such as Ustaoglu et al., (2021) emphasizing co-production of knowledge. These locally identified criteria could then become the direct inputs for an MCDA framework. In this case, the AHP method would be used as a tool to facilitate community deliberation on the relative importance of their values, moving beyond the extraction of preferences from predefined expert criteria. The resulting GIS-based suitability maps would serve not as final, authoritative solutions but as dynamic tools fostering ongoing community dialogue—acting as boundary objects that support participatory decision-making. This integrative approach directly addresses a key limitation of the spatial planning cluster: its tendency toward top-down, technocratic planning. By fostering genuine collaboration, this method holds the promise of generating plans that are more

likely to be embraced, effectively implemented, and sustainably maintained by the communities they aim to serve.

Secondly, integrating Empirical Field-Studies with Systems & Nexus Modelling could address the most significant weakness of the Systems & Nexus Modelling cluster: the risk that its complex simulations are disconnected from the biophysical reality of urban farms. The Empirical & Field-Based Studies cluster provides the essential "ground-truthing" data needed to build, calibrate, and validate these models. For example, context-specific soil and nutrient data from Grard et al., (2018) can be used to parameterize and calibrate a top-down SDM or ABM model. The primary advantage is a noticeable increase in the credibility, accuracy, and local relevance of the systems models. It could support creating policy-support tools that decision-makers can trust, as the model's projections are not based on generic literature values but on measured data from their own city or region. However, the generation of high-quality, long-term empirical data is expensive, slow, and labour-intensive. A robust systems model requires years of calibration data, which conflicts with the short timelines of most academic and funding cycles. There is also a significant challenge in scaling up. How does one translate data from a few experimental plots into parameters that are representative of the vast heterogeneity of hundreds of gardens across a city? This is not just a technical problem of averaging but a conceptual one of dealing with complexity and

diversity. Navigating this tension requires a careful and reflexive understanding of what each method can and cannot claim to know.

Finally, integrating quantitative and socio-ecological assessments is notably challenging. A genuinely integrated approach should not simply append social indicators to a traditional LCA; it would start with socio-ecological inquiry. Using methods like interviews and participatory workshops, stakeholders would help define key social impact categories early on. Instead of relying on generic Social-LCA frameworks, the process would develop context-specific indicators for concepts such as community cohesion, food sovereignty, and cultural heritage. While this approach holds promise for a multi-dimensional sustainability assessment that addresses both technical efficiency and human values, it faces significant obstacles. Quantifying social impacts in a way that aligns with environmental metrics could dilute social critiques by the dominant technocratic paradigm. For instance, how do we measure a "unit of social cohesion" or ethically compare it to CO₂ emissions? A more honest approach, as suggested by some scholars, may be to resist full integration and instead present the social and environmental assessments as parallel, incommensurable narratives, forcing decision-makers to grapple with the trade-offs in a deliberative, rather than a purely technical, manner (Munda, 2004).

2.4.4 Managing the uncertainty across UFGs assessments

In a complex domain like UFGs and other NbS, where biophysical and social dynamics intertwine, different research methods employ distinct strategies to navigate the limits of their knowledge. These approaches reflect core assumptions about what can be known and how. Empirical field studies, for example, rely on statistical analysis, controlled experiments with replication, and multi-site studies to reduce statistical error and enhance the generalizability of their findings. Similarly, modelling approaches like LCA treat uncertainty as a technical challenge of data variability. Researchers working with LCA often use sensitivity analysis to test key assumptions and frame results through comparative scenarios, shifting the focus from absolute certainty to a more defensible comparative advantage.

When research aims to inform planning, it must grapple with the uncertainties of social values and future needs. To navigate subjective values, participatory methods like the AHP play a central role in systematically determining what defines a "good" outcome (Ustaoglu et al., 2021), treating this uncertainty as a question for social negotiation rather than technical calculation. To address an unknowable future, scenario-based planning helps design strategies that are robust across multiple possibilities (Fanfani et al., 2022), a common approach in Spatial Planning & Optimization and System & Nexus modelling.

Other methods embrace uncertainty as an inherent feature of the systems they study. Complex systems modelling, for instance, moves beyond prediction to explore possibilities. It uses stochastic simulations to generate a range of potential outcomes and exploratory scenarios to test for resilience (Pueyo-Ros et al., 2024). In parallel, qualitative and socio-ecological research addresses interpretive uncertainty—the trustworthiness of its findings. Rigor is established through triangulation of data sources, rich contextual detail ("thick description"), and researcher reflexivity, aiming not for statistical generalizability but for "transferability" that allows others to assess the relevance of the findings to their own contexts (Siegener et al., 2020). This methodological pluralism provides a sophisticated toolkit for investigating urban agriculture, where different forms of uncertainty demand different approaches to produce credible and useful knowledge.

2.4.5 Re-grounding the methodologies with a soil dimension

Despite its foundational role in UFGs, soil dimension is under-theorized in the research landscape, often being a soil-based system. Our systematic review of 58 papers reveals that while 20 papers (34.5%) include a soil-related parameter, it is predominantly treated as a passive substrate or a "black box". Soil often appears as a static variable in spatial models, an abstracted item in life cycle assessments, or is absent from nexus simulations altogether. Authors agree with Lal, (2017) that

soil is not external to the WEF nexus but should be placed at its very centre, forming a "WEFS" (Water-Energy-Food-Soil) nexus. We argue that integrating a dynamic soil dimension is a necessary corrective that can address the limitations inherent in current UA research by grounding abstract models in material reality. Integrating a sophisticated understanding of soil can bridge the current divide between its empirical treatment as a living system and its abstraction in other research paradigms. Such integration offers direct pathways to enhance existing methodologies.

- **Quantitative Sustainability Assessment:** Current LCAs effectively track emissions from inputs like compost but largely overlook soil's critical capacity for carbon sequestration. Incorporating dynamic soil carbon models would account for practices that build soil organic matter, potentially reframing gardens from carbon sources to net sinks and offering a more holistic climate impact assessment.
- **Spatial Planning & Optimization:** For spatial planning, a soil-centric view is a matter of environmental justice. Models that treat urban land as a blank slate ignore the legacy of industrial contamination, which is often concentrated in the low-income and minority communities targeted for greening initiatives. Integrating high-resolution soil contamination data is therefore essential for creating planning tools that are not only efficient but also safe and equitable.

- **Systems & Nexus Modelling:** In nexus modelling, soil is the nexus—the dynamic interface where water, nutrient, and energy cycles converge. Current WEF models often treat soil as a simple container, missing key feedback loops. A dynamic soil health module could simulate how practices like composting enhance soil organic matter and water-holding capacity, thereby reducing irrigation demands and improving system resilience. This approach positions soil as the central regulating node of the urban agroecosystem.
- **Socio-Ecological Assessment:** Soil provides a tangible, material anchor for abstract social concepts. Building on findings that practitioners consider "soil health a central organizing principle", methods like participatory soil health assessment can be used. This transforms soil from a simple biophysical parameter into a site for knowledge co-creation, grounding concepts like "stewardship" in the shared, material practice of community care.

2.4.6 Limitations of this study

This systematic review offers a structured overview of the field; however, its findings should be interpreted with awareness of certain methodological limitations. The following are some limitations that authors were able to identify. First, the review's scope was limited to the Web of Science core collection, which,

while comprehensive, excludes potentially relevant literature from other databases, as well as grey literature. Additionally, the review was restricted to peer-reviewed articles published in English, which may underrepresent research from non-Anglophone regions and omit important insights present in other languages or formats outside traditional academic journals. A pivotal definitional choice was the classification of UFGs as primarily non-commercial; this exclusionary criterion intentionally omitted studies on profit-oriented models (e.g., vertical farms). As a result, the findings and identified methodological clusters may not comprehensively reflect assessment tools used within this rapidly evolving sector of NbS.

Furthermore, the five methodological clusters proposed are analytical constructs devised by the author to bring together the body of literature. While this framework is useful, the boundaries between clusters are somewhat permeable, as evidenced by overlap analyses; some studies could logically fall into multiple categories. Therefore, this categorization represents one of many possible approaches to organizing the field, and readers should consider its tentative and heuristic nature.

2.5 CONCLUSION

In seeking to map the methodological landscape of nexus-based assessment for urban food gardens, this systematic review has addressed three central questions.

Our analysis of 58 research papers reveals a field structured around five distinct, albeit interconnected, methodological clusters: Quantitative Sustainability Assessment, Spatial Planning & Optimization, Systems & Nexus Modeling, Socio-Ecological & Community Assessment, and Empirical & Field-Based Studies.

Emerging from this analysis is a characterization of the field, defined by a tension between the quantitative, model-driven paradigms that dominate at the city scale and the qualitative, empirical approaches that provide deep, contextualized knowledge at the plot and community levels. This review suggests that while each cluster operates with a coherent internal logic and a specialized toolkit, their inherent limitations are profound, ranging from the reductionism of LCA in capturing complex social benefits to the risk of spatial models perpetuating injustice if not critically applied.

Looking forward, this work points toward two critical frontiers for future research. The first is the pressing need for greater methodological integration. The path forward appears to lie in intentionally building bridges between these paradigms—grounding abstract models with empirical data and humanizing quantitative assessments with participatory values. The second gap identified is the systemic underrepresentation of the soil dimension. Our findings suggest that re-grounding research by integrating dynamic soil processes, from carbon sequestration to contamination legacies, is essential for addressing the core limitations of current assessment paradigms. Ultimately, this review proposes the

need for robust analytical tools to guide the planning of urban food gardens.

Authors would like to point out that the most fruitful path lies in fostering a more integrated, transdisciplinary, and materially-aware science—one capable of navigating the complex nexus of interactions not just between food, water, and energy, but between data and values, between models and communities, and between the city and the living ground upon which it is built.

3

UNTANGLING THE FOOD-WATER-SOIL NEXUS

DISCLAIMER

A part of this chapter has been submitted as an opinion paper. It is currently under review in the journal *Earth: Environmental Sustainability*, under the title “*Untangling the Food-Water-Soil nexus through the lens of Ecosystem Services and Disservices in urban gardens*”. The paper was authored by:

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Abstract

This chapter introduces the Food-Water-Soil (FWS) nexus as a novel conceptual framework tailored for the analysis of soil-based, low-energy urban systems, with a specific focus on urban food gardens (UFGs). Traditional nexus frameworks often marginalize the foundational role of ecosystems, particularly soil, treating it as a static variable rather than a dynamic capital asset. In contrast, the FWS nexus positions soil at the core of the system, mediating the interactions between water availability and food (biomass) production. We employ the dual lenses of Ecosystem Services (ES) and Ecosystem Disservices (EDS) to serve as the primary analytical tools for evaluating the tangible benefits and detriments that emerge from these nexus interactions. This approach moves beyond simple material flow analysis to capture the qualitative performance of the garden as a living, multifunctional system. The chapter systematically categorizes the diverse ES and EDS co-produced within UFGs, linking them directly to biophysical processes within the FWS nexus. Furthermore, we critically apply a management hierarchy to develop a robust framework for the adaptive governance of the FWS nexus. Finally, building on these conceptual foundations, we propose a practical, indicator-based FWS Sustainability Index, adapted from the MESMIS methodology, designed to operationalize the nexus approach and guide management decisions toward optimizing the net contribution of urban gardens to urban well-being and fostering long-term socio-ecological resilience.

Keywords: urban garden, ecosystem services, ecosystem disservices, food-water-soil nexus

3.1 INTRODUCTION

Current resource management paradigms, notably the Water-Energy-Food (WEF) nexus, have provided valuable frameworks for understanding systemic interdependencies. However, their applicability diminishes when examining soil-based, low-energy urban systems such as urban food gardens (UFGs). In these contexts, energy is often a minimal input, while soil is not merely a passive substrate but the foundational living infrastructure that mediates the system's core functions. The WEF nexus and its conceptual descendants frequently marginalize ecosystems, treating them either as an additional node, an endpoint for impact assessment (e.g. Ravar et al., 2020), or an external driver influencing adaptation strategies (e.g. Muthee et al., 2021). This under-theorization of soil's central, dynamic role represents a critical gap in nexus research (Hatfield et al., 2017; Lal et al., 2017).

To address this gap, we propose and develop the Food-Water-Soil (FWS) nexus as a more fitting conceptual framework for UFGs. This framework repositions soil as the central mediating component, a capital asset whose health governs the interactions between water availability and food (biomass) production. To analyze the performance of this nexus, we employ the dual lenses of Ecosystem Services (ES) and Ecosystem Disservices (EDS). ES refers to the direct and indirect benefits humans derive from ecosystems (IPBES, 2019; Millennium Ecosystem Assessment, 2005), a concept that has become central to valuing urban nature-based solutions

(NbS) for their contributions to citizen well-being (Bush and Doyon, 2019; Castellar et al., 2021; Puppim de Oliveira et al., 2022). While some frameworks have begun integrating ES to clarify policy trade-offs (Fürst et al., 2017), they often overlook the co-produced harms, or EDS. As human–ecosystem interactions inherently involve nuisances (Lyytimäki and Sipilä, 2009; Pereira et al., 2023), a comprehensive assessment requires an integrated view of both positive and negative outcomes.

The necessity of a soil-centric approach is increasingly recognized at the policy level, with initiatives like the “Mission ‘A Soil Deal for Europe’” highlighting the urgency of addressing soil health challenges (Panagos et al., 2024). Such programs acknowledge that the success of cultivated NbS requires a multi-stakeholder, adaptive approach to innovation and management (Mumaw and Mata, 2022). The FWS nexus provides the holistic and transdisciplinary perspective needed to guide such an approach.

This chapter, therefore, systematically builds the case for the FWS nexus, progressing from theory to practice. First, we establish the conceptual basis of the FWS nexus and detail the core biophysical interactions that drive it. Second, using the analytical lenses of ES and EDS, we categorize the beneficial and detrimental outputs generated by these interactions. Third, we present a hierarchical framework for the adaptive management and governance of these outputs. Finally, to render this conceptual work operational, we introduce a novel FWS

Sustainability Index—an assessment framework adapted from the MESMIS methodology. It is designed to translate the complexities of the nexus into a practical tool for evaluating and improving the sustainability of urban gardens.

3.2 FOOD – WATER – SOIL INTERACTIONS

3.2.1 Soil – Plant interactions

Soil serves as the fundamental medium for plant life, performing three critical functions. It provides physical anchorage for roots, offers a porous structure for water and gas exchange between roots and the atmosphere, and hosts a biotic community that decomposes organic matter, recycles nutrients, and stores resources. The architecture of the soil, known as its structure, is paramount to these functions. This arrangement of soil particles dictates the nature of the pore space, which in turn governs the soil's ability to hold and transport water, air, and dissolved substances (Bronick and Lal, 2005). Soils rich in organic matter are more likely to possess this optimal structure and, as a result, a greater capacity to store plant-available water (Huntington, 2020). This environment also fosters the biological breakdown of materials like plant litter and manure by bacteria and fungi, a process that drives decomposition and carbon sequestration (Mueller et al., 2024). Conversely, physical disturbances such as compaction or tillage degrade this structure, leading to nutrient loss, surface crusting, and a reduction in water

and air for roots. Soil compaction presents a significant physical barrier that can severely limit root growth (Cogger, 2005). Furthermore, erosion by wind or water strips away the nutrient-rich topsoil, diminishing the land's productivity.

Vegetation is not a passive inhabitant of the soil; it actively engineers its environment through the actions of its roots and the contribution of its litter. As roots navigate the soil, they influence the formation and turnover of soil aggregates, create channels or macropores, and modify the physical structure of their immediate surroundings. This root growth can exert mechanical stress, sometimes compact the soil and reducing porosity in the immediate vicinity of the root. However, depending on the existing soil conditions, it can also increase porosity at the root surface (Mueller et al., 2024). This is a critical distinction: in well-structured soils, roots often follow paths of least resistance, growing into existing large pores with minimal disturbance, whereas in compacted soils, they are forced to expend significant energy to physically displace the soil matrix. At a finer scale, roots release carbon-rich compounds (rhizodeposits) and extend fine root hairs that bind soil particles together, forming a cohesive layer known as a rhizosheath that stabilizes the soil structure (Materechera et al., 1992). This rhizosheath is a key adaptive trait, especially under drought conditions, as it maintains crucial contact between the root and soil, ensuring continued water uptake from a drying environment. The area around the roots, teeming with microbial life, further enhances this process. These microbial communities, which

are more abundant and diverse in the rhizosphere, produce sticky "glues" like extracellular polysaccharides that adhere to soil particles, strengthening aggregation (Redmile-Gordon et al., 2014). This cycle completes as decaying roots disintegrate, contributing to the buildup of soil structure and the formation of persistent soil organic carbon (Mueller et al., 2024; Neurath et al., 2021). Research indicates that this belowground carbon input from roots is disproportionately important for building the most stable forms of soil carbon, as it is deposited directly within the mineral soil, where it can be protected from rapid decomposition (Costa et al., 2018).

Aboveground plant litter is another crucial link between plants and soil, playing a primary role in shaping the physical structure of the topsoil (Ponge, 2003). This structure emerges from the interplay between vegetation and soil biota, particularly the burrowing and fragmenting activity of soil fauna. The decomposition of litter releases nutrients, boosts the soil's capacity to hold onto them, and increases its water-holding capacity (Frouz, 2024; Six et al., 2004). The long-term result of these decomposition processes gives rise to distinct humus forms (e.g., Mull, Moder, and Mor), which serve as visible indicators of the ecosystem's functional state, reflecting the speed of nutrient cycling and the dominant decomposer communities. In many urban farm and garden settings, where natural litterfall is limited, materials like mulch, wood chips, and compost are often added to fulfil this essential role.

While these soil-plant interactions are universal, they face unique pressures in urban environments:

- *Soil Compaction*: Urban soils are frequently compacted from construction and traffic, which destroys pore space, limits water infiltration, and creates a physical barrier that can stunt root systems, leading to shorter, thicker roots with fewer lateral branches (Alessa and Earnhart, 2000).
- *Contamination*: Geogenic origins, industrial activities, lead-based paint, and leaded gasoline have left many urban soils contaminated with heavy metals, particularly lead. These contaminants pose health risks through direct ingestion of soil or consumption of produce grown in it (Bechet et al., 2018; Jean-Soro et al., 2015; Paltseva et al., 2022).
- *Urban Heat Island (UHI) Effect*: The prevalence of heat-absorbing surfaces like asphalt and concrete raises ambient temperatures, which in turn increases the rate of evaporation from soil, depleting moisture more rapidly and placing additional stress on plants (Zipper et al., 2017).

These stressors often create a negative feedback loop: compaction and heat lead to poor plant growth, which means less organic matter is returned to the soil, preventing the natural rebuilding of the soil structure that would otherwise help

mitigate these very issues. Management strategies in urban gardens, therefore, must focus on actively breaking this cycle by importing clean soil for raised beds, incorporating large amounts of compost to rebuild structure and organic matter, and adopting practices like no-tillage and cover cropping to protect and continuously build soil health (van Es and Magdoff, 2009).

The most effective strategy for improving water dynamics in urban soils is to increase soil organic matter. Adding amendments like compost directly counteracts compaction by lowering bulk density and rebuilding soil structure (Frouz, 2024; Six et al., 2004). Organic matter acts like a sponge; it is hydrophilic and enhances the formation of soil aggregates, which creates a network of pores that both improves infiltration and increases the soil's water-holding capacity (Huntington, 2020). According to the USDA, a 1% increase in soil organic matter can help an acre of soil hold up to 20,000 additional gallons of water (Keller and Handler, 2024).

3.2.2 Soil – Water Dynamics

The principles of water balance in UFGs mirror those in other urban green spaces and agricultural settings. The system gains water through precipitation and irrigation and loses it via infiltration into the soil, surface runoff, and evapotranspiration. This "green water"—the moisture held within the soil profile for plant use—is sometimes supplemented by stored water from rain barrels or

tanks. While tap water and harvested rainwater are the primary irrigation sources for UFGs, some research highlights the use of greywater from kitchens and bathrooms, particularly in arid regions facing water scarcity (Al-Ismaïli et al., 2017; Faruqui and Al-Jayyousi, 2002).

Water is more than just a resource for hydration; it is a dynamic agent within the soil. It facilitates the transport of nutrients to roots and serves as the medium for essential chemical and biological reactions. In this role, soil moisture acts as a critical nexus, linking hydrological cycles with biogeochemical processes and mediating the interactions between climate, soil, and plant life. However, excessive water movement through the soil profile can be detrimental, leading to the leaching of fertilizers and reducing their availability to plants.

The urban environment alters this water balance. A primary challenge is soil compaction, often a consequence of construction and traffic, which destroys the soil's porous structure (van Es and Magdoff, 2009). This reduces the soil's ability to absorb water. Studies have shown that compaction from construction can decrease infiltration rates by 70% to 99% (Gregory et al., 2006), consequently turning soils into surfaces that shed water rather than absorb it. This not only increases stormwater runoff but also prevents rainwater from replenishing moisture deep in the root zone, making plants more susceptible to drought (Bronick and Lal, 2005). This issue is compounded by the UHI effect mentioned in

the previous section 3.3.1, further increasing the overall water demand of garden plants.

The source of irrigation water presents another layer of complexity. Stormwater runoff from impervious surfaces like roads and rooftops often carries pollutants, including heavy metals like lead (Pb), zinc (Zn), and copper (Cu) from vehicle wear and building materials (Guo et al., 2020). When this runoff enters a garden, these contaminants can accumulate in the soil, posing potential health risks (Tom et al., 2014). The use of untreated greywater, while a good conservation strategy, also carries risks (Shqerat and Al-Tabbal, 2025). Long-term application can lead to the buildup of salts and sodium from detergents, increasing soil salinity and sodicity (Rahman et al., 2022). High sodicity is particularly damaging, as it can destroy soil structure, clog pores, and create an impermeable surface layer that prevents water infiltration (Al-Hamaiedeh and Bino, 2010). Greywater may also contain pathogens, though the risk is lower than with blackwater.

A study from Lin et al. (2018) demonstrates that foundational improvement in soil health should be paired with water-wise practices to improve the soil-water positive feedback loop:

- *Mulching*: Applying a layer of organic material like wood chips or straw to the soil surface insulates the soil, reduces evaporation, and suppresses weeds that compete for water.

- *Efficient Irrigation*: Using drip irrigation delivers water slowly and directly to the root zone, minimizing losses to evaporation and runoff. This method can reduce water use by 20% to 50% compared to overhead sprinklers (Ignaciuk et al., 2015). Watering deeply but infrequently also encourages plants to develop deeper, more resilient root systems.
- *Raised Beds*: For sites with severe contamination or compaction, raised beds filled with a high-quality engineered soil mix of compost and other materials offer a practical solution (Kessler, 2013; Le Guern et al., 2018). However, these contained systems often dry out faster and require more frequent watering.

3.2.3 Food – Water Interconnections

For crops, a steady supply of water is indispensable. It fuels essential metabolic processes and drives evaporative cooling through transpiration, a function that can moderate the local microclimate (Lin et al., 2018a). This cooling service becomes increasingly vital in urban gardens, where rising ambient temperatures intensify water demand. Effective vegetation management is therefore key to conserving this precious resource by influencing plot-level factors like surface temperature and evapotranspiration rates to preserve soil moisture (Lin, 2010). Strategic planting designs, such as increasing the canopy coverage and structural diversity of crops, can significantly improve water retention within the garden's

micro-ecosystem. Additionally, applying ground covers like mulch or straw is a powerful technique. These materials shield the soil from wind and water erosion and dramatically cut evaporation rates. This practice directly boosts water use efficiency (i.e., the amount of yield produced per unit of water consumed) and can ultimately lead to greater harvests (Wang et al., 2021).

Plants are not passive water consumers; they actively regulate their use through opening and closing stomata, the microscopic pores on leaves (Pirasteh-Anosheh et al., 2016). Low water availability would trigger the stomatal closure, which reduces water loss from transpiration. However, this process also limits the intake of carbon dioxide, slowing photosynthesis and potentially reducing growth and final yield. Different vegetable species exhibit distinct strategies for managing this trade-off. Under drought, snap beans, for example, may increase their stomatal density, while green peas show little change (Nemeskéri and Helyes, 2019). Tomatoes, which are highly sensitive to drought, respond acutely by altering both the size and number of their stomata. If water stress persists, plants may resort to more drastic measures, such as shedding older leaves or producing smaller new ones to reduce their total leaf area and overall water demand.

A key goal in sustainable agriculture is to improve water use efficiency, effectively producing "more crop per drop". In a garden setting, water use efficiency is enhanced by minimizing non-productive water loss (evaporation from the soil) and maximizing productive water use (transpiration by the plant) (Pollard et al.,

2018). Practices like mulching and drip irrigation are highly effective because they target evaporation (discussed in previous sections). A meta-analysis confirmed that mulching significantly increases both crop yield and water use efficiency, with some studies showing gains of over 20% and 30%, respectively (Li et al., 2025). Interestingly, the relationship between water and food quality is not always linear. While severe drought is damaging, mild and controlled water stress can sometimes trigger physiological responses in certain crops that concentrate sugars, organic acids, and beneficial compounds like antioxidants. This can result in produce with more intense flavour and enhanced nutritional value, though often at the cost of slightly reduced size or total yield (Jin et al., 2023; Naeem, 2025).

A resilient urban garden begins with smart plant selection and design. Below are some recommendations from improving Food-Water nexus in UFGs:

- *Drought-Tolerant Crops:* Vegetables like pole beans, Swiss chard, eggplant, hot peppers, and certain varieties of tomatoes and squash are excellent choices. They often possess adaptive traits like deep root systems that can access more soil moisture. Amaranth and chickpeas are also exceptionally resilient. Conversely, water-intensive cool-season crops such as most brassicas (broccoli, cauliflower), lettuce, and celery are best avoided during peak summer heat.

- *Planting Design*: Arranging plants in dense blocks rather than traditional single rows allows their leaves to touch as they mature. This creates a "living mulch" that shades the soil, suppresses weeds, and reduces evaporation. Selecting compact "bush" or "determinate" varieties of plants like tomatoes and squash further enhances this effect (UC Agriculture and Natural Resources, n.d.).
- *Efficient Irrigation*: The most effective watering method is to irrigate deeply but infrequently. This encourages plants to develop deep, robust root systems that are less vulnerable to surface drying. As discussed previously, Drip irrigation can reduce water use by up to 50% compared to overhead sprinklers by minimizing evaporation and runoff (Ignaciuk et al., 2015).

3.3 FWS NEXUS THROUGH THE LENS OF ES AND EDS

A crucial insight from ecological economics is that ecosystem services do not flow directly and passively from this natural capital to benefit people. Instead, both services and disservices are co-produced through the dynamic interaction between the cultivated nature capital (in this dissertation, it is viewed as urban gardens) and the other three capital stocks: social, human, and built. **Figure 3.1** establishes the key assets and actors—the four capitals—involved in this co-production (Costanza et al., 2014).

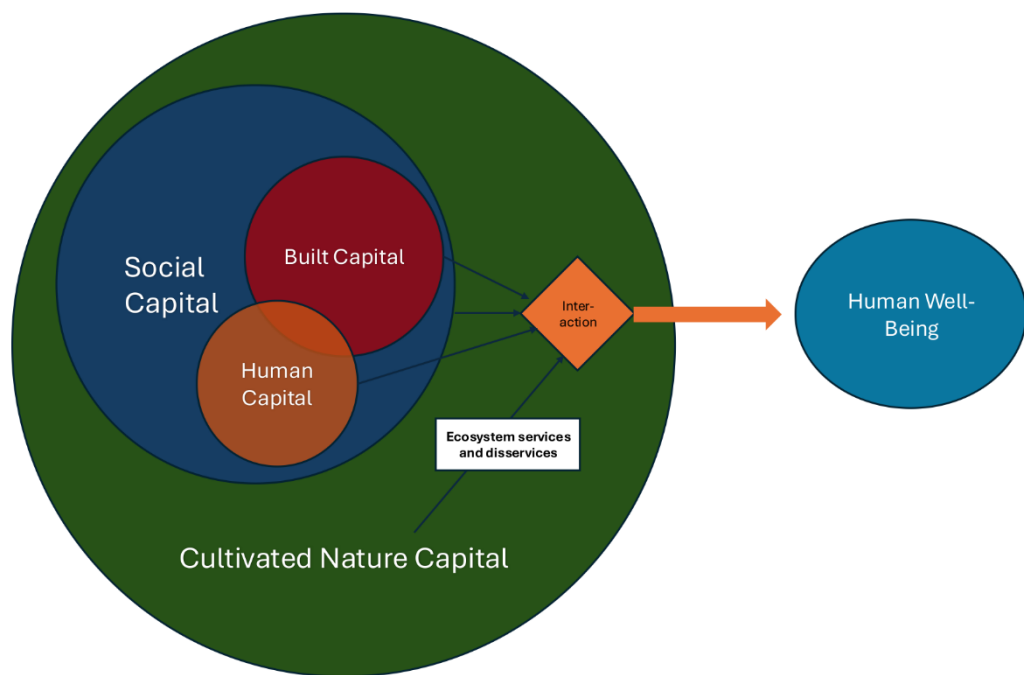


Figure 3.1. Interactions between nature capital and social capital affecting human well-being.

Modified from Costanza et al., (2014).

Let us consider the provisioning of food. The natural capital of urban gardens (i.e., soil, biodiversity, and climatic conditions) holds the potential to generate produce. This potential is only realized through the application of human capital (a gardener's knowledge and physical labour), which is aided by built capital (tools, irrigation systems, raised beds), and often structured by social capital (community garden rules, shared knowledge networks, and trust among members). The resulting harvest is a co-produced ecosystem service. Conversely, the very same interactions can generate ecosystem disservices. A gardener's lack of knowledge (human capital) combined with an overreliance on external inputs (built capital) can deplete soil organic matter, resulting in nutrient runoff and pollution of local waterways.

In contrast, a successful garden that enhances well-being can build social capital (stronger community ties) and human capital (new gardening skills). Thus, the ultimate contribution of the urban gardens to human well-being is the net outcome of these complex, co-produced services and disservices, which are mediated by the constant interplay of all four forms of capital. It thus becomes clear that ESs and EDSs are not an 'add-on' but rather the enabling and disabling features that define the nexus system's potential and its limits. ESs and EDSs emerge at the dynamic interface between the managed ecology of an urban food garden and the surrounding urban social system (Gómez-Baggethun and Barton,

2013), and represent the ultimate locus of interaction among the food, water, and soil dimensions of this nexus.

The ES-centric framework, proposed by Sambo et al. (2024), represents a significant step in centring ecosystems within the WEF nexus. It clarifies the complex interactions among multiple nexus sectors, as well as the pressures and responses of WEF to the ecosystem processes and functions that deliver their services. However, this framework neither offers the integration of the EDSs nor does it contain the soil dimension. Thus, the author adopted the framework and modified it to express the interdependence between the FWS nexus and the ESs and EDSs that are generated. **Figure 3.2** illustrates the dynamic process through which their interactions generate specific services, disservices, and trade-offs in the nexus. In this cascade model, the experience of “*nuisances*” (e.g., a pest outbreak) leads to societal “*values*” (e.g., a new appreciation for pest-regulating insects), which in turn drives “*responses*” (e.g., planting pollinator-friendly species), altering the “*ecosystem structure*” for the next cycle.

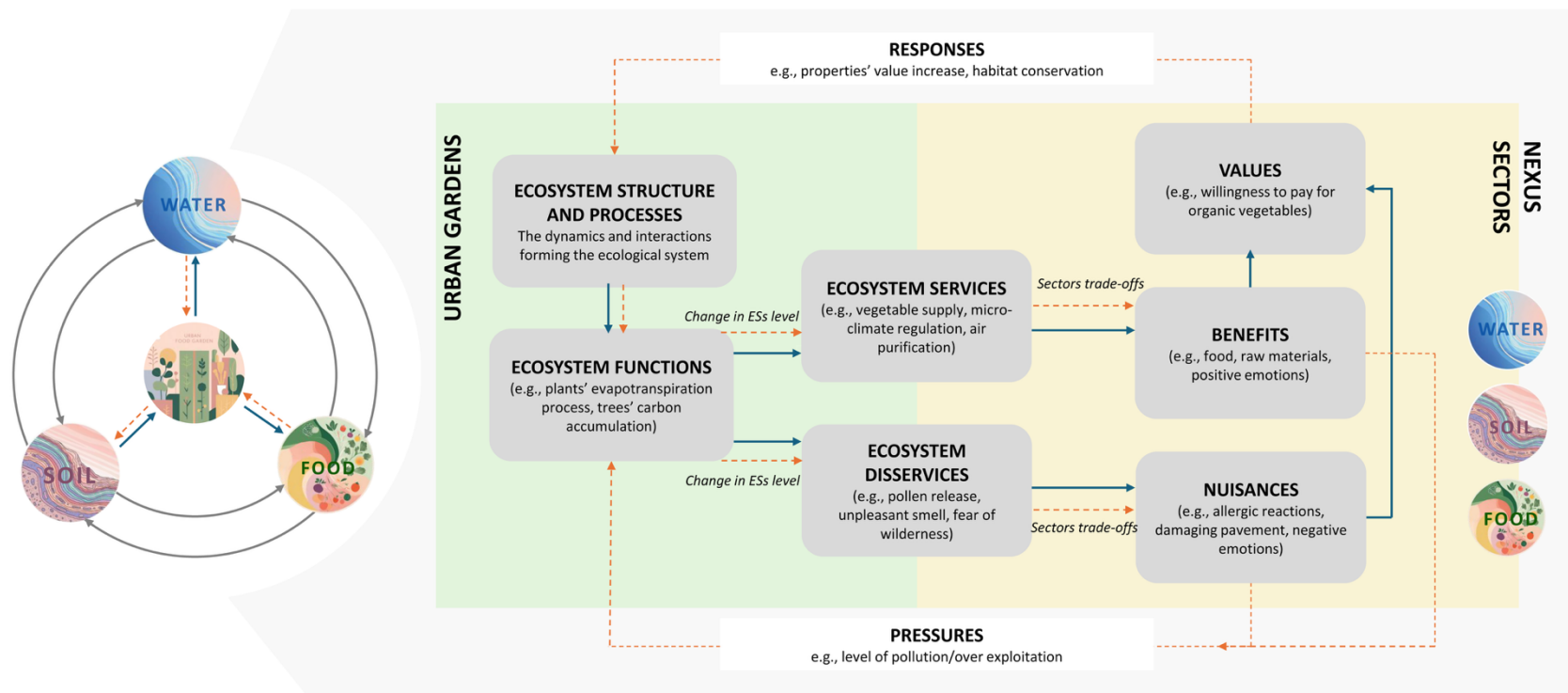


Figure 3.2. Integrating the ESs and EDSs model in the FWS nexus framework. Adapted from Sambo et al., 2024

The intention of clarifying the FWS framework through the ES and EDS lenses in this dissertation is rooted in the dynamic and systemic perspective. Quantitative assessments, such as material flow analysis, often work best with resources that are easily quantified in standardized units (e.g., cubic meters of water, tonnes of grain, gigajoules of energy). Soil, however, defies such straightforward accounting. Unlike water flowing through a pipe or stored in buckets, many of soil's most critical functions are services rather than physical flows. The "flow" of nitrogen mineralization or disease suppression is a biological process rate, not a mass transfer in the traditional sense of material flow analysis. While soil mass can erode and be transported (as in material flow), this represents a system degradation rather than a managed resource flow. The critical resource we aim to manage is not soil mass, but rather soil health or soil quality—a complex, multi-attribute characteristic that encompasses physical, chemical, and biological properties (Séré et al., 2024). A single metric, such as tonnes or joules, cannot capture this quality. Hence, soil is often overlooked in traditional urban nexus assessments that prioritize engineered infrastructure. By framing the functions of soil, water, and plants as services and disservices, we shift the analytical focus from a simple accounting of material stocks to an assessment of the performance of the urban gardens as a living, multifunctional piece of infrastructure. The degradation of this infrastructure incurs real costs, not just in lost productivity, but in diminished regulatory capacity (e.g., stormwater management) and resilience.

Following this framing, we argue that both ESs and EDSs must be incorporated into the center of the FWS nexus as the implicit functions that mediate the links between its components, rather than as simple outputs. The role of ESs/EDSs in facilitating synergies and, more critically, in exposing trade-offs becomes explicit when considering the system's underlying biophysical state (Séré et al., 2024). The soil's biophysical structure and microbial processes are the foundation that underpins ecosystem functions. These functions are essential for delivering services valued by urban dwellers, such as fresh food, aesthetic beauty, or microclimate regulation.

The FWS nexus is best understood by characterizing the distinct but interdependent roles of its components. Water serves as the dynamic medium, facilitating primary flow and nutrient transport, which in turn supports plant growth. Food, or more broadly biomass, represents the productive output and the principal objective of management interventions. At the core sits soil, the foundational living infrastructure of the system. Unlike water or food, soil is a capital asset whose function is determined by its state. We can distinguish between its inherent properties (e.g., texture, mineralogy), which define the system's baseline capability, and its dynamic properties (e.g., soil organic carbon, aggregate stability, biota), which reflect its current health and are directly modified by management (Robinson et al., 2009). It is the management interventions targeting food production that alter these dynamic properties,

thereby mediating the soil's relationship with water. For example, a decision to apply compost to increase crop yield enhances the dynamic property of soil organic carbon, which in turn improves the soil's capacity to store water, demonstrating a synergy between the food, water, and soil dimensions. It is these changes in soil-mediated function—driven by the interactions between food, water, and soil—that ultimately manifest as the ecosystem services and disservices valued by society.

3.3.1 Ecosystem services from UFGs

The previous sections have established the foundation of clarifying the FWS nexus and how FWS nexus interactions generate ESs and EDS. In this section, ESs and EDSs in existing research literature are summarized and categorized to provide a clearer look into how ESs and EDSs relate to FWS. The Common International Classification of Ecosystem Services (CICES) was used to categorize common ESs into provisioning, regulating and maintenance, and cultural groups. Table 3.1 presents the list of perceived ESs streaming from UFGs identified from Italy, Spain, Europe, and Southeast Asia, based on Langemeyer et al. (2016). Camps-Calvet et al. (2016), Sanyé-Mengual et al. (2018), and Lourdes et al. (2021). While much can be discussed about the ESs and EDSs' perception, assessment methodology, and applications, this section focuses on using them to clarify how the FWS nexus interacts among its components that shape the flow of ESs and EDSs toward

human well-being capital" that is essential for managing the FWS nexus effectively.

Table 3.1. List of common perceived ecosystem services from urban gardens

Provisioning Services	Food (edible parts of vegetative plants)
	Material (fuels, raw materials)
	Medicinal and aromatic plants
Regulating and Maintenance	Air quality and microclimate regulation
	Biological control (pets and pathogen)
	Biodiversity and Habitat provision
	Carbon storage and sequestration
	Erosion control and soil fertility conservation
	Flood regulation and mitigation (drainage and buffer)
	Water quality and water regulation
	Waste decomposition
Cultural Services	Pollination
	Cultural history, identity, and heritage
	Social cohesion, integration, community building
	Aesthetics, spirituality, religion, inspiration, recreation
	Sense of place, political realization
	Learning, education and research
	Physical exercise, stress reduction, mental health

3.3.1.1 Provisioning Services

The most direct and tangible service provided by UFGs is the production of food.

This service is the primary output of the FWS nexus, where soil fertility and structure (S) mediate water availability (W) to support the growth of edible crops

(F). Beyond subsistence, this local food production can generate significant economic outcomes, particularly in low socio-economic areas, by supplementing household food budgets and fostering local economic development (Mensah, 2023). This service directly addresses concerns of urban food security and sovereignty by providing communities with greater control over their food and medicinal supply.

3.3.1.2 Regulating and Maintenance Services

A majority of research related to regulating services emerges from the garden's capacity to modulate local environmental conditions (Haase et al., 2014; Langemeyer et al., 2016). While UFGs and other urban green spaces only account for a small fraction of the global biosphere, their contribution to local environmental resilience and human well-being is of significant importance and well-documented (Cabral et al., 2017b; De la Sota et al., 2019; Jha et al., 2023).

A key service is *stormwater management* and *flood mitigation*, a direct outcome of the Soil-Water dynamic. Healthy, uncompacted urban garden soils, rich in organic matter, function like sponges. They exhibit higher hydraulic conductivity and water-holding capacity compared to typical compacted urban soils or impervious surfaces. This enhanced infiltration reduces surface runoff, mitigating local flooding and decreasing the load on municipal stormwater systems. Conversely, urban soil compaction can reduce infiltration rates by as much as 70%

to 99% (Gregory et al., 2006), effectively turning permeable land into a runoff-generating surface.

UFGs also provide microclimate regulation, an ES that arises from Food-Water interconnections. The canopy of garden plants provides shade, which can significantly reduce surface temperatures. Simultaneously, the process of evapotranspiration, fuelled by water drawn from the soil, releases water vapor and reduces the temperature of the surrounding air. This localized cooling helps mitigate the UHI effect. The effectiveness of this cooling is influenced by the density and structural complexity of the plant canopy; multi-layered plantings of trees, shrubs, and groundcover are more effective than single-layer vegetation (Egerer et al., 2019).

Furthermore, UFGs are hubs for waste recycling and nutrient cycling, embodying a circular economy model (Salomon et al., 2020). The practice of composting organic waste—such as food scraps and yard trimmings—and applying it to garden soil is a central FWS interaction. This process diverts waste from landfills and transforms it into a valuable amendment that improves soil structure (S), enhances fertility for food production (F), and increases water retention capacity (W).

UFGs contribute significantly to biodiversity maintenance in otherwise fragmented and ecologically simplified urban landscapes (Jha et al., 2023). By

cultivating a diversity of flowering plants and creating structurally complex habitats, these gardens serve as crucial refuges and ecological corridors for urban wildlife, particularly pollinators like bees and butterflies, as well as birds (Speak et al., 2015). The health of the soil ecosystem (S) and the diversity of cultivated plants (F) underpin this service, which in turn supports other services like pollination for food production.

3.3.1.3 Cultural Services

Perhaps the most deeply felt benefits of UFGs are the cultural services they provide. These gardens are strongly associated with improved health and well-being (Rao et al., 2022). They offer opportunities for physical activity and increase access to fresh, nutritious food. Moreover, numerous studies have documented the profound mental health benefits of gardening, including significant reductions in stress, anxiety, and depression, and improvements in mood and overall life satisfaction (Genter et al., 2015).

UFGs are also unique urban green spaces that foster social cohesion and community building (Delshad, 2022). They act as neutral grounds where diverse residents can interact, share labor, solve problems collectively, and build relationships, thereby strengthening neighborhood trust and social capital. This process of community building enhances the social resilience of neighborhoods.

Finally, UFGs serve as powerful platforms for education and skills development. They are living classrooms where participants of all ages can gain practical knowledge about ecology, food production, and nutrition (Gregory et al., 2016), directly enhancing the "human capital" that is essential for managing the FWS nexus effectively.

3.3.2 Ecosystem disservices from UFGs

The definition of EDS, in contrast to ES, is *"functions of ecosystems that are perceived as negative for human well-being."* Many are concerned about the EDS concept's potential to compromise conservation efforts or hamper the advancement of nature planning in cities. However, people inevitably experience nuisances and harm from ecosystems in their daily lives; various forms of EDS are commonly discussed by the media. Acknowledging these potential harms is critical for balanced management and governance. These disservices can be categorized as ecological/material, health-related, and economic/social.

Table 3.2. List of common urban ecosystem disservices (Vaz et al., 2017)

Human health disservices <i><infectious disease; toxicity and allergenicity; other pollution effects></i>	Animal and insect bites
	Contaminated food (contamination accumulation)
	Zoonotic diseases and pathogens
	Skin irritation and pollen-induced allergies
Ecology and Resource disservices	Exposure to soil contamination through inhaling
	Water depletion (used for watering)
	Water quality – fertilizer and pesticide runoff

<i><direct and indirect ecological and environmental costs and impacts></i>	Soil nutrient inputs – altered nutrient cycles
	Green waste
	Air pollution emissions from decomposition
	Introduction of invasive species
Economic and Social disservices <i><disrupting physical, personal, financial stabilisation; impacts on mental/cultural interactions with nature; prevention of physical interaction with nature></i>	Maintenance cost (planning, pruning, mulching, irrigation, fertilization)
	Safety hazard from tree/branches fall
	Species and landscapes perceived as irritating or unpleasant
	Green gentrification
	Sounds and smells produced by ecosystem functions
	Presence of weeds, pets, mosquitoes
	Habitats associated with the unknown, remoteness or wilderness considered unpleasant for outdoor activities
	Fear and risk of wilderness (animals and lack of light in dense vegetative area)

3.3.2.1 Ecological and Material Disservices

A primary disservice is the high consumption of potable water. The F-W nexus presents a direct trade-off: maximizing food production often requires intensive irrigation, which, in many cities, relies on treated municipal water. This places a direct strain on urban water supplies, particularly in arid or drought-prone regions

Poor management can also lead to nutrient runoff and water pollution (Wielemaker et al., 2019). The over-application of fertilizers or nutrient-rich compost (an F-S interaction), combined with inefficient irrigation (W) or compacted soil with poor infiltration capacity (S), can cause excess nitrogen and phosphorus to leach into groundwater or be carried by runoff into urban

waterways. This nutrient pollution contributes to eutrophication and degrades aquatic ecosystems.

Finally, UFGs generate green waste. Non-edible plant parts, pruned branches, and end-of-season crop residues constitute a waste stream that requires management. As UFGs and other urban green spaces increase worldwide, leading to a higher generation of green waste (Viretto et al., 2021). If not composted effectively on-site, this green waste can place an additional burden on municipal waste collection and processing systems.

3.3.2.2 Health Disservices

The presence of urban flora and fauna exerts subtle yet significant influences on human health, often operating beneath the threshold of conscious perception. A significant health concern stems from soil and water contamination. Urban soils (S) are frequently contaminated with heavy metals such as lead (Pb), zinc (Zn), and cadmium (Cd) from legacy sources like leaded gasoline, industrial emissions, and lead-based paint (Ashrafzadeh et al., 2018; Jean-Soro et al., 2015; Leitão et al., 2018). Using contaminated stormwater runoff (W) or locating gardens near high-traffic roads can introduce these pollutants into the garden environment. This creates multiple exposure pathways, including direct ingestion of soil (especially by children), inhalation of contaminated dust, and consumption of produce (F) that has absorbed heavy metals, posing serious long-term health risks.

The risk of pathogen exposure is another health disservice. The use of untreated or inadequately treated greywater (W) for irrigation can introduce a range of pathogens, including *E. coli* and *Salmonella*, into the garden (Shqerat and Al-Tabbal, 2025). Similarly, applying improperly managed or "cold" composted animal manure (S) can contaminate soil and produce harmful bacteria, creating a direct risk of foodborne illness

Gardens can also be a source of pests and allergens. They can provide habitats for pests such as rodents and certain insects, which may become a nuisance to gardeners and adjacent residents. Furthermore, the cultivation of numerous plant species, particularly wind-pollinated varieties, can release pollen that triggers allergic reactions in susceptible individuals, contributing to local air quality concerns for some. Additionally, there are numerous ornamental, crop, and weedy plants that may induce contact dermatitis in certain individuals, e.g., dandelion, ginkgo trees, common ivy; read more in Douglas et al. (2020).

3.3.2.3 Economic and Social Disservices

UFGs are not without costs. Maintenance costs, including the financial outlay for tools, seeds, water, and soil amendments (built capital), as well as the significant investment of time and labour (human capital), can be a substantial barrier to participation, particularly for individuals and families with limited resources

The shared nature of many community gardens can also lead to social conflicts. Disagreements over plot maintenance standards, water use, the introduction of "weedy" or unkempt plants, and differing aesthetic visions for the space can create tension among gardeners and with the surrounding community, undermining the very social cohesion the garden is meant to foster (Haase and Gaeva, 2023).

Green gentrification exemplifies a nuanced and troubling form of social injustice. Although urban greening initiatives provide essential ecosystem services, such as aesthetic appeal, social cohesion, and community resilience. Their effectiveness can unintentionally heighten the attractiveness of neighbourhoods to affluent residents and developers (Marche, 2015). This often leads to rising property values and rents, forcing long-standing, low-income populations to displace, despite benefiting from these environmental enhancements (Quinton et al., 2022). Consequently, the development of localized ecological assets can paradoxically intensify social and economic disparities, underscoring a critical drawback of urban greening policies that demands re-evaluation.

3.3.3 EDS and ES feedback loops on FWS

The relationship between ES and EDS is not static; it is part of a dynamic feedback loop that influences management and the state of the FWS nexus over time. As illustrated in Figure 2, the societal experience of a disservice can trigger a response

that alters the ecosystem's structure and function, thereby changing the future flow of both services and disservices (refer to **Figure 3.2**).

For example, a pest outbreak (an EDS) in a community garden might initially be met with the application of chemical pesticides. This response, while addressing the immediate nuisance, could generate further disservices, such as contaminating the soil (S), harming beneficial pollinating insects (reducing a supporting service), and posing health risks to gardeners. The negative experience of these secondary disservices may lead to a new societal valuation—a greater appreciation for ecological pest control. This new value could drive a different response: the adoption of integrated pest management practices, such as planting pollinator-friendly species to attract predatory insects. This action directly alters the garden's ecosystem structure (increasing plant diversity) for the next cycle, aiming to enhance the ES of natural pest regulation while minimizing the EDS associated with chemical use

This cascade demonstrates that disservices are not merely negative endpoints but are also critical signals within a social-ecological system. They reveal unsustainable practices and create opportunities for learning and adaptation. Recognizing and responding to these signals is fundamental to the adaptive management of the FWS nexus, allowing gardeners and planners to iteratively refine their practices to minimize harms and optimize the net positive contribution of the UFG to urban well-being.

Urban gardens incorporate practices that utilize natural processes to enhance agricultural productivity sustainably, while also promoting soil health, effective water management, and biodiversity conservation within the urban environment. (Brauman et al., 2020; Nassary, 2025). It is at this critical interface of natural assets and human well-being that our FWS framework becomes operational. Urban gardens can be understood as a form of cultivated natural capital, a hybrid system that is neither entirely natural nor purely an artifact of human creation (Hernández-Blanco and Costanza, 2018). Urban gardens, such as allotments or communal gardens, require minimal energy input for production (Caputo et al., 2023). In contrast, soil is the foundational medium where critical interactions occur, and managing soil health through sustainable agricultural practices is of significant importance for long-term urban resilience (Nassary, 2025); therefore, soil should be placed at the centre of nexus thinking, especially in planning and evaluating the effectiveness of soil-based NbS.

3.4 IMPLICATIONS FOR NEXUS GOVERNANCE AND ADAPTIVE MANAGEMENT

Urban gardens are often considered a type of NbS (Cabral et al., 2017a; van der Jagt et al., 2017). Adopting a dual ES and EDS perspective is fundamental to navigating the complexities of the FWS nexus and achieving sustainable outcomes

in urban gardens. Rather than simply maximizing benefits, this balanced lens aims to optimize the net contribution of these systems to urban well-being.

First, this approach offers a far more comprehensive and accurate understanding of the garden's internal dynamics. The goal of FWS nexus thinking is to develop integrated management practices that recognize the profound interdependence of food production, water use, and soil health. The soil, in particular, acts as a living ledger—its health is a direct reflection of past management choices and a predictor of future potential. Spatially and temporally mapping the flows of both services (e.g., food provision, stormwater infiltration) and disservices (e.g., potable water consumption, nutrient runoff) allows us to identify critical interdependencies and time-lag effects. For example, the decision to maximize yield in one season through intensive irrigation and fertilization can create a “yield drag” in subsequent years by degrading soil structure and depleting organic matter, thus reducing its water-holding capacity (Hatfield et al., 2017). The FWS lens makes this causality visible, transforming our management from a series of isolated actions into a conscious dialogue with the garden's ecological memory (Cortinovis et al., 2021).

Second, the ES/EDS perspective fosters a more rational and inclusive dialogue among stakeholders. All too often, urban greening initiatives are promoted with a uniformly positive narrative that can obscure real-world conflicts and trade-offs. The FWS framework provides a common, transparent language that makes not

only the benefits but also the potential risks and nuisances of urban gardens visible and debatable. This moves the conversation beyond green aesthetics, enabling gardeners, residents, and city planners to openly address conflicts over issues like water rights, pest management strategies, the risk of allergens, or even gentrification pressures associated with new green spaces (Lyytimäki, 2015). By quantifying and qualifying both sides of the ledger, we encourage evidence-based negotiation and the co-design of solutions that reflect the shared goal of minimizing harms while securing benefits. This process enhances the legitimacy of urban greening policies and fosters a more resilient social fabric surrounding them (Webb et al., 2018).

Furthermore, this dual perspective is a powerful antidote to the fragmented, siloed governance that plagues most municipalities. Acknowledging that urban gardens co-produce a wide array of services and disservices reveals their deep entanglement with multiple domains of urban policy (Haase et al., 2014). The FWS framework allows different city departments to understand the interconnected impact of a single garden, e.g., *public health* (nutrition vs. allergens), *urban hydrology* (infiltration vs. contaminated runoff), *waste management* (composting vs. green waste & odor), and *planning and development* (community building & aesthetic improvement vs land-use conflicts).

Finally, this approach is crucial for moving beyond a mindset of resource extraction and toward a culture of ecological stewardship. It reorients our ultimate goal from

simply “*producing food*” to “*maintaining the health and functionality of the garden as a long-term urban asset.*” Assessing the outcomes of different management practices on the net provision of ESs over EDSs provides a direct feedback loop for continuous learning and improvement. By making the costs of disservices clear, it justifies investments in knowledge, skills, and infrastructure (i.e., human and built capital). This perspective helps us value the garden not just for the harvest it provides, but for its ongoing capacity to function (van der Jagt et al., 2017). It aligns with the “One Health” concept, recognizing that the health of soil, plants, and people is deeply interwoven and indivisible. Ultimately, governing urban gardens through an FWS-ES/EDS lens is an act of foresight—a commitment to managing these vital urban spaces not for a single season’s yield, but for their enduring contribution to a resilient and livable city for generations to come.

3.5 MANAGEMENT HIERARCHY TO THE FWS NEXUS IN UFGs

Governing urban gardens through an FWS-ES/EDS lens requires moving beyond a simple goal of maximizing benefits and toward a more nuanced approach of optimizing the net contribution of these systems to urban well-being. This necessitates a structured framework for decision-making that is both comprehensive and adaptive. The management hierarchy proposed by Vaz et al. (2017), which refines the traditional precautionary principle, offers such a

framework. It provides a systematic, step-by-step process for managing complex social-ecological systems by integrating the identification, prevention, and mitigation of harms alongside the protection and restoration of benefits. This hierarchy consists of five sequential but interconnected steps: (1) Identification and Evaluation, (2) Protection and Avoidance, (3) Minimization, (4) Restoration, and (5) Compensation and Adaptation. Figure 3 illustrates the main components and principles of the management hierarchy for ESs and EDSs, modified for UFGs and FWS nexus. Applying this hierarchy provides an actionable pathway for the adaptive management of the FWS nexus in urban gardens.

Each step of the hierarchy addresses a different aspect of managing the services and disservices that arise from FWS interactions. The application of this hierarchy

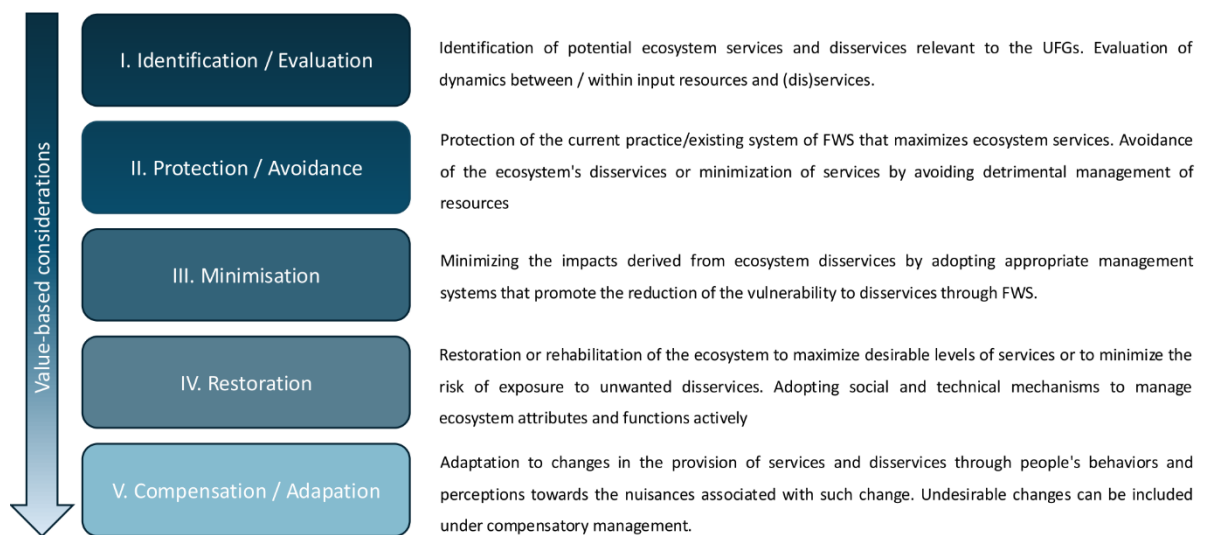


Figure 3.3. Management hierarchy for strategic management of ecosystem services and disservices, adapted from Vaz et al. (2017).

is not a linear, one-time process. Instead, it forms the engine of a continuous cycle of adaptive management. The "Evaluation" in Step 1 and the "Adaptation" in Step 5 create a crucial feedback loop. After a management adaptation is implemented (e.g., changing crop types), the system must be re-evaluated to assess its new performance and identify any new or altered ES and EDS that may have emerged. This iterative cycle of planning, acting, monitoring, and learning is the essence of managing a complex social-ecological system like an urban garden for long-term resilience and net positive impact. Furthermore, this governance framework makes it clear that managing the FWS nexus cannot be a purely technical or ecological endeavour confined within the garden's boundaries. The identification of large-scale social disservices like green gentrification demands that nexus governance extends to the municipal policy level. Minimizing such disservices requires planners and advocates to employ tools beyond the scope of garden management, such as inclusionary zoning, community land trusts, rent stabilization policies, and other anti-displacement strategies. Therefore, a holistic application of the Vaz et al. (2017) framework compels a broader interpretation of "social-ecological management" that fully integrates on-site biophysical practices with engagement in urban planning and housing policy

The following Table 3.3 provides a practical summary of how this management hierarchy can be operationalized to address common challenges within the FWS nexus of urban gardens

Table 3.3. Application of the Management Hierarchy to FWS Nexus challenges in Urban Gardens

FWS Challenge	1. Identification/Evaluation	2. Protection/Avoidance	3. Minimization	4. Restoration	5. Compensation/Adaptation
Soil Contamination	Conduct soil tests for heavy metals; research site's land-use history.	Avoid siting gardens on former industrial land or near high-traffic roads.	Use mulch to reduce dust/splash; wash produce thoroughly; practice good personal hygiene (gloves, handwashing).	Build raised beds with clean, imported soil; amend existing soil with compost/biochar to immobilize Pb.	Grow non-edible plants (ornamentals); implement public health advisories and educational programs.
Water Scarcity	Assess local climate data, water sources, and costs.	Select drought-tolerant crop species and varieties.	Use efficient drip irrigation systems; apply a thick layer of mulch to reduce evaporation.	Amend soil with compost to significantly increase its water-holding capacity.	Install rainwater harvesting systems (e.g., rain barrels); adapt planting schedules to align with rainy seasons.
Pest Outbreaks	Monitor crops for pests and identify beneficial insects.	Plant pest-resistant varieties; create diverse	Use Integrated Pest Management techniques: companion planting,	Actively introduce beneficial insects (e.g., ladybugs) if natural	Adapt crop selection to species that are less

		habitats to support natural predators.	mechanical removal (hand-picking), biological controls.	populations are insufficient.	susceptible to prevalent local pests.
Social Conflict	Facilitate community meetings to identify shared values, goals, and potential points of conflict.	Establish clear, written garden rules, bylaws, and a transparent governance structure from the project's outset.	Implement formal and informal conflict mediation processes; create shared communication channels (e.g., message boards, group chats).	Host community-building events (e.g., potlucks, workshops) to restore social cohesion and trust.	Periodically review and adapt garden rules and governance structures based on community feedback and evolving needs.
Green Gentrification	Analyse local housing market data (rents, property values); engage with residents about displacement concerns.	Prioritize gardens in neighbourhoods with high rates of homeownership or existing affordable housing protections.	Advocate for and implement anti-displacement policies (rent control, inclusionary zoning) alongside greening projects.	Establish Community Land Trusts to secure land for permanent affordability and community control.	Develop "right to return" policies for displaced residents; ensure new jobs created by greening prioritize local hiring.

3.5.1 Step 1: Identification and Evaluation

This initial diagnostic phase is the cornerstone of adaptive management. It involves a comprehensive assessment to identify the potential ES and EDS relevant to a specific garden context. This is not a purely technical exercise; it requires participatory engagement with all relevant stakeholders to understand their diverse values, needs, and perceptions of risk. (e.g., gardeners, adjacent residents, community leaders, and municipal officials). Key activities in this stage include:

- *Site History and Context Analysis:* Researching the past land use of the site to identify potential legacy issues, such as industrial contamination
- *Biophysical Assessment:* Conducting soil tests to determine baseline fertility, structure, and, crucially, the presence of contaminants like heavy metals (S). Water sources (W) must also be evaluated for quality, whether they are municipal tap water, harvested rainwater, or potentially contaminated stormwater or greywater.
- *Trade-Off Analysis:* Making the inherent trade-offs within the FWS nexus explicit. For example, evaluating how plans to maximize food production (F) would impact water demand (W) and the potential for nutrient leaching (S). This step ensures that a full accounting informs decisions of both positive and negative potential outcomes.

3.5.2 Step 2: Protection and Avoidance

This proactive step prioritizes preventing negative impacts before they occur and preserving existing ecological assets. It is the most effective and often the least costly stage of management.

- *Protection*: This involves safeguarding valuable existing resources. For instance, if a proposed garden site contains a patch of high-quality, undisturbed topsoil or hosts a remnant population of native pollinators, management plans should prioritize their conservation.
- *Avoidance*: This entails making strategic choices to prevent the introduction of risks. Examples include siting gardens away from high-traffic corridors to avoid the atmospheric deposition of vehicle-related pollutants, prohibiting the use of materials known to be sources of contamination (such as chromated copper arsenate-treated wood for raised beds), and selecting drought-tolerant crops in water-scarce climates to avoid exacerbating water stress.

3.5.3 Step 3: Minimization

When a disservice cannot be completely avoided, the goal shifts to minimizing its magnitude and impact. This step focuses on implementing best management practices to reduce ongoing harms.

- *Minimizing Water Consumption:* To address the disservice of high potable water use, gardens can implement water-wise strategies. These include the use of efficient drip irrigation systems and the application of mulch to the soil surface to reduce evaporative losses dramatically
- *Minimizing Pest-Related Disservices:* To reduce reliance on chemical pesticides (which cause disservices of soil/water contamination and harm to non-target species), gardens can adopt Integrated Pest Management (IPM). IPM strategies include companion planting to repel pests, creating insectary plantings to attract beneficial predatory insects, and using mechanical controls like hand-picking pests or using traps.

3.5.4 Step 4: Restoration

This step involves active intervention to repair degraded components of the FWS nexus and rebuild the system's capacity to provide ecosystem services.

- *Restoring Soil Health:* Urban soils are often severely compacted and may be contaminated. Restoration is a critical management action. For sites with heavy contamination or intractable compaction, this may involve constructing raised beds and importing a clean, high-quality engineered soil mix. For less degraded soils, restoration can be achieved by amending the soil with large quantities of compost. This practice directly counteracts

compaction, rebuilds soil structure, increases soil organic matter, dramatically improves water-holding capacity (W), and can help immobilize certain heavy metal contaminants, reducing their bioavailability.

3.5.5 Step 5: Compensation and Adaptation

This is the final level of the hierarchy, addressing unavoidable negative impacts that persist after all other steps have been taken

- *Compensation*: This is less common at the scale of a single garden but is relevant in a city-wide planning context. For instance, if a community garden is unavoidably lost to development, a compensatory policy might require the developer to fund the establishment of a new, equivalent garden space elsewhere in the neighbourhood
- *Adaptation*: This involves changing human behaviours, expectations, and social structures to cope with unavoidable disservices. A primary example is adapting to water scarcity by shifting crop selection (F) away from water-intensive vegetables like lettuce and toward drought-tolerant alternatives such as pole beans, Swiss chard, or amaranth. Adaptation also includes social strategies, such as developing clear governance structures and conflict resolution protocols to manage social disagreements, or

implementing public health education campaigns to ensure gardeners follow safe practices (e.g., hand washing, produce washing) to adapt to the risks of soil contaminants.

4

ECOSYSTEM SERVICES AND DISSERVICES PERCEIVED BY NANTES URBAN GARDENERS

DISCLAIMER

This chapter is a novel contribution to the dissertation. It will be submitted to the *Ecosystem and People* journal or *Ecosystem Service* journal, under the title “*Perceived ecosystem service and disservice synergies, trade-offs and bundles in urban gardens of Nantes, France.*”. The paper is authored by:

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Data and code availability

The dataset and all R scripts used for the statistical analyses presented in this chapter have been permanently archived. The materials are publicly available on GitHub and can be accessed at: <https://doi.org/10.5281/zenodo.17379253>

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Abstract

This chapter deconstructs the worldviews of 54 urban gardeners in Nantes Nord, a district targeted for urban regeneration, to inform sustainable governance. Using a multi-phase statistical analysis of survey data on perceived ecosystem services (ES) and disservices (EDS), the research reveals that gardener perceptions are organized into four fundamental bundles: (1) Nuisance, (2) Ecological Stewardship, (3) Pollution, and (4) Well-being. The analysis demonstrates that these perceptions are shaped more profoundly by social identity (person) than by physical location (place), with significant gendered differences in valuing care-related services and perceiving risks. The findings underscore the necessity of integrating user perceptions into urban planning, showing how a granular understanding of ES/EDS dynamics can lead to more effective, equitable, and resilient management of urban green infrastructure.

Keywords: *urban garden, ecosystem services, ecosystem disservices, trade-offs, synergies.*

4.1 INTRODUCTION

The 21st century is unequivocally the urban century. With a majority of the global population now residing in cities (United Nations, 2019), the pursuit of sustainable urban development has become a paramount global challenge. In response, urban planners, policymakers, and researchers are increasingly turning to Nature-Based Solutions (NbS) as a strategic framework for enhancing urban resilience, conserving biodiversity, and improving human well-being (Frantzeskaki, 2019; Laforze et al., 2018; Nesshöver et al., 2017). Urban nature delivers multiple ecosystem services (ES) necessary for human health and well-being (Daily et al., 1997; Millennium Ecosystem Assessment, 2005). These solutions, which leverage the functions of natural ecosystems, represent a shift away from purely engineering-based approaches towards more integrated, adaptive, and multifunctional urban interventions. Inevitably, natural ecosystem also generates ecosystem disservices (EDS) (Lyytimäki and Sipilä, 2009; Pereira et al., 2023; von Döhren and Haase, 2015).

While ES or nature's contribution to people has been extensively studied (IPBES, 2019), EDS are often less considered and less discussed, yet equally important for urban ecosystem management (Lyytimäki et al., 2008; Schaubroeck, 2017). Previous work has shown that EDS can have a profound impact on economic loss and people's well-being in home gardens if not appropriately managed (Herd-Hoare and Shackleton, 2020). Risks from soils contaminated with heavy metals

from geogenic origins (Jean-Soro et al., 2015) or industrial inheritance (Howard et al., 2013) are frequently discussed without necessarily being categorized as EDS.

In the previous chapter, we established the theoretical construction of the ES and RDS model in the resource management framework (FWS). To understand how the EDS and EDS play their roles in the dynamic feedback loop that influences management and the state of urban garden resources over time, this chapter investigates the perceptions, values, beliefs, and concerns of the people who interact with these spaces daily. Understanding their relationships, synergies, and trade-offs can determine levels of engagement, stewardship, and community ownership in urban gardens.

The primary objective of this chapter is to systematically deconstruct and model the perceptual framework of urban gardeners in Nantes Nord. To achieve this, the analysis proceeds through a multi-phase, sequential methodology. We begin by establishing a detailed socio-demographic and behavioural profile of the gardeners, followed by quantifying their perceptions of the benefits (Ecosystem Services – ES) and drawbacks (Ecosystem Disservices – EDS) associated with their gardens, exploring the degree of consensus and disagreement within the community. Subsequently, it investigates the potential drivers of these perceptions, comparing views across different garden sites and between genders. Finally, we map the inter-relationships between individual perceptions to synthesize them into a coherent, underlying model of the gardeners' viewpoint.

4.2 METHODOLOGY

4.2.1 Study area

The city of Nantes serves as the administrative capital of the Pays de la Loire region in France. Covering an area of 65.19 km², it ranks as the sixth largest city in France by population, with 325,070 inhabitants recorded in 2022 (Insee, 2025). Nantes is also the principal municipality within Nantes Métropole. Historically, the city has been renowned for its horticultural and botanical activities, a tradition that has been strengthened through robust political support and active public engagement. The city features an extensive network of rivers, parks, and green corridors, specifically along the Loire, Erdre, Cens, Gesvres, Chézine, and Sèvre rivers, which enhance its biodiversity and facilitate pedestrian movement. Notably, approximately 60% of Nantes's land comprises natural and agricultural areas. Among these green spaces are 27 community gardens dispersed throughout the city, totalling 1,163 garden parcels ranging from 50 to 150 m² each. These gardens are managed collectively by volunteer organizations of gardeners, which maintain a formal connection with the Nantes municipal government via the Department of Nature and Gardens.

This study focuses on Nantes Nord, a district situated in the northern part of the Nantes municipality, covering approximately 7.6 km² (760 hectares). Although often regarded as Nantes's greenest district, with 62 m² of green space per inhabitant, it also exhibits the highest density of social housing, with 37% of residents living in such accommodations. This combination presents complex socio-economic, security, and health challenges (Bodénan and Béchet, 2023). In response, Nantes Métropole has initiated the "Projet Global Nantes Nord" to enhance living conditions, promote social cohesion, and support economic development in the area. As part of this initiative, the city partnered with the URBiNAT project in 2016 to explore and implement urban nature-based solutions (NbS), fostering cross-sector collaboration aimed at addressing urban complexities. Community gardens serve as a prominent example of these NbS. Currently, the district features six collective gardens with a total of 205 plots

(Nantes Métropole, 2025). Our research focuses on five of these gardens: Angle Chaillou (AC), Amande (AM), Bruyères (BR), Eglantiers (EG), and Cressonnière (CR).

First, background information related to previous and ongoing projects focusing on urban gardens in the study area was collected, alongside regulations and policy documents about urban gardening in Nantes. Perceptions were measured on a 4-point ordinal Likert scale, ranging from "Not important/concerned" (1) to "Highly important/concerned" (4). Data were collected through a structured survey that recorded demographic information, gardening behaviours (e.g., visit frequency and duration), and perceptions of 10 ES and 11 EDS. The survey questions are listed in Appendix B.1. We focused on those most relevant to gardeners and combined similar ES, such as aesthetic and mood enhancement, into broader

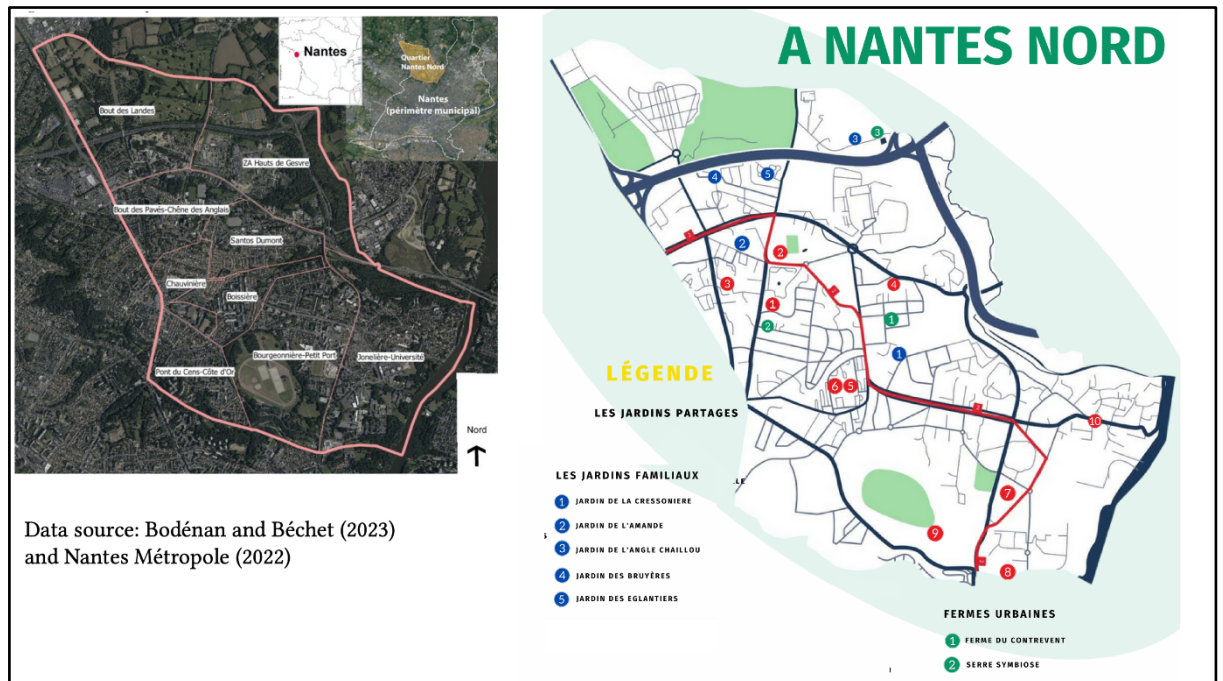


Figure 4.1. Location of studied gardens

categories like mental health to reduce gardeners' confusion. Additionally, during the interview process, we excluded ES and EDS, for which several gardeners expressed their lack of sufficient understanding to rate accurately, such as the global climate regulation service. Table 4.1 summarizes the final list of ES and EDS in this study.

Table 4.1. Typologies of ecosystem services and disservices in this study

Services and Disservice from urban garden	Qualitative proxies	References
Ecosystem service		
Air Quality	<i>Air filtering pollution and particulates from vegetation</i>	
Biodiversity	<i>Richness and diversity of plants and animals that contribute to healthy urban garden systems</i>	
Education	<i>Knowledge acquired about the structure and functioning of the current and past natural environment</i>	
Food Provision	<i>Vegetative parts that are harvested and used for human consumption</i>	
Medicinal & Aromatic	<i>Flowers, herbs, and other parts of plants are used for cooking and medicinal purposes</i>	Camps-Calvet et al., 2016; Evans et al., 2022; Langemeyer et al., 2016
Mental Health	<i>Mood improvement and stress reduction associated with working and leisure time spent in gardens</i>	
Physical Health	<i>Gardening as part of the physical exercise, improving bodily functions</i>	
Social Cohesion	<i>Increments in social interactions and relations through time spent in gardens</i>	
Soil Conservation	<i>Protecting soil from water and wind erosion</i>	
Water Regulation	<i>Water storage, purification, and groundwater recharge from soil and vegetation</i>	
Ecosystem disservices		
Air Pollution	<i>Pollution linked to volatile compounds emitted by plants</i>	

Allergies	<i>Pollen release that provokes allergic reactions or intoxications</i>	
Pet/Insect Bites	<i>Animal or insect bites on humans</i>	
Food Contamination	<i>Risk of contaminated vegetables, herbs, and fruits being consumed by humans</i>	
Maintenance Cost	<i>Costs associated with installing and maintaining gardens (seeds, equipment, materials, rent)</i>	
Fear of Wilderness	<i>Fear and risks of wilderness due to lack of light or attack by animals</i>	Vaz et al., 2017; von Döhren and
Unpleasant Odor	<i>Smells produced by animals, plants, or decomposing matter, such as green waste</i>	Haase, 2015
Water Consumption	<i>Amount of water used for plant growth</i>	
Soil Pollution	<i>Contaminated soil that can affect human health (inhalation or swallowing), or vegetation quality</i>	
Water Pollution	<i>Contaminated water that can affect human health (inhalation or swallowing) or vegetation quality</i>	
Green Waste	<i>Organic waste from plants (leaf litter, roots, branches, etc)</i>	

4.2.2 Semi-structured interview

Between April and July 2025, we conducted interviews using non-participant and participant observation techniques (Bessette, 2004). During this period, we engaged with gardeners in their gardens through tasks such as wall painting and weed clearing, where possible. We interviewed participants from five community gardens in Nantes Nord, with each session lasting 45 to 90 minutes, either at community rooms or in their gardens. Initially, interviews focused on garden association board members. We expanded our sample through a snowball approach, asking these members to refer additional gardeners. The study concluded with a total of 54 gardeners (see Table 4.2). Additionally, we held meetings with Nantes' urban planning practitioners (Nantes Métropole) and two non-profit organizations, MAAD and ECOS, which support urban garden initiatives. These meetings provided broader insights into how urban gardens integrate within green space policies and Nantes' strategic development.

Table 4.2. Breakdown of conducted surveys

Garden	N of interview	Total plot	Surface area (m ²)
AC	10	58	14469,12
AM	8	21	4754,45
BR	11	30	11306,78
CR	7	9	1176,76
EG	23	92	23222,69

4.2.3 Statistical analysis

Descriptive statistics (frequencies, means, medians) were employed to understand the socio-demographic and behavioural profile of the gardeners. To test for homogeneity in gardening behaviour across demographic groups, non-parametric inferential tests were used: the Mann-Whitney U test to compare time commitment between the two gender groups, and Spearman's rank correlation to assess the monotonic relationship between age group and time commitment.. This approach was chosen due to the nature of the primary data, which was collected using Likert scales. Likert-scale data is fundamentally ordinal; it possesses a natural order (e.g., "Highly important" is greater than "Moderately important"), but the intervals between the categories cannot be assumed to be equal (South et al., 2022). Treating such data with parametric tests (e.g., t-tests, ANOVA), which assume interval-level data and a normal distribution, can lead to statistically invalid and misleading conclusions. Non-parametric methods circumvent this issue by analysing the ranks of the data rather than the raw scores, providing a more robust and appropriate framework for ordinal variables. Given the ordinal nature of the primary outcome variables and the demographic data, the statistical analyses in the initial phases rely on non-parametric methods. These tests, such as the Mann-Whitney U test, Spearman's rank correlation, and the Kruskal-Wallis H test, analyse the ranks of the data rather than assuming a specific distribution, making them more robust and appropriate for this type of data.

Analysis was conducted using R (R Core Team, 2025), the *tidyverse* package (Wickham et al., 2019), and the *psych* package (Bernaards and Jennrich, 2005).

4.2.3.1 Comparative analysis

The Mann-Whitney U is the non-parametric equivalent of the independent samples t-test. It is the appropriate method for comparing the distributions of an ordinal dependent variable between two independent groups (McKnight and Najab, 2010). In this study, it was used to compare perceptions between genders (F/M). The test combines all observations from both groups and ranks these combined observations from lowest to highest. The test statistic, U, is calculated based on these rank sums. It essentially tests the null hypothesis that the probability of an observation from one group being higher than an observation from the other group is 0.5. A small p-value indicates this is unlikely, and the distributions are therefore different.

The Kruskal-Wallis H test, the non-parametric equivalent of a one-way ANOVA, was used to determine if perceptions differed significantly across the five community garden sites. The test assesses the null hypothesis that the medians of the groups are equal by pooling all data, ranking it, and comparing the mean rank of each group to the overall mean rank (Kruskal and Wallis, 1952) by the test statistic H.

$$H = \frac{12}{N(N + 1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N + 1)$$

When N is the total sample size, k is the number of groups, R_i is the rank sum for the group i , and n_i is the sample size of group i . A statistically significant result ($p < 0.05$) indicates that at least one group's median perception differs from the others.

Statistical significance (p-value) alone does not convey the practical importance or magnitude of an observed difference. Therefore, for all comparative tests, an appropriate rank-based effect size was calculated. For the Kruskal-Wallis test, Epsilon-squared (ϵ^2) was used to measure the proportion of variance in the ranks explained by group membership. For the Mann-Whitney U test, the *rank-biserial correlation* (r) was calculated. These metrics provide a crucial, standardized measure of the size of the effect, independent of sample size.

In cases where a Kruskal-Wallis test yields a significant result, a post-hoc test is required to conduct pairwise comparisons and identify which specific groups differ. Dunn's test is the appropriate follow-up procedure as it maintains the rank-based structure of the Kruskal-Wallis test. Crucially, it incorporates a p-value adjustment (e.g., Bonferroni correction) to control for the family-wise error rate—the increased probability of a Type I error (false positive) that occurs when conducting multiple statistical tests simultaneously.

4.2.3.2 Relational Analysis

To accurately model the associations between the ordinal Likert-scale items, a polychoric correlation matrix was calculated. This technique is specifically designed for ordinal variables, as it assumes an underlying continuous latent trait for each ordinal variable and estimates the relationship between these latent traits, thereby avoiding the underestimation of the true relationship strength (Holgado–Tello et al., 2010; Olsson, 1979). In this study, the procedure estimates "thresholds" on this underlying normal distribution that correspond to the boundaries between the observed categories (1, 2, 3, 4). For each pair of variables, it assumes that their two latent continuous variables follow a bivariate normal distribution. Using Maximum Likelihood Estimation, the algorithm finds the correlation coefficient (ρ) for the underlying latent variables that is most likely to have produced the observed contingency table of two ordinal variables.

To reveal the latent structure within this matrix, an agglomerative hierarchical clustering algorithm (*hclust*) was applied to reorder its rows and columns. This data-driven process groups variables with similar correlation patterns together, causing conceptually related items to form visible blocks or "clusters" in the resulting heatmap visualization (Eisen et al., 1998).

4.2.3.3 Structural Analysis

We employed Exploratory Factor Analysis (EFA) to synthesize the complex web of correlations into a parsimonious model of the underlying latent constructs. EFA is a multivariate statistical technique designed to uncover the latent structure of a set of observed variables by identifying groups of items that are highly intercorrelated. EFA is based on the common factor model, which posits that the variance in any observed variable (Z_i) can be partitioned into two components: Common variance (variance shared with other variables, attributable to the underlying common factors, F_m) and Unique Variance (variance specific to that variable alone, U_j). EFA represented by the equation:

$$z_i = a_{j1}F_1 + a_{j2}F_2 + \dots + a_{jm}F_m + d_jU_j$$

Where factor loadings (a_{jm}) are the correlation between the observed variable and the latent factor (Schreiber, 2021). The analysis was performed on the polychoric correlation matrix, the methodologically preferred approach for ordinal data (Holgado-Tello et al., 2010). The number of factors to retain was determined using Horn's Parallel Analysis, which compared the eigenvalues from the actual data to those from randomly generated data (Horn, 1965). This analysis provided an unambiguous recommendation to extract four factors. An oblique rotation (oblimin) was used, which allows the factors to be correlated—a realistic assumption for psychological constructs (Fabrigar et al., 1999).

Before conducting the analysis, two diagnostic tests were performed to assess the data's suitability: Bartlett's Test of Sphericity and the Kaiser-Meyer-Olkin (KMO) test. The first one evaluates the null hypothesis that the correlation matrix is an identity matrix (i.e., all variables are uncorrelated). For factor analysis to be viable, there must be patterned relationships to uncover, so this null hypothesis must be rejected ($p < 0.05$). KMO test then assesses the proportion of variance among variables that might be common variance. The KMO statistic ranges from 0 to 1, with values greater than 0.60 generally considered the minimum threshold for a satisfactory factor analysis.

4.3 RESULT AND ANALYSIS

4.3.1 Nantes Nord gardeners' profiles

The initial analysis establishes a distinct and consistent profile of the gardeners in the 54 surveyed plots. As summarized in Table 4.3, the community is predominantly composed of older adults; the largest age groups are 55-64 years (21 gardeners) and over 65 years (25 gardeners), meaning a remarkable 85% of the sample is aged 55 or older. This demographic suggests that the community gardens serve as important sites for post-retirement activity, social engagement, and well-being.

Table 4.3. Profile of the Nantes Nord Gardener Community (n=54)

Demographic/Behavioural Variable	Frequency	Percentage
<i>Age group</i>		
25-34	1	1.9%
35-54	7	13%
55-64	21	38.9%
65+	25	46.3%
<i>Daily time</i>		
< 1 hour	3	5.6%
1 – 2 hours	8	14.8%
2 – 3 hours	21	38.9%
> 3 hours	22	40.7%
<i>Weekly Frequency</i>		
1-2 days per week	3	5.6%
3-5 days per week	31	57.4%

Everyday	20	37%
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Engagement with the gardens is intense and represents a significant and regular time investment rather than a casual hobby. The most common weekly frequency of visits is 3-5 days (31 gardeners), and the most common duration per visit is more than 3 hours (22 gardeners). The mean waiting time for a plot is 1.87 years, with a median of 1.5 years (**Figure 4.2**). This demand is not uniform across the city; gardens such as BR and EG exhibit significantly longer average waiting times (approximately 2.2 years), indicating pressure on these specific urban green resources. The size of the garden plot ranges from over 50m² to 190 m².

Waiting Time to Acquire a Garden Plot

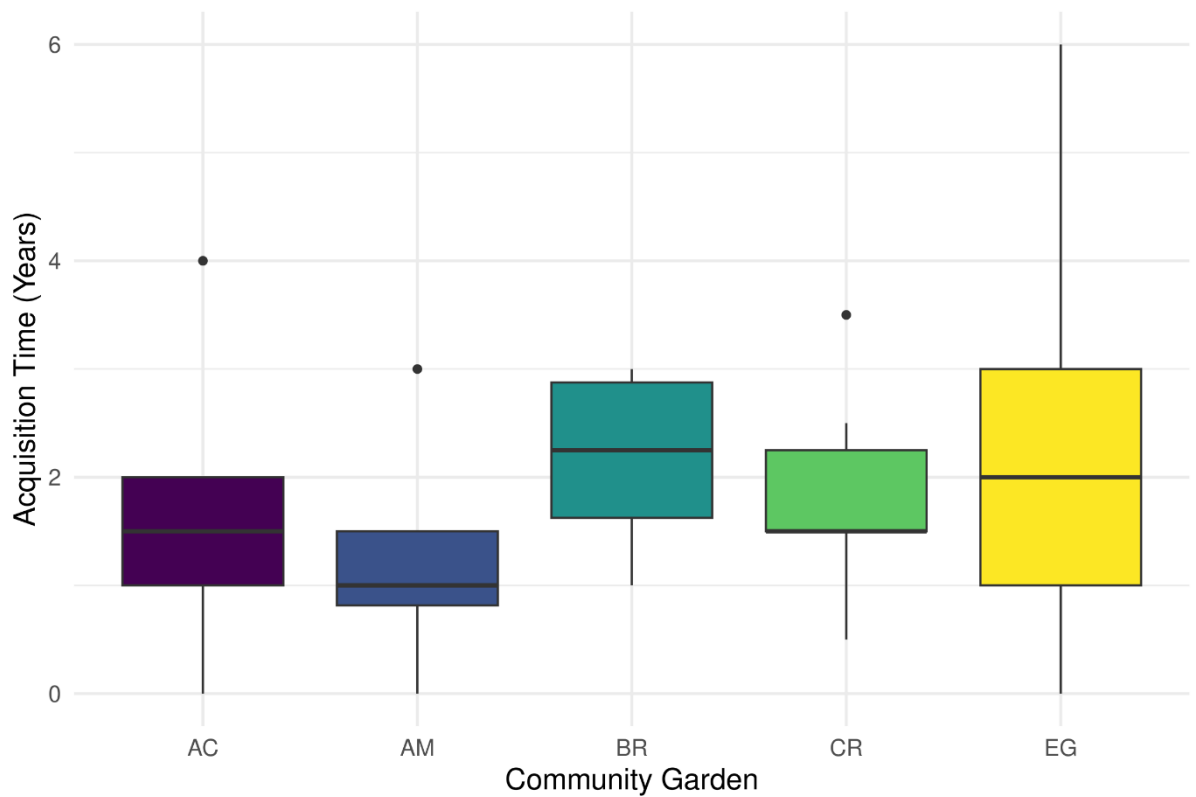


Figure 4.2. Average acquisition time for a garden plot in different gardens

Inferential tests revealed a lack of statistically significant relationships between these demographic characteristics and gardening behaviour. Mann-Whitney U tests showed no difference between male and female gardeners in terms of weekly visits ($p = 0.743$), or hours spent per visit ($p = 0.919$). Similarly, Spearman's rank correlations found no significant association between age group and time commitment. The results suggest that the high level of engagement is a shared cultural norm within this community, independent of gender or age.

4.3.2 Perceived ecosystem services and disservices

The analysis of gardeners' perceptions reveals a clear distinction between their views on benefits (ES) and their concerns about drawbacks (EDS). While the valuation of services is marked by high importance and strong consensus, the perception of disservices is characterized by a mix of specific, shared concerns and broad areas of deep disagreement.

4.3.2.1 Ecosystem services importance perceived by gardeners

There is a remarkably high and consistent appreciation for a wide range of ecosystem services. As shown in **Figure 4.3**, most bars that represent different ES are skewed heavily to the right, showing an overall positive sentiment towards ES perceived by the gardeners and high agreement that ES are “moderately” or “highly” important.

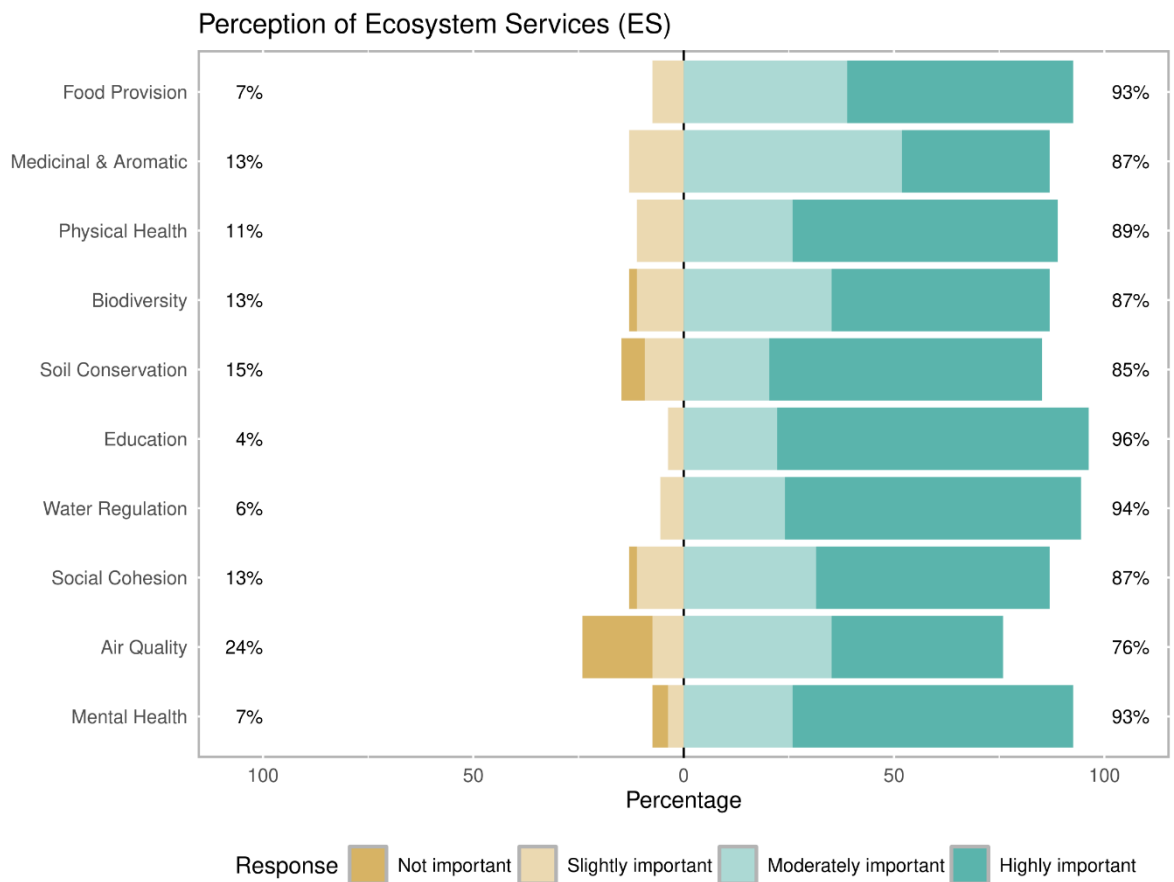


Figure 4.3. Gardeners' perception towards ecosystem service

Table 4 shows a precise numerical summary of the results. An impressive seven out of the ten services have a median rating of 4 ("Highly important"), including *Food Provision*, *Physical Health*, *Biodiversity*, *Soil Conservation*, *Social Cohesion*, *Water Regulation*, and *Mental Health*. The low Interquartile Ranges (IQR) for these items, consistently at 1.00, signify a high level of consensus among respondents. This indicates a strong, shared belief that these services are of significant importance. The service of *Education* stands out as the most uniformly agreed-upon benefit, with a median of 4 and the lowest IQR of 0.75, suggesting a powerful and near-universal belief in the educational value of the gardens. Gardeners often

express their appreciation for learning about the plants' interactions with weather, insects, and each other during interviews. Over 74% of respondents rated *Education* as "Highly important".

Table 4.4. Collective Perceptions of Ecosystem Services

Ecosystem service	Not important (%)	Slightly important (%)	Moderately important (%)	Highly important (%)	Median	IQR
<i>Food Provision</i>	0,0	7,4	38,9	53,7	4	1
<i>Medicinal & Aromatic</i>	0,0	13,0	51,9	35,2	3	1
<i>Physical Health</i>	0,0	11,1	25,9	63,0	4	1
<i>Biodiversity</i>	1,9	11,1	35,2	51,9	4	1
<i>Soil Conservation</i>	5,6	9,3	20,4	64,8	4	1
<i>Education</i>	0,0	3,7	22,2	74,1	4	0,75
<i>Water Regulation</i>	0,0	5,6	24,1	70,4	4	1
<i>Social Cohesion</i>	1,9	11,1	31,5	55,6	4	1
<i>Air Quality</i>	16,7	7,4	35,2	40,7	3	1
<i>Mental Health</i>	3,7	3,7	25,9	66,7	4	1

Note: Responses were coded into a Likert scale system, with 1 = not important; 2 = slightly important, 3 = moderately important, 4 = highly important.

4.3.2.2 Ecosystem disservices' importance perceived by gardeners

In stark contrast, the landscape of concern over EDS is far more varied and reveals important areas of polarization. *Water Consumption* is the standout issue, being the only disservice with a median of 4 ("High concern") and a low IQR of 1.00, signalling a strong and widely shared worry about water usage. Conversely, a strong consensus exists that several potential issues are not problems. Five of the

eleven disservices have a median of 1 ("Not concerned"), including *Unpleasant Odour*, *Fear of Wilderness*, and *Air Pollution* (as shown in **Table 4.5**).

Table 4.5. Collective Perceptions of Ecosystem Disservices

Ecosystem disservice	Not concerned (%)	Low concern (%)	Moderate concern (%)	High concern (%)	<i>Median</i>	<i>IQR</i>
<i>Food Contamination</i>	22,2	24,1	25,9	27,8	3	2
<i>Green Waste</i>	55,6	11,1	5,6	27,8	1	3
<i>Water Consumption</i>	3,7	9,3	25,9	61,1	4	1
<i>Water Pollution</i>	44,4	7,4	11,1	37,0	2	3
<i>Soil Pollution</i>	37,0	9,3	9,3	44,4	3	3
<i>Air Pollution</i>	53,7	20,4	11,1	14,8	1	1,75
<i>Allergies</i>	51,9	5,6	24,1	18,5	1	2
<i>Maintenance Cost</i>	46,3	22,2	25,9	5,6	2	2
<i>Pest/Insect Bites</i>	38,9	18,5	18,5	24,1	2	2
<i>Fear of Wilderness</i>	59,3	14,8	22,2	3,7	1	1,75
<i>Unpleasant Odour</i>	66,7	14,8	13,0	5,6	1	1

Note: Responses were coded into a Likert scale system, with 1 = not concerned; 2 = low concern, 3 = moderate concern, 4 = high concern.

The most critical finding in this phase, however, is the evidence of division on other environmental issues. Items such as *Green Waste*, *Water Pollution*, and *Soil Pollution* all have extremely high IQRs of 3.00. For *Green Waste*, while the median response is "Not concerned" (1), the IQR of 3.00 indicates that the central 50% of respondents span almost the entire scale, from "Not concerned" to "High concern". This reveals a possibility of a highly polarized community on these

topics, a finding that a simple measure of central tendency would have completely obscured (illustrated in **Figure 4.4**).

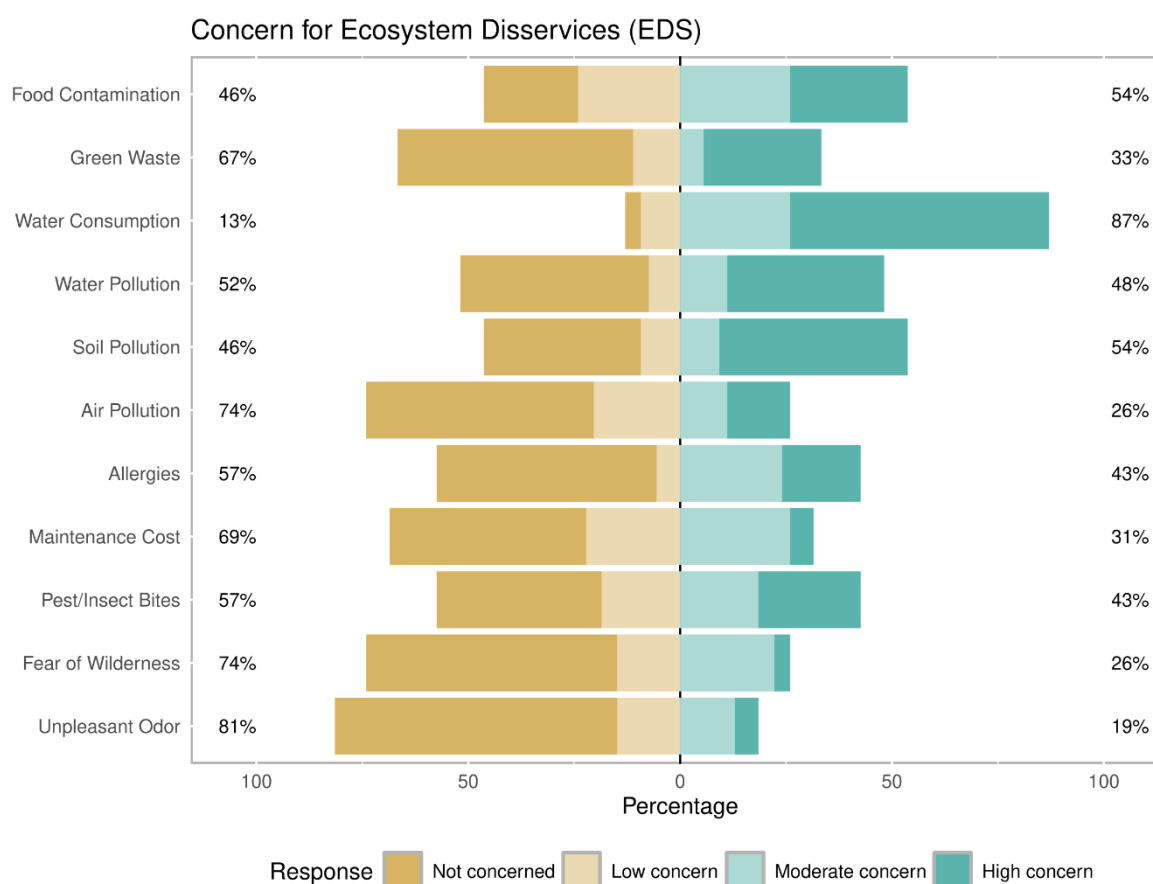


Figure 4.4. Gardener perceptions towards ecosystem disservices

4.3.2.3 Perception of ES and EDS across gardens

A series of Kruskal-Wallis H tests were conducted to determine if perceptions of the 21 ES and EDS items differed significantly across the five community garden sites. The tests yielded a clear and unambiguous result: there are no statistically significant differences in perception for any of the 21 items across the five

different community garden sites (see Appendix A6). All p-values were well above the 0.05 significance threshold, with the closest being for *Pest/Insect Bites* at $p = 0.06$. Furthermore, the calculated effect sizes were uniformly small or negligible, reinforcing the conclusion that the practical significance of any minor variations between gardens is trivial.

This consistency can be explained by several factors well-documented in urban ecological and sociological research. First, community gardeners are often a self-selecting group, drawn to the activity for similar reasons and from similar socio-economic backgrounds, leading to a pre-existing homogeneity in values regardless of the specific garden they join (Kingsley and Townsend, 2006). Second, gardeners within a single city often participate in a shared culture, exchanging information and practices through social learning networks, which leads to a convergence of beliefs (Krasny et al., 2014; Tidball and Krasny, 2007). Finally, the gardens are likely governed by a common municipal or institutional framework that standardizes rules and management practices, further minimizing site-specific differences (Colding and Barthel, 2013). This robust finding of homogeneity indicates that the specific physical location of gardens in Nantes Nord does not seem to be a primary driver of how its members perceive its benefits and drawbacks. This suggests that factors transcending individual sites (e.g., shared gardening culture, common municipal oversight, or the self-selecting nature of the gardeners themselves) are more influential.

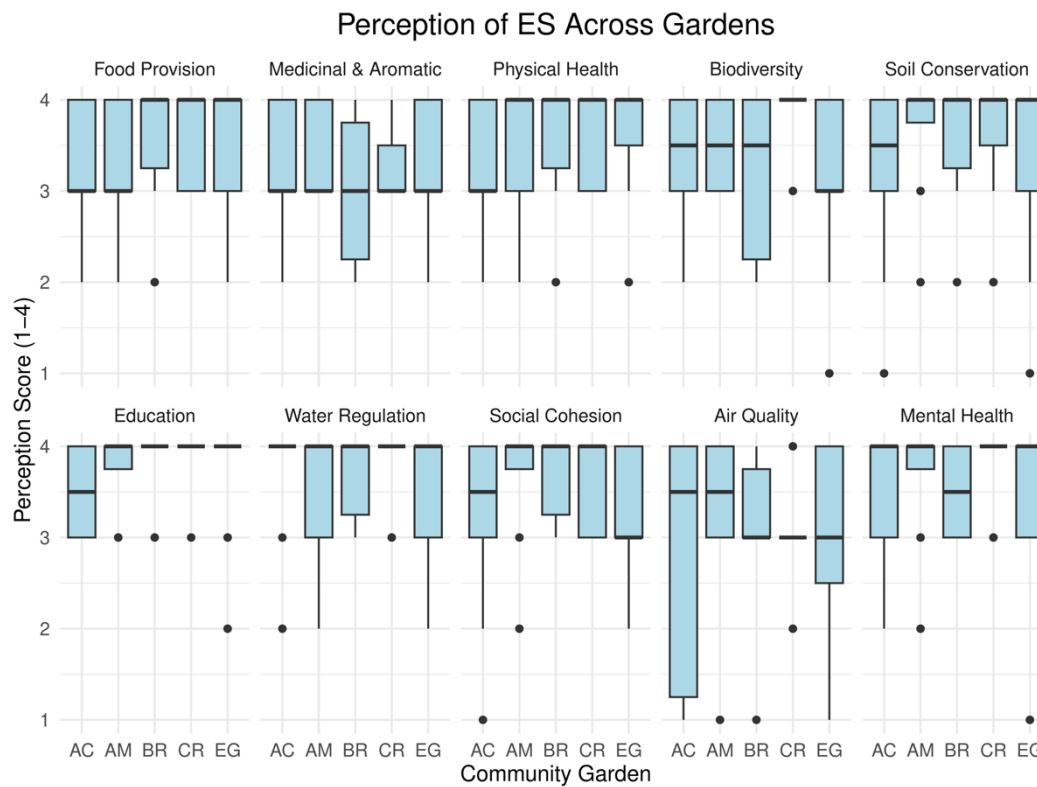


Figure 4.5. Gardeners' perception towards ES across 5 community gardens

Figure 4.5 illustrates the perception scores for all of the ES from the Kruskal-Wallis H test across five gardens. The medians (darker horizontal lines) are all roughly at the same height, while the boxes (representing the middle 50% of responses) show a high degree of overlap. It reinforces the argument that the distribution of responses for any given item is nearly identical regardless of the garden site.

Similarly, **Figure 4.6** shows a homogeneity across garden sites in the perceptions of gardeners towards EDS. Across 11 EDS, the box plots of different gardens show nearly identical medians and heavily overlapping interquartile ranges. The plots confirm the universal concern of *Water Consumption* in all study sites, and portray

a similarity in the low concern levels towards *Air Pollution*, *Unpleasant Odor*, *Fear of Wilderness*, and *Green Waste*.

Among the ecosystem disservices, the concern for *Soil Pollution* emerges as a point

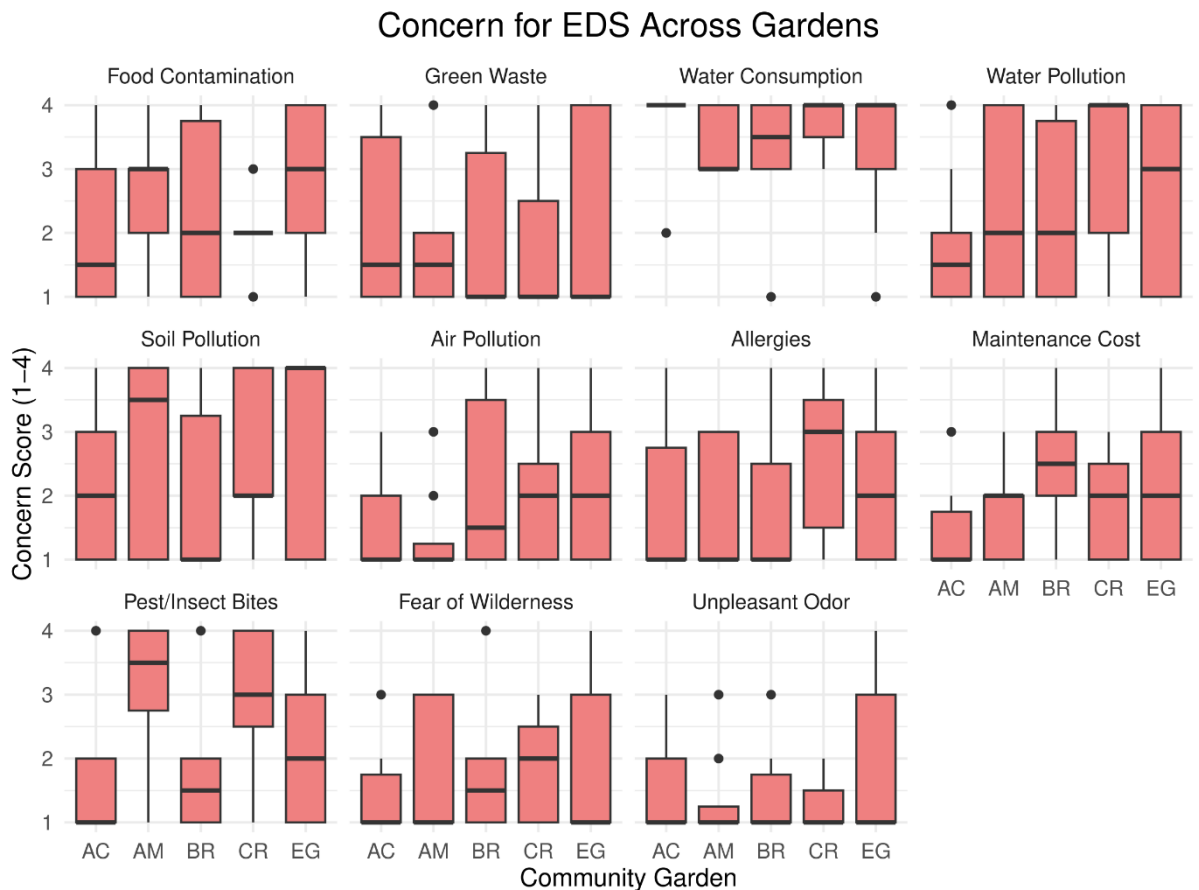


Figure 4.6. Gardeners' perception towards EDS across 5 community gardens

of notable community division, evidenced by a highly polarized response pattern with an IQR of 3.00. While this polarization is present across the sample, it can be explained by a well-documented, site-specific issue at the EG garden, which has a known history of trace element contamination. Notably, despite remediation actions undertaken by Nantes Métropole, gardeners at this site continue to

express a high level of concern, illustrating the persistent psychological legacy of environmental contamination, where perceived risk can endure long after technical mitigation has been implemented

4.3.2.4 Perception of ES and EDS by gender

In sharp contrast, the Mann-Whitney U tests revealed that gender is a significant factor in shaping a specific, but not universal, subset of perceptions. While most views were shared, statistically significant differences emerged for five items, falling into clear thematic categories related to care and risk (**Figure 4.7**).

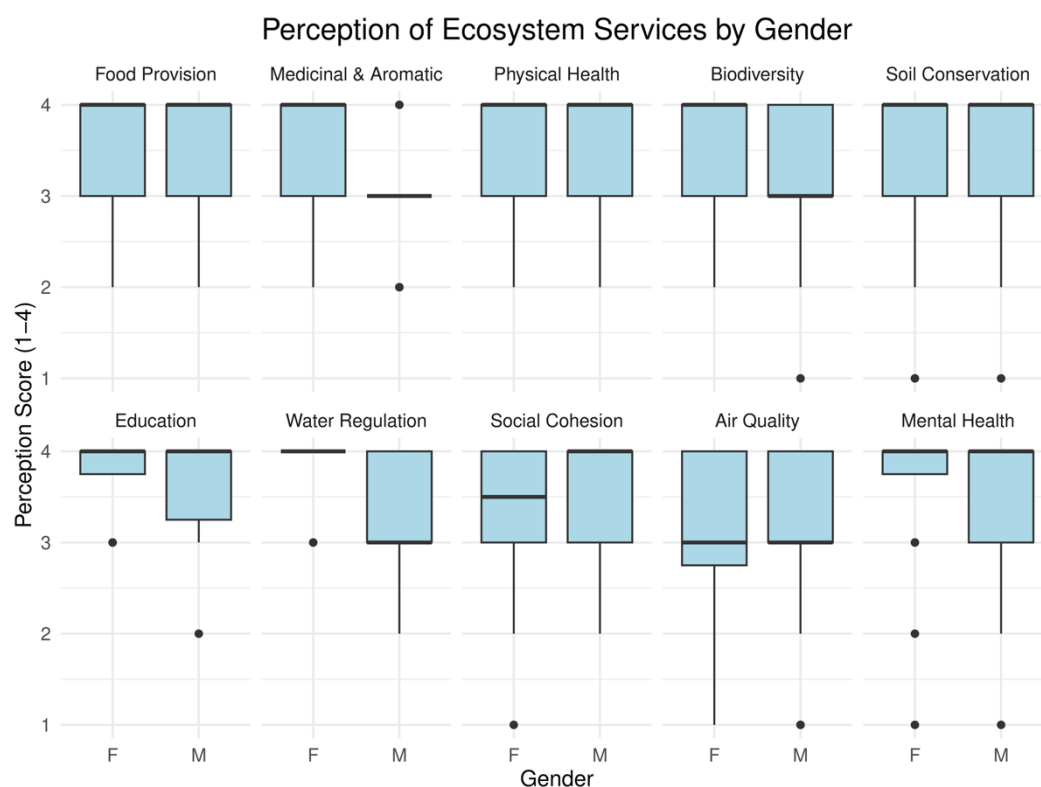


Figure 4.7. Gardeners' perception towards ES by different gender

Women were found to place a significantly higher value on services associated with an "ethic of care." The difference was most pronounced for *Medicinal & Aromatic Plants*, where the effect size was large ($r = 0.620$, $p < .00001$). A large effect was also found for *Water Regulation* ($r = 0.51$, $p = .00018$) and a moderate effect for *Biodiversity* ($r = 0.365$, $p = .0075$). This pattern is consistent with a large body of research suggesting that gendered socialization can foster different ways of relating to the environment, often linking women more closely to household care and pro-environmental behaviours (Wut et al., 2021).

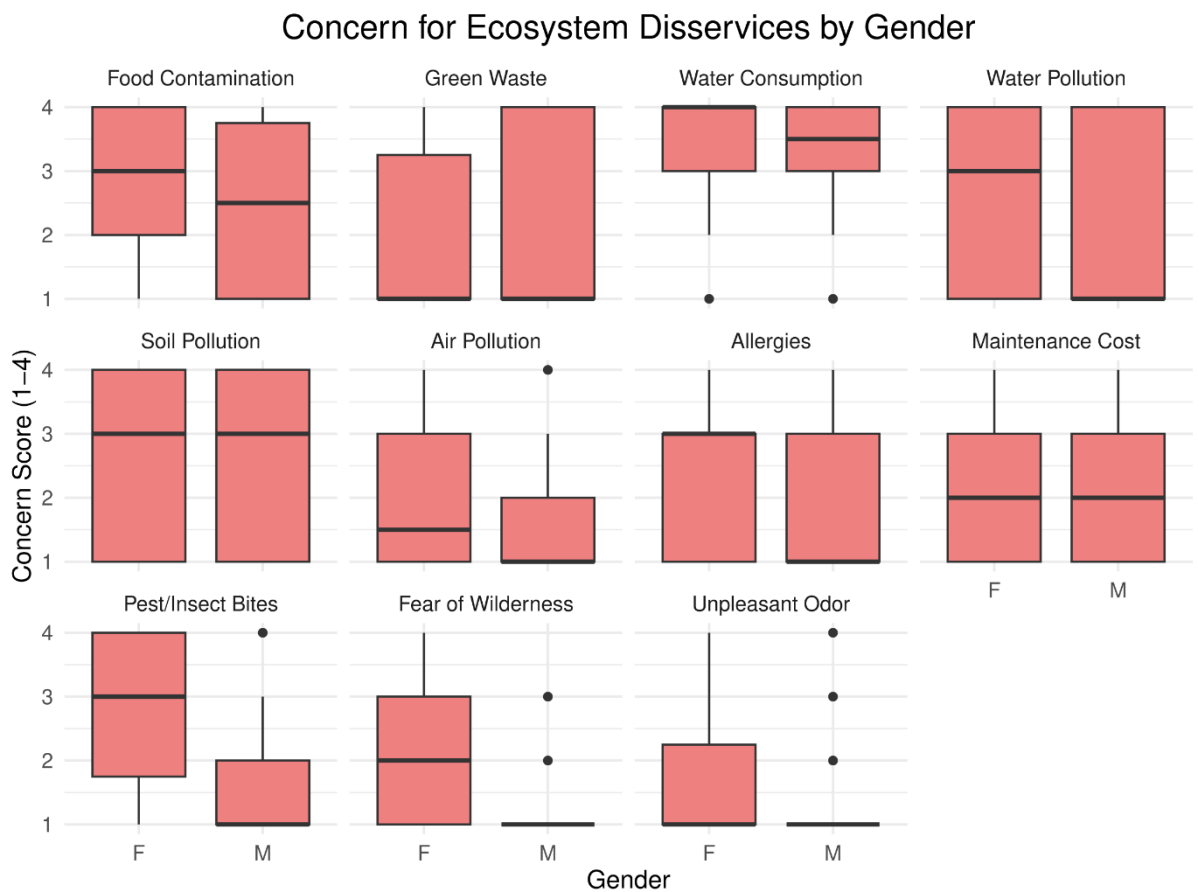


Figure 4.8. Gardeners' perception towards EDS in by different gender

Conversely, women reported significantly higher levels of concern for disservices related to personal risk and fear (shown in Figure 4.8). Moderate effect sizes were found for both *Pest/Insect Bites* large ($r = 0.366, p = .00739$). and *Fear of Wilderness* large ($r = 0.363, p < .00782$). Details of the statistical summary are in Appendix C7. This finding aligns with extensive literature showing that women tend to report higher perceptions of risk, particularly for health and safety, a pattern often attributed to social factors like a lower sense of physical control and higher personal vulnerability rather than innate traits (Skiba and Abramiuk, 2025).

It is equally important to note where no differences were found. Perceptions of core pragmatic items, such as *Food Provision*, *Maintenance Cost*, and the top concern of *Water Consumption*, were not significantly different between genders.

4.3.3 Synergies and trade-offs among ES and EDS

By employing a polychoric correlation matrix, we attempted to model the associations between these ordinal items, revealing the implicit structure of gardeners' beliefs, thus uncovering the latent connections between the 21 perception variables. Figures 4.10 and 4.11 visualize the correlation matrices of ES and EDS, respectively, as heatmaps. The strength of the correlation is indicated by the colour's intensity. Deep greens represent strong positive correlations, while deep red indicates strong negative correlations. White or pale colours signify a

weak or non-existent relationship. The heatmap is dominated by several blocks of strong positive correlations, indicating that certain beliefs are conceptually linked in the gardeners' minds.

4.3.3.1 Synergies among ES group and among EDS group

There is a strong positive correlation between the valuation of *Biodiversity*, *Soil Conservation*, *Water Regulation*, and *Education*, highlighted by a strong link between *Biodiversity* and *Soil Conservation* ($r = 0.74$). This correlation suggests that gardeners with a strong pro-environmental ethic perceive these services as integral components of a single, functional garden ecosystem. The tight coupling with *Education* ($r = 0.74$ with *Biodiversity*) indicates that these gardeners embody a "civic ecology" ethic (Krasny and Tidball, 2012), viewing the garden not merely as a space to conserve nature, but as a living pedagogy for understanding it (correlation matrix is illustrated in **Figure 4.9**)

A second, equally strong cluster connects *Mental Health*, *Physical Health*, and *Social Cohesion*. The strong correlation between mental and physical well-being ($r = 0.54$) confirms that gardeners experience these benefits holistically. It suggests that the physical act of gardening is inseparable from its stress-reducing mental benefits and that these are fundamentally mediated through community interaction. *Social Cohesion* is deeply integrated into this cluster, strongly correlated with both *Mental Health* ($r = 0.51$) and *Physical Health* ($r = 0.42$).

The garden is a social prescription as much as an ecological one, reinforcing its role
in building social capital

Correlations Within Ecosystem Services

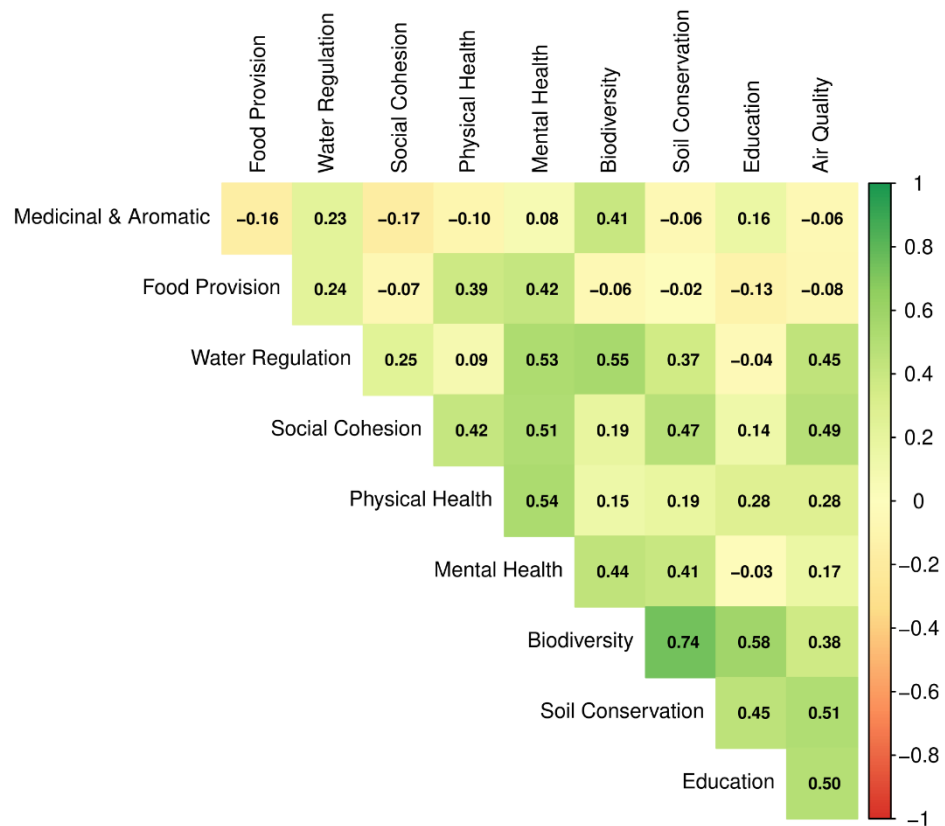


Figure 4.9. Ecosystem service correlation in urban garden

Within the ecosystem disservices (**Figure 4.10**), a coherent cluster of concerns related to contamination is visible. Strong positive correlations link the anxieties around *Soil Pollution*, *Water Pollution*, *Air Pollution*, and *Food Contamination*. Among this group, the connection between *Water Pollution* and *Soil Pollution* is highest, with $r = 0.87$. It seems to indicate a generalized "contamination anxiety", since being concerned about one pollution pathway is likely to be concerned about others. Another less intense, positive correlation links *Allergies* to *Green Waste* and *Unpleasant Odour* ($r = 0.66$ and $r = 0.65$, respectively).

Correlations Within Ecosystem Disservices

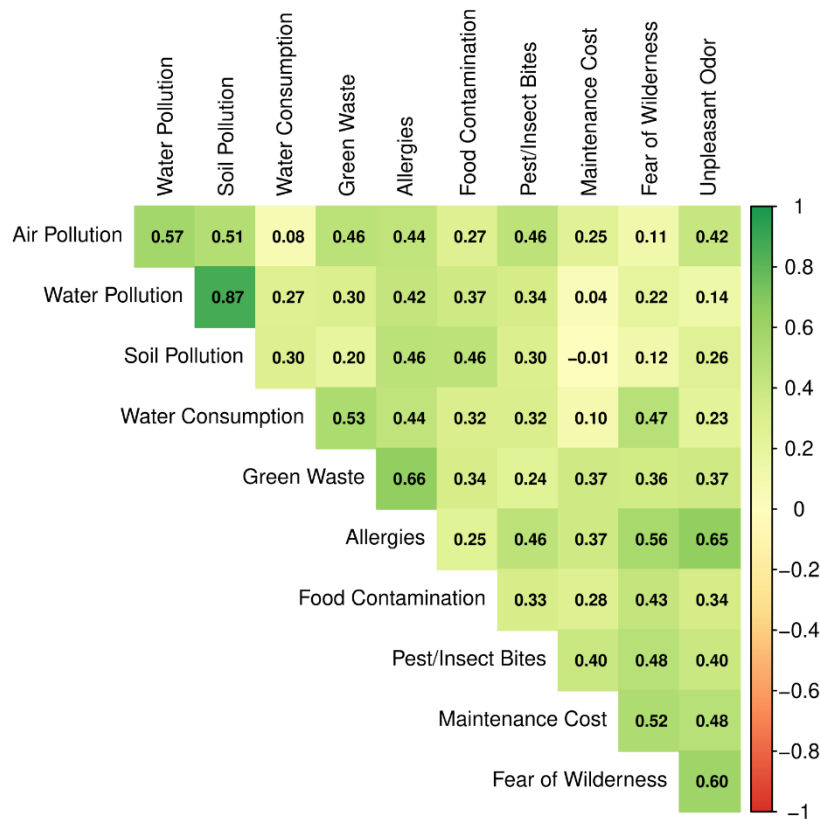


Figure 4.10. Ecosystem disservice correlation in urban garden

4.3.3.2 Synergies and Trade-offs between ES and EDS

While synergies dominate the belief system, the analysis of negative correlations is equally important, as it highlights the central conflicts that gardeners must navigate. Figure 11 presents the comprehensive heatmap of 21 ES and EDS (numeric data is presented in Appendix A8). The most critical trade-off identified in the data is the significant negative correlation between the value placed on *Food Provision* (an ES) and the concern towards EDS, such as *Water Pollution* ($r = -0.42$), or *Soil Pollution* ($r = -0.31$). *Food Provision* is also negatively

correlated with concerns about *Green Waste* ($r = -0.33$) and *Allergies* ($r = -0.32$). This might reinforce the idea of a utilitarian focus. Gardeners prioritizing food production seem less troubled by issues of waste management or personal discomforts like allergies, possibly viewing them as secondary to the primary goal of the harvest.

A different type of trade-off appears between the benefits of community and health versus minor discomforts. There is a moderate negative correlation between the value of *Social Cohesion* and concern for *Unpleasant Odour* ($r = -0.30$). Similarly, a negative correlation exists between *Physical Health* and *Unpleasant Odour* ($r = -0.32$). This suggests that gardeners who highly value the social and health-related aspects of their plots have a higher tolerance for sensory nuisances. They may perceive things like the smell of compost or manure not as a problem, but as a necessary and acceptable part of a healthy community garden.

One of the interesting findings concerns the relationship between ecological values and personal comfort. The initial interpretation of this study suggested a potential trade-off, with a negative correlation expected between the value for *Biodiversity* and the concern for *Pest/Insect Bites* and *Fear of Wilderness*. However, the empirical data reveals the opposite: there are moderately strong positive correlations between *Biodiversity* and *Pest/Insect Bites* ($r = 0.45$) and between *Biodiversity* and *Fear of Wilderness* ($r = 0.46$). The correlations represent not a conflict but a synergy of tolerance and ecological literacy. It

suggests that the gardeners who most value a complex, biodiverse ecosystem are also more aware of, and accepting of, its "wilder" elements, including a wide variety of insects. This reflects a sophisticated eco-centric mindset where an appreciation for ecological complexity comes with an acceptance of its associated realities.

Finally, the analysis highlights how effective policy can resolve potential disservices. *Green Waste*, which was identified previously as a point of community polarization, is shown in the relational analysis to be correlationally isolated, with very weak links to other concerns. The existence of effective municipal composting and organic waste collection infrastructure has likely removed the problem of green waste disposal from the gardeners' network of anxieties, demonstrating how top-down structural support can eliminate a potential disservice at its source.

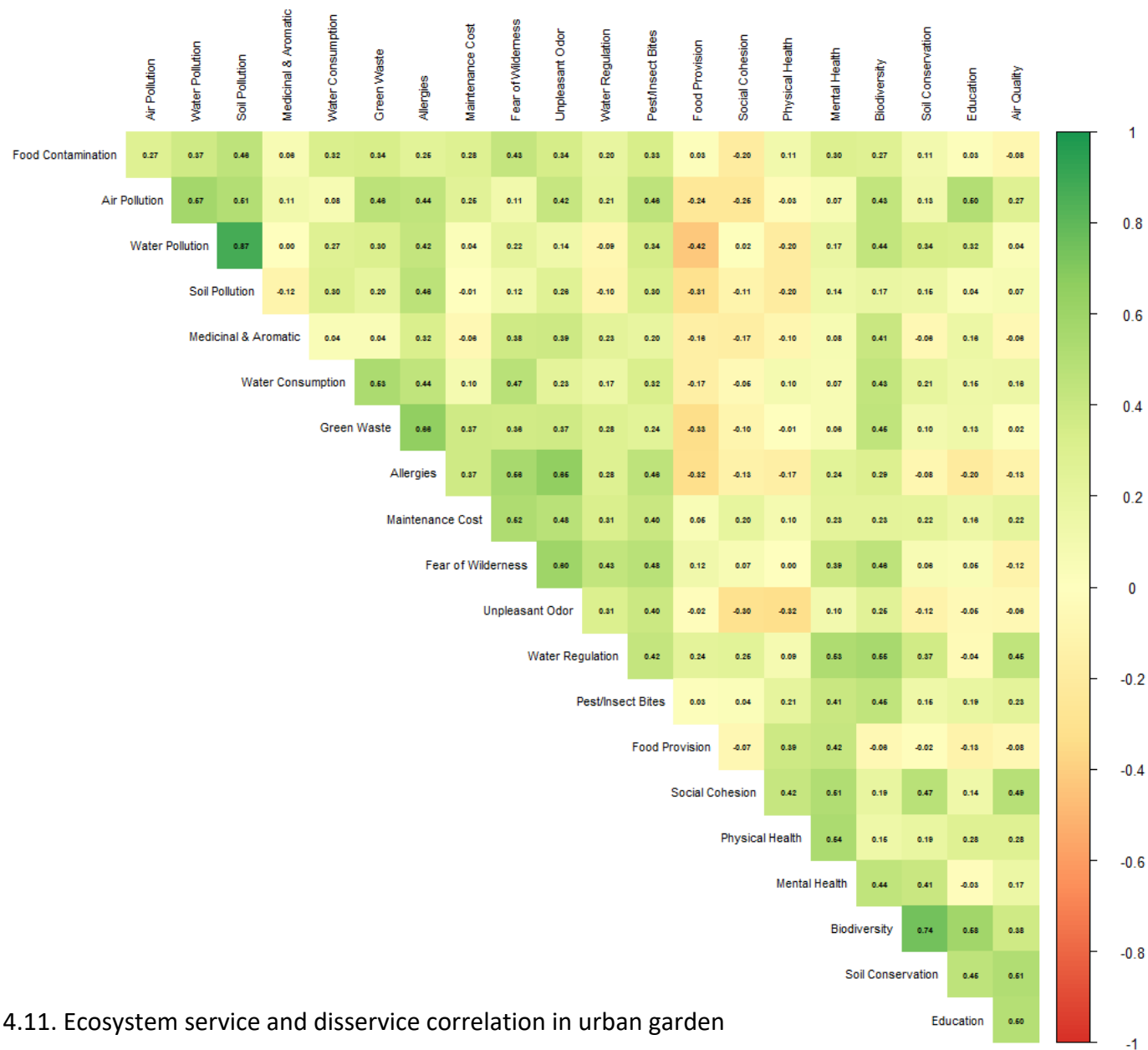


Figure 4.11. Ecosystem service and disservice correlation in urban garden

4.3.4 Bundles of ES and EDS from urban gardens

The EFA extracted four distinct and conceptually coherent factors that collectively account for 56% of the total variance in the data. The low inter-factor correlations ($r < 0.27$) confirm that they represent relatively independent dimensions of perception. The final rotated factor loading matrix is presented in **Table 4.6**, with weak loadings suppressed for clarity (only loadings with an absolute value > 0.40 are shown for clarity; details are in Appendix A9).

Table 4.6. Rotated Factor Loading Matrix for Gardener Perceptions

Ecosystem services and disservices	Factor 1: Nuisance	Factor 2: Ecological Stewardship	Factor 3: Pollution	Factor 4: Well- being
<i>Unpleasant Odour</i>	0.85			
<i>Allergies</i>	0.76			
<i>Fear of Wilderness</i>	0.76			
<i>Green Waste</i>	0.53			
<i>Medicinal & Aromatic</i>	0.47			
<i>Water Regulation</i>	0.45			
<i>Pet/Insect Bites</i>	0.45			
<i>Maintenance Cost</i>	0.44			
<i>Education</i>		0.79		
<i>Biodiversity</i>		0.75		
<i>Air Quality</i>		0.68		
<i>Soil Conservation</i>		0.66		
<i>Soil Pollution</i>			0.98	
<i>Water Pollution</i>			0.91	
<i>Mental Health</i>				0.91
<i>Physical Health</i>				0.53
<i>Food Provision</i>				0.51

- **Factor 1 - Nuisance:** This factor is primarily defined by a collection of EDS related to the immediate, visceral experience of being in the garden. High loadings from *Unpleasant Odour* (0.85), *Allergies* (0.76), *Fear of Wilderness* (0.76), and *Pest/Insect Bites* (0.45) characterize it as a dimension of personal sensitivity to the less pleasant aspects of a semi-wild environment. The inclusion of *Maintenance Cost* (0.44) suggests that economic burdens are perceived as a form of practical nuisance, grouped with these other discomforts
- **Factor 2 - Ecological Stewardship:** This is a strong factor that perfectly aligns with the synergistic cluster identified in section 4.3.3. It is defined exclusively by high loadings from ES that represent an appreciation for the garden as a functional ecosystem: *Education* (0.79), *Biodiversity* (0.75), *Air Quality* (0.68), and *Soil Conservation* (0.66). This factor represents a coherent eco-centric worldview, where the garden is valued for its intrinsic ecological functions and its role as a space for environmental learning.
- **Factor 3 - Pollution:** This is the most statistically pure factor, defined by near-perfect loadings from *Soil Pollution* (0.98) and *Water Pollution* (0.91). It represents a singular, focused anxiety about chemical contamination in the gardening environment. This concern is clearly distinct from the

general "nuisance" of Factor 1, pointing to a specific awareness of and concern for anthropogenic pollutants.

- **Factor 4 - Well-being:** This factor clearly represents the direct, personal benefits gardeners derive from their plots. It is strongly anchored by *Mental Health* (0.91), with significant contributions from *Physical Health* (0.53), *Food Provision* (0.51), and *Social Cohesion* (0.49). This statistically demonstrates that gardeners perceive these benefits holistically, integrating the psychological, physical, social, and provisioning aspects of their experience into one coherent dimension of personal well-being.

The results from EEA suggests that urban gardeners' perceptions are organized along two primary axes of benefits (Ecological Stewardship and Personal Well-being) and two primary axes of concerns (Nuisance/Discomfort and Pollution).

4.4 DISCUSSION

4.4.1 Self-Selected Community of Practice

One of the most significant findings of this study is the remarkable homogeneity of perceptions across the five different garden sites. The lack of statistically significant differences suggests that the specific physical location or micro-culture of an individual garden is not the primary determinant of how its members view

its benefits and drawbacks. From the author's perspective, this result should not be seen as a failure to find a difference, but rather as a positive discovery of a shared global view. The explanation for this homogeneity likely lies not in the gardens themselves, but in the social and administrative processes that populate them.

Community gardeners are not a random sample of the general population; they are a self-selecting group (D'Amore et al., 2026). The demographic profile revealed in this study—predominantly older, highly committed individuals—is consistent with findings from other Western cities that show community gardeners often share similar life stages and motivations (Grebitus, 2021). In Nantes, this self-selection is further structured by municipal policy. The official regulations for "jardins familiaux" stipulate that applicants must reside in Nantes and, critically, must not already own or rent a property with a garden. This policy effectively filters the applicant pool, selecting for individuals who lack private green space and are motivated enough to place their names on a waiting list that averages nearly two years.

This process creates a community of practice that is relatively homogeneous in its values and circumstances before its members even set foot in a specific garden. They share a common starting point: a strong desire for a connection to nature and a willingness to invest significant time and effort to achieve it. This shared background and motivation appear to be a more powerful shaping force on

perception than the unique characteristics of any single garden site. Research on urban allotment gardens has similarly found that overarching institutional frameworks are key in shaping their social and ecological outcomes, often leading to a convergence of practices and beliefs that minimizes site-specific differences (Colding and Barthel, 2013). Therefore, the homogeneity observed here is likely a direct reflection of a social system where demographic trends (an aging population seeking meaningful post-retirement activity) intersect with specific municipal policies that curate a community with a pre-existing shared ethos.

4.4.2 The Interplay of Governance and Grassroots Action

The perceptual landscape of the Nantes gardeners is not a static phenomenon but is actively co-produced by the dynamic interplay between top-down municipal governance and bottom-up civil society action. The data reveals specific patterns that can be understood by examining this interaction.

Gardeners in Nantes show a notably strong and consistent appreciation for a broad range of ES, especially those linked to health, nature, and community learning. The data highlights a clear consensus on these values. Seven out of ten surveyed services received a median importance rating of 4, the highest score on a 4-point scale, meaning at least half the respondents regard them as "Highly important." These top services include *Physical Health*, *Biodiversity*, *Soil*

Conservation, Water Regulation, Social Cohesion, and Mental Health. The low interquartile ranges (IQR) of 1.00 for most of these emphasize a shared and deeply rooted belief system. This pattern reflects more than just preferences; it mirrors the priorities built into Nantes' civic and political frameworks. The city's strategic plans and active local associations foster an environment where these benefits are promoted, experienced, and valued. For example, the importance given to *Social Cohesion* aligns with the city's participatory governance approach through the "Dialogue Citoyen" platform, which encourages collective action and engagement. Complementing this top-down effort are grassroots initiatives like MAAD, which aims to strengthen community ties through projects such as "potagers solidaires" (solidarity gardens). The high valuation of social cohesion relates to the lived experience within a civic infrastructure designed to support it. Similarly, appreciation for ecological services like *Biodiversity* and *Soil Conservation* supports Nantes' environmental objectives. The Projet Alimentaire Territorial seeks to develop food production that advances ecological transition. This goal is reinforced by municipal garden rules, which ban pesticides ("l'utilisation de produits phytosanitaires étant interdite"), prompting gardeners to adopt natural, regenerative practices such as composting, companion planting, and fostering beneficial insects—activities that directly promote soil health and biodiversity. Their strong valuation stems from direct, practical engagement with practices mandated by city policies.

The most striking finding, however, is that Education has the highest level of consensus, boasting a median of 4 and an exceptionally low IQR of 0.75. This near-universal agreement on the educational value of the gardens is evidence of the success of a multi-level effort to frame these spaces as pedagogical tools. The city's official regulations explicitly state that a key objective of the gardens is to "DÉVELOPPER une pédagogie" (develop a pedagogy), particularly for children. This official mandate is brought to life by local associations like ECOS, which actively mobilizes the urban garden as a "lieu d'échange, de créativité et d'apprentissage" (place of exchange, creativity, and learning). ECOS offers a rich program of public courses and workshops on natural gardening techniques, wild plant identification, and seasonal cooking, effectively transforming the gardens into living classrooms. The gardeners' perception that education is a "Highly important" service is therefore not just a belief, but a lived reality, validated by the rich educational opportunities provided by the city's vibrant civic ecosystem.

For EDS, a notable example is the case of "*Green Waste*." The data for this item is anomalous: it has a low median concern of 1 ("Not concerned") but the highest possible IQR of 3, indicating extreme polarization. Furthermore, in the relational analysis, it is conceptually isolated, showing weak correlations with other disservice concerns. This pattern could be a direct reflection of successful municipal policy. The City of Nantes' regulations mandate composting and provide infrastructure for the collection of green waste, with the city itself responsible for

emptying the collection bins. This top-down intervention has effectively solved the systemic problem of green waste disposal, "decoupling" it from the gardeners' network of anxieties about pollution or resource management. The residual polarization likely reflects individual-level disagreements over compliance or the tidiness of compost areas, rather than a systemic concern. This provides a powerful illustration of how effective governance can reshape gardener perception by designing a problem out of the system (Fox-Kämper et al., 2018).

On the other hand, despite municipal regulations that mandate the installation of rainwater collection tanks and provide access to tap water for irrigation, these top-down measures have evidently failed to resolve a strong concern surrounding water resources. Our data confirms this tension unequivocally: "*Water Consumption*" is the only ecosystem disservice to register a median of 4 ("High concern") with a low IQR of 1.00, indicating it is the most pressing and uniformly shared worry within the community. This concern manifests in practice, particularly during the increasingly arid final months of summer, compelling many gardeners to rely on tap water to ensure crop survival. This behaviour creates a direct feedback loop, contributing to the disservice of "*Maintenance Cost*" as higher collective consumption translates into increased fees. As global climate change intensifies, regional projections for Pays de la Loire, where Nantes is situated, forecast a gradual yet consistent rise in average temperatures (Raisson-Victor et al., 2022). This escalating climatic pressure will inevitably place a higher

demand on water resources, intensifying the strain on the food-water linkage. This dynamic is likely to amplify the conflict between the desire for productive harvests and the necessity of sustainable water use, posing a significant long-term challenge for both gardeners and municipal governance.

4.4.3 Navigating the synergies and trade-offs in the human-nature relationship

The relational analysis reveals the complex negotiations gardeners make between competing values. A particularly insightful finding is the moderate positive correlation between the value placed on *Biodiversity* and the concern for *Pest/Insect Bites* ($r \approx 0.45$). This counterintuitive result refutes a simple trade-off between ecology and comfort. Instead, this suggests that the gardeners who are most invested in the ecological health of their plots are also the most attentive to potential threats. This "informed concern" can be understood in several ways. First, a gardener actively cultivating a biodiverse plot is exposed to a greater variety of insect life, leading to a more realistic awareness of both beneficial organisms and potential pests. Second, this correlation may identify a highly engaged "steward" for whom valuing biodiversity and managing pests are two sides of the same coin of attentive management. Their concern is not a generalized fear, but a focused, managerial awareness of specific threats that could disrupt the ecosystem they are carefully cultivating.

This synergy of vigilance contrasts sharply with a more fundamental trade-off shown in the data. The analysis shows a significant negative correlation between the value of *Food Provision* and concern for *Water Pollution* ($r = -0.42$), a tension between utilitarian goals and abstract environmental concerns. It suggests that gardeners with a strong "productionist" mindset, who are highly focused on the

pragmatic goal of the harvest, tend to have a lower level of concern for broader, less immediate environmental issues. Their focus on yield may overshadow anxieties about the potential downstream impacts of their practices.

These dynamics are also inflected by gender. The finding that women report significantly higher concern for *Pest/Insect Bites* aligns with a large body of research on risk perception, which consistently finds that women tend to report higher perceptions of risk related to health and personal safety. This difference is widely attributed to social factors rather than innate traits, suggesting that women may be socialized to be more alert about potential physical threats in their environment. In the context of the garden, this heightened awareness of pests suggests that gender can influence the intensity and focus of that attentiveness, placing a greater emphasis on the personal safety aspect of garden management.

4.4.4 Limitations of this study

One of the primary statistical weaknesses of this study is the small sample size. The sample-to-variable ratio of approximately 2.5:1 (54 participants for 21 perception items) is below the conservative ideal of 10:1 (Watkins, 2018). This issue affected the EFA process, where the correlation matrix required statistical smoothing and was not positive definite. However, the data used in this study are part of a larger questionnaire that requires approximately 2 hours to complete; thus, the author did not have the time to conduct more interviews. We follow

Barrett and Kline's (1981) affirmation that a minimum of 50 observations could produce a clear, recognizable factor pattern. Future studies should aim for a larger sample size, which would produce a stable correlation matrix and support more complex modelling.

Our sample is heavily skewed towards older adults (85% > 55 years old), in specific communities that are already motivated to participate in community gardening, which could affect the generalizability of the results. Future research could employ a stratified sampling strategy to capture a broader range of age groups, socio-economic backgrounds, and experience levels.

4.5 CONCLUSION

This comprehensive analysis has systematically deconstructed the complex worldview of 54 urban gardeners in Nantes Nord. Through a multi-phase statistical approach, we have moved from a broad description of the community to a synthesized and nuanced model of their underlying belief structures. The research demonstrates that the gardeners form a mature, highly committed, and behaviourally homogeneous community, whose perceptions are shaped more by their shared social identity and the overarching institutional context than by the specific location of their garden plot.

The entire system of 21 distinct perceptions of ES and EDS can be robustly summarized by a four-factor model. This model reveals that the gardeners' psychological reality is organized around four fundamental and largely independent constructs: (1) a concern for Nuisance, (2) a value for Ecological Stewardship, (3) a tension about Pollution, and (4) an appreciation for Well-being.

Ultimately, a granular understanding of the synergies and trade-offs between ecosystem services and disservices elevates urban green planning from a simple act of provision to a sophisticated practice of socio-ecological negotiation. By identifying synergies, such as the link between a concern for pollution and an appreciation for the ecosystem's regulatory functions, planners can design interventions that resonate with the community's existing mental models. Conversely, making trade-offs explicit is essential for avoiding policy failure, most notably the tension between fostering biodiversity and ensuring personal comfort. It forces a move away from one-size-fits-all greening strategies towards nuanced, co-designed solutions that proactively balance competing values, such as pairing habitat creation with improved pathway design or targeted ecological education. Furthermore, this analysis reveals that not all disservices are inherent. The case of '*Green Waste*,' a concern effectively neutralized by Nantes' municipal infrastructure, demonstrates that some of the most intractable conflicts can be 'designed out' of the system through targeted investment in resource management. This allows planners to distinguish between conflicts that require

social adaptation and those that require better infrastructure. In this way, the ES/EDS framework becomes an essential diagnostic tool, enabling planners to manage not just green assets but the complex web of human-nature relationships that determine their ultimate success and sustainability

5

THE FOOD-WATER-SOIL SUSTAINABILITY INDEX: AN INTEGRATED FRAMEWORK FOR ASSESSMENT AND GOVERNANCE

5.1 INTRODUCTION

This dissertation has, in its preceding chapters, established the urgent need for effective, context-specific assessment tools to measure the performance and long-term viability of Urban Food Gardens (UFGs) as a key form of Nature-based Solution (NbS).¹ As cities worldwide increasingly adopt UFGs to tackle a myriad of sustainability challenges, a significant gap persists between their widespread promotion and their evidence-based implementation. The preceding analysis has charted this gap, revealing a landscape of fragmented assessment methods and a dominant Water-Energy-Food (WEF) nexus paradigm that consistently fails to adequately theorize the central, foundational role of soil in these unique urban ecosystems.¹ In these specific contexts, energy is often a minimal input, while soil functions not as a passive substrate but as the foundational living infrastructure that mediates the system's core functions.¹

This systemic under-theorization of soil's dynamic role represents a critical gap in nexus research that this dissertation seeks to address.¹ In Chapter 3, I proposed and developed the Food-Water-Soil (FWS) nexus as a more fitting conceptual framework, one that repositions soil as the central mediating component—a capital asset whose health governs the interactions between water availability and food production.¹ This chapter directly addresses the critical deficiency in assessment by presenting, justifying, and operationalizing a novel framework that

translates this theory into practice: the Food-Water-Soil (FWS) Sustainability Index.¹

This framework is specifically designed to be soil-centric, context-aware, and methodologically robust, offering a more relevant and accurate paradigm for evaluating the sustainability of soil-based urban ecosystems. It represents the culmination of the theoretical and empirical work presented earlier, translating the conceptual FWS nexus into a practical tool for strategic management and governance. The development of this index is not an arbitrary exercise; it is the synthesis phase of this dissertation's scientific method. It demonstrates how the theoretical work of Chapter 3 and the empirical investigation of stakeholder perceptions in Nantes from Chapter 4 converge to produce a tangible, applied output.

The chapter unfolds in a structured progression. It begins by deconstructing the conceptual architecture of the FWS framework, tracing its methodological lineage to the established MESMIS methodology and critically positioning it as a necessary evolution from the ill-fitting WEF nexus paradigm. The analysis then delves into the biophysical and social underpinnings of the urban FWS nexus, justifying the selection of specific indicators through the dual lenses of core ecological processes and the Ecosystem Services (ES) and Disservices (EDS) they generate. The central challenge of indicator weighting is then addressed through an exhaustive analysis that integrates expert judgment with the empirically-derived values of the Nantes

gardener community. Finally, the chapter concludes by linking the assessment framework to an adaptive governance cycle, discussing its profound implications for fostering evidence-based urban greening policies and promoting a more resilient approach to urban management.

5.2 CONCEPTUAL ARCHITECTURE FOR FWS

The development of a robust assessment framework requires a solid methodological foundation. The FWS Sustainability Index is not a creation *ex nihilo*; rather, it is a deliberate and critical adaptation of an established sustainability assessment methodology, re-engineered to address the specific ecological and social realities of UFGs. This section details this conceptual architecture, first by establishing its heritage in the MESMIS framework, then by critically interrogating the dominant WEF nexus paradigm to demonstrate its inadequacy, and finally by formally presenting the MESMIS-FWS framework as a necessary and superior alternative for this context

5.2.1 Methodological Heritage from MESMIS

The FWS framework's methodological lineage is rooted in the Indicator-based Framework for Evaluation of Natural Resource Management Systems (MESMIS).¹ Developed in the mid-1990s for the context of peasant-managed Natural Resource Management Systems, MESMIS is more than a simple tool; it represents a comprehensive philosophical approach to sustainability assessment that is fundamentally systemic, participatory, and flexible.¹ Its selection as the backbone for our framework was a deliberate choice, reflecting a deep alignment between its core philosophy and the nature of the system under study.

The core strength of MESMIS, and the primary reason for its adaptation here, lies in its use of systemic attributes to structure the evaluation. Instead of relying on a static checklist of indicators, MESMIS proposes that any sustainable system can be evaluated based on a set of seven fundamental dynamic properties: Productivity, Stability, Reliability, Resilience, Adaptability, Equity, and Self-reliance.¹ These attributes provide a robust, scale- and discipline-independent foundation for evaluation, moving beyond a simple accounting of inputs and outputs to assess the deeper, functional characteristics of a system's long-term viability.

Operationally, the MESMIS methodology is structured as an iterative, six-step evaluation cycle designed for continuous learning and adaptive management:

1. **System Definition:** Characterizing the system to be evaluated, its components, and its socio-environmental context.
2. **Identification of Critical Points:** A diagnostic step where stakeholders collaboratively determine the key factors or processes that enhance or constrain the system's sustainability attributes.
3. **Selection of Strategic Indicators:** Deriving and selecting case-specific indicators that are linked to the critical points and the overarching sustainability attributes.
4. **Measurement and Monitoring:** Assessing the selected indicators using a variety of quantitative and qualitative methods. Integration and
5. **Presentation of Results:** Synthesizing the information from diverse indicators, often using graphical tools like "amoeba" diagrams to compare the strengths and weaknesses of different systems.
6. **Conclusions and Recommendations:** Formulating recommendations for improving the system, which then initiates a new evaluation cycle.

This cyclical, "spiral" process contrasts sharply with static evaluation methods and aligns perfectly with the continuous learning loop required for the adaptive management of complex social-ecological systems like UFGs. The inherent participatory ethos of MESMIS is a direct philosophical match for the nature of

UFGs. The methodology presupposes the importance of user perception, a premise that the empirical data from the Nantes case study in Chapter 4, which revealed the profound influence of gardeners' values and social dynamics, later confirms.¹ A purely top-down, expert-driven assessment framework would represent a methodological mismatch for the system being studied; the choice of MESMIS, therefore, reflects a pre-emptive alignment of methodological philosophy with the social reality of UFGs.

5.2.2 WEF Nexus to the Soil-Centric FWS Nexus

The WEF nexus concept has provided a powerful and influential paradigm for understanding large-scale resource security challenges.¹ Originating from high-level policy discussions, its primary utility has been in addressing cross-sectoral interdependencies at national and transboundary levels.¹ However, as with any powerful concept, its application across different scales and contexts requires careful consideration. When applied to the hyper-local, socially embedded ecosystem of an urban garden, the WEF framework reveals critical limitations that necessitate a context-specific adaptation.¹

A primary consideration is the issue of scale. The WEF nexus was largely conceived to analyse and govern massive, formalized resource flows through national grids and river basins.¹ This macro-scale, flow-based logic is not a natural fit for the small-scale, often informal resource dynamics of a community garden, where

resource flows are mediated by social relationships as much as by infrastructure.¹

Perhaps the most significant reason for adaptation, however, is the treatment of ecosystems. In many WEF frameworks, the environment and soil are positioned as external factors or as an additional, sometimes secondary, component (see chapter 2). This approach, while valid in some contexts, does not fully capture the reality of soil-based systems like UFGs, where the ecosystem is not an add-on but the foundational living infrastructure upon which the entire system depends.¹

The FWS nexus is therefore proposed not as a rejection of the WEF paradigm, but as a logical and necessary re-grounding of nexus thinking for the specific context of soil-based, low-energy urban ecosystems. The conceptual shift is straightforward yet significant: it replaces the '**Energy**' component with '**Soil**'.¹ The core innovation of the FWS nexus is its reframing of soil not as a simple resource *flow* but as a *capital asset*.¹ The health of this asset—its physical, chemical, and biological properties—is a qualitative state that determines the system's capacity to perform its functions.¹ This is why traditional nexus models based on material flow analysis are often insufficient; the critical resource to manage is not soil mass, but soil health, a complex quality that cannot be captured by a single metric.¹ This reframing necessitates a move towards a performance-based assessment that evaluates the quality of the system's functioning, a task for which the MESMIS framework is uniquely equipped.

5.2.3 MESMIS-FWS Framework Adaptation

Our adaptation of MESMIS for the FWS nexus is a principled application of its core philosophy, building upon its demonstrated methodological flexibility.¹ Notably, the "Nexus Pampa" project successfully adapted the methodology for assessing livestock production systems by replacing the traditional MESMIS tripod of social, economic, and agro-ecological dimensions with the axes of the WEF nexus.¹ This project provides a direct and powerful precedent, validating the methodological approach of integrating a nexus-based dimensional framework within the overarching structure of the MESMIS systemic attributes.¹ This act of substitution reveals that the MESMIS framework is process-oriented, not content-specific, providing a promising pathway for operationalizing nexus thinking.¹

Building upon this precedent, the MESMIS-FWS framework involves a fundamental adaptation: the substitution of MESMIS's broad evaluative dimensions with the specific, interconnected axes of our proposed nexus—Food, Water, and Soil.¹ This creates a hybrid framework that is both methodologically robust and theoretically coherent with the dissertation's central thesis. The logical justification for this shift is rooted in the specific context of UFGs. In these systems, external energy inputs are often minimal, whereas the biophysical interactions between soil quality, water cycles, and plant growth are the dominant drivers of the system's productivity, resilience, and overall sustainability.¹ The FWS framework is thus tailored to this reality, providing an analytical lens that aligns

with the system's actual ecological and physiological dynamics. Table 5.1 summarizes this conceptual evolution, illustrating the distinct logic and application of each framework.

Table 5.1. A comparative analysis of sustainability assessment framework

Feature	MESMIS Framework	NEXUS-MESMIS (WEF)	Food-Water-Soil (FWS) Nexus
<i>Core Dimensions</i>	Social, Economic, Environmental	Water, Energy, Food	Food, Water, Soil
<i>Central Logic</i>	Balancing three pillars of sustainability	Managing interdependencies and trade-offs between sectors	Mediating food production and water cycles through soil health
<i>Typical Application</i>	Rural/Peasant Natural Resource Management Systems (NRMS)	Regional/National resource security (e.g., river basins)	Soil-based, low-energy agroecosystems (e.g., urban gardens)
<i>Treatment of Soil</i>	A component of the 'Environmental' dimension	Often externalized as 'Land' or a passive substrate	A central, dynamic, and mediating capital asset
<i>Primary Output</i>	A holistic sustainability assessment (radial chart)	Optimization of resource allocation across sectors	Assessment of socio-ecological resilience and performance

5.3 INDICATOR SELECTION

Following the establishment of the overarching MESMIS-FWS framework, the next critical step in its operationalization is the selection of strategic indicators. This

process moves from the general system definition to the specific, measurable variables that will be used to assess sustainability. The selection of indicators is a crucial step in the MESMIS cycle, as it determines what is measured and, therefore, what is managed.

5.3.1 Justifying the Indicators through ES/EDS lens

A robust indicator must measure something that is both biophysically meaningful within the garden ecosystem and socially relevant to the stakeholders who manage and benefit from it. To structure this selection process, we re-introduce the dual analytical lenses developed in Chapter 3: the core biophysical interactions of the soil-plant-water system and the emergent Ecosystem Services (ES) and Ecosystem Disservices (EDS) that these interactions generate. The performance of the FWS nexus is most accurately assessed through these qualitative and functional lenses, which evaluate the system's net contribution to human well-being by considering both the benefits it generates and the potential harms it produces. Therefore, the chosen indicators for the FWS Sustainability Index are designed to representing the optimal the state or rate of a biophysical process that is known to underpin a key ES or EDS. This approach ensures that the index is not a disconnected list of abstract metrics but an integrated diagnostic system grounded in both ecological reality and human values.

The FWS Sustainability Index is composed of three primary Axes—Food, Water, and Soil—which serve as the main pillars of sustainability for a soil-based UFG. Each axis is further divided into "scopes" that capture different facets of sustainability, and each scope contains a set of representative indicators

5.3.1.1 Soil Axis Indicator

Positioned as the foundational axis, the Soil Axis evaluates the condition of the soil as a capital asset, the pressures leading to its degradation, and the stewardship practices employed to maintain its health. The indicators selected for this axis directly measure the soil properties that govern its capacity to function.

- **Condition** Scope: Indicators such as *Bulk density*, *Top-soil depth*, and *Total organic carbon* are not abstract metrics. They are direct measures of the soil's physical and chemical state, which were identified in Chapter 3 as being paramount to its functions. For instance, low bulk density and high organic carbon are directly linked to improved soil structure, which enhances the soil's ability to provide the regulating service of stormwater infiltration (an ES) and resist compaction (avoiding an EDS). Indicators for *Trace elements* and *Organic pollutants* directly assess the health-related disservice of soil contamination, a key concern in urban environments.

- **Degradation** Scope: Indicators such as *Perception of the erosive process* and *Fertilizer usage* measure the negative pressures on the soil system. The overuse of inorganic fertilizers, for example, is directly linked to the EDS of nutrient runoff and water pollution.
- **Stewardship** Scope: Indicators such as *Disturbance minimization* (e.g., no-tillage) and *Preservation practices* (e.g., crop rotation, vegetation cover) quantify the positive human interventions that build and maintain soil capital. These practices were identified in Chapter 3 as essential for breaking the negative feedback loops of urban soil degradation.

This structure, separating the assessment into **Condition**, **Degradation**, and **Stewardship**, mirrors established ecological pressure-state-response models. The "Condition" indicators measure the state of the ecosystem asset. The "Degradation" indicators measure the negative pressures acting upon it, while the "Stewardship" indicators measure the positive management responses. This transforms the index from a simple report card into a sophisticated diagnostic tool. A low score in "Condition" is the symptom; the scores in "Degradation" and "Stewardship" provide the diagnosis, suggesting the appropriate management intervention.

5.3.1.2 Water Axis Indicators

F The Water Axis encompasses the management of water resources for both human use and crop production, focusing on efficiency, conservation, and quality.

- **Production** Scope: The indicator for *Susceptibility to drought*¹ directly addresses the core Food-Water interconnection, where water availability is a primary limiting factor for plant growth.
- **Conservation** Scope: Indicators such as *Irrigation system* and *Soil moisture protection methods* (e.g., mulching)¹ are justified by their direct impact on water use efficiency. As discussed in Chapter 3, practices like drip irrigation and mulching are highly effective at minimizing non-productive water loss to evaporation.¹ The inclusion of these indicators is powerfully validated by the findings from Chapter 4, where *Water Consumption* was identified as the single most pressing and universally shared concern among the Nantes gardeners.
- **Human Consumption** Scope: Inherited from the original MESMIS framework, while important for gardener health, the volume and systemic impact of water for direct human consumption at the garden site are minor compared to the volumes used for production and managed for conservation. This scope did not emerge as a significant concern or value in the survey data, supporting its lower relative weight compared to production and conservation.

5.3.1.3 Food Ais Indicator

The Food Axis represents the productive outputs of the garden, its role in community well-being, and the institutional and technological environments that govern it.

- **Consumption** Scope: Indicators include *Self consumption* and *Close network sharing* directly measure the primary provisioning service of the UFG—providing fresh food—which was identified in Chapter 4 as a highly valued ES.
- **Organizational and Institutional Environment** Scope: The inclusion of an indicator for *Social and associative participation*³ is a critical feature that reflects the understanding of UFGs as deeply embedded socio-ecological systems. This indicator is justified by its direct link to the crucial cultural service of *Social Cohesion*, which Chapter 4 identified as not only a highly valued ES but also a core component of the statistically derived "Well-being" bundle, proving that gardeners perceive social interaction as integral to the garden's benefits.

5.3.2 Integrating expert judgment with stakeholder value

A central challenge in constructing any composite sustainability index is the assignment of weights to its constituent indicators.¹ This process is inherently value-laden, and a transparent, robust justification for the chosen weighting scheme is essential for the credibility and utility of the final index. For the FWS Sustainability Index, we have developed a multi-layered justification that integrates literature-based expert judgment with the empirically-derived values and concerns of the stakeholder community from our Nantes case study. This approach ensures that the index is not only theoretically sound but also socially relevant and responsive to the lived realities of its intended users.

The proposed FWS Sustainability Index is structured around the three core axes of Food, Water, and Soil, each disaggregated into several "scopes" that capture different facets of sustainability. The initial weighting assigned to these scopes is presented in Table 5.2, followed by a detailed justification that explicitly links these weights to the findings of the preceding chapters.

This detailed justification demonstrates a powerful convergence of theory and practice. The weighting scheme is not merely an expert-driven proposition; it is an empirically validated structure that reflects the priorities, anxieties, and holistic worldview of the urban gardening community in Nantes. For instance, the decision to assign the highest weight within the Water Axis to "Production" is directly justified by the finding that "Water Consumption" was the most significant concern for gardeners. Similarly, the balanced weighting of the "Organizational"

and "Productive/Technological" scopes within the Food Axis is not an arbitrary compromise but a direct reflection of the EFA results, which showed that gardeners statistically bundle social benefits like "Social Cohesion" with pragmatic outcomes like "Food Provision" into a single, integrated concept of "Well-being". This process combine the weighting scheme of "expert-proposed", that many were originated from the original framework, with "stakeholder-validated," a critical step that enhances its legitimacy and aligns with the participatory ethos of the MESMIS philosophy.

Table 5.2. Justification of scope weights within the FWS index

Axis	Scope	Weight	Literature-Based Justification	Empirical Justification from Nantes Case Study
Soil	Condition	40%	Represents the foundational capital stock. Soil's physical, chemical, and biological properties determine its capacity to function and provide all other services. It is the primary determinant of potential productivity and resilience.	The high value placed on the "Ecological Stewardship" bundle, which statistically grouped Soil Conservation and Biodiversity , demonstrates that gardeners perceive the intrinsic health of the ecosystem as a core, integrated value.
	Degradation	30%	Represents the negative pressures (e.g., erosion, contamination) that directly reduce soil capital. Managing degradation is a primary goal of sustainable land use.	The emergence of a distinct "Pollution" factor, defined by high concern for Soil Pollution and Water Pollution , validates the need to explicitly assess and weight these degradation pressures.
	Stewardship	30%	Represents the positive human interventions (e.g., conservation practices, amendments) that build and maintain soil capital. It reflects the active management required for sustainability.	The near-universal agreement on the importance of Education (Median=4, IQR=0.75) reflects a community deeply engaged in learning and applying stewardship practices.
Water	Production	50%	Water is the most critical input for biomass production (food). Its availability and efficient use directly govern yield, which is a primary objective of UFGs.	Water Consumption was the <i>only</i> EDS to receive a median concern level of 4 ("High concern") with a low IQR of 1.00, making it the single most pressing and uniformly shared

			anxiety among gardeners. This high weight directly reflects this primary stakeholder concern.
	Conservation	40%	Reflects the dual importance of on-site water use efficiency (reducing demand on municipal supplies) and the key ecosystem service of stormwater management for urban resilience.
			Women placed a significantly higher value on Water Regulation ($r=0.51$, $p<.001$), indicating that conservation is a key value for a significant portion of the gardening community.
	Human Consumption	10%	While important for gardener health, the volume and systemic impact of water for direct human consumption at the garden site are minor compared to the volumes used for production and managed for conservation.
			This scope did not emerge as a significant concern or value in the survey data, supporting its lower relative weight compared to production and conservation.
Food	Consumption	40%	Addresses the primary provisioning and cultural services of UFGs: providing nutritious food, enhancing food security, and promoting health and well-being.
			Food Provision was a top-rated ES with a median of 4 ("Highly important"), confirming its central importance to the gardeners' experience.
	Organizational	30%	Acknowledges that UFGs are socio-ecological systems. Social cohesion, governance, knowledge sharing, and community participation are critical for long-term viability and resilience.
			The "Well-being" bundle from the EFA statistically grouped Social Cohesion with Food Provision , Mental Health , and Physical Health , proving that gardeners perceive the social infrastructure as an inseparable component of the garden's overall benefit.

Productive/Tech.	30%	Encompasses the technical aspects of cultivation (e.g., crop choice, pest management, diversification) that determine the efficiency and stability of food production.	The equal weighting with the "Organizational" scope reflects the holistic nature of the "Well-being" bundle, where pragmatic production is valued alongside social and health benefits.
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5.4 CONCLUSION

This chapter has detailed the development, justification, and operationalization of the FWS Sustainability Index, a novel framework designed to address a critical gap in the assessment of urban food gardens. By moving beyond the limitations of the conventional WEF nexus, this index provides a more relevant and robust tool for understanding and managing the complex socio-ecological dynamics of these vital urban green spaces.

The contribution of the FWS Sustainability Index is threefold, reflecting the integrated logic of this dissertation. First, it is **theoretically grounded** in the soil-centric FWS nexus, a paradigm that correctly identifies soil not as a passive input but as the central, dynamic capital asset governing the system's performance. This provides a coherent conceptual basis for assessment that aligns with the biophysical realities of UFGs.

Second, the index is **methodologically sound**, building upon the proven, process-oriented MESMIS framework. This heritage ensures that the assessment is systemic, cyclical, and philosophically aligned with the principles of adaptive management and participatory evaluation that are essential for complex systems.

Third, and most critically, the index is **empirically informed**. Its structure, indicators, and weighting scheme are not merely abstract constructs but are

validated by the expressed values, concerns, and holistic worldview of the stakeholder community in Nantes. The direct linkage between the gardeners' perceived priorities—such as their profound concern for water consumption and their integrated view of social and productive well-being—and the final weights assigned within the index demonstrates a powerful synthesis of top-down expertise and bottom-up social reality.

Ultimately, the FWS Sustainability Index is more than an assessment tool; it is a governance instrument. By providing a data-driven engine for an adaptive management cycle, it offers a practical pathway for moving urban greening policy beyond simple aesthetics or provision. It enables a more integrated, evidence-based approach that fosters a continuous dialogue between managers, gardeners, and the ecosystem itself, thereby enhancing the long-term socio-ecological resilience of our cities.

6 CONCLUSION AND RECOMMENDATION

6.1 INTRODUCTION

This chapter serves as the capstone of my doctoral research, weaving together the theoretical, methodological, and empirical threads of the preceding chapters into a cohesive whole. It revisits the central problem that motivated this thesis: Urban Food Gardens (UFGs) are increasingly recognized as vital Nature-Based Solutions (NbS) for enhancing urban resilience, yet the dominant assessment paradigms, particularly the Water-Energy-Food (WEF) nexus, are fundamentally ill-suited for these low-energy, soil-based systems. As my research has demonstrated, these frameworks often marginalize the foundational role of soil, treating it as a static substrate rather than the dynamic capital asset it truly is.¹ This oversight represents a significant blind spot in sustainability science, hindering our ability to effectively manage and govern these crucial pieces of urban green infrastructure.

The primary aim of this dissertation was to address this critical gap by formulating, developing, and validating a novel Food-Water-Soil (FWS) nexus framework.¹ This

dissertation argues that the FWS nexus and its operational tool, the FWS Sustainability Index, represent a significant contribution to both sustainability science and urban governance. By re-grounding nexus thinking in the material reality of the urban agroecosystem, this work offers a more accurate, relevant, and socially-informed approach to assessment. It provides a means to move beyond anecdotal accounts of garden benefits towards a quantitative, diagnostic understanding of their performance, thereby enabling more strategic management and evidence-based policy.

This introduction provides a clear roadmap for the reader. The chapter will progress through a series of structured reflections, beginning with a recapitulation of the research journey, tracing the logical arguments from problem identification in the literature to the creation of a practical, validated tool. This is followed by a critical discussion of the research's broader relevance, its inherent strengths, and, just as importantly, its limitations. I will then present a forward-looking outlook with concrete implications for future research and for policy and practice, with a specific focus on the city of Nantes. Finally, a concluding statement will crystallize the core message of this work: that the path to a resilient urban future depends on re-grounding our sustainability assessments in the living soil and in the lived experiences of the communities who steward it.

6.2 RESEARCH TRAJECTORY AND KEY FINDINGS

This section retraces that trajectory, demonstrating how the systematic identification of a research gap led to the development of a new theoretical framework, which was then empirically grounded and finally operationalized into a practical assessment tool.

The research began in Chapter 2 with a systematic review of the landscape of integrated assessment tools for UFGs. The analysis mapped the field into five distinct methodological clusters: Quantitative Sustainability Assessment, Spatial Planning & Optimization, Systems & Nexus Modeling, Socio-Ecological & Community Assessment, and Empirical & Field-Based Studies. While this mapping provided a useful typology, its most crucial finding was the exposure of a systemic weakness across all paradigms. The review revealed a critical and persistent failure of existing frameworks, including the popular WEF nexus, to adequately "re-ground" assessments in the dynamic, living dimension of soil. Across the 58 papers analyzed, soil was consistently treated as a passive substrate, a static variable in a spatial model, an abstracted input in a LCA, or was absent from nexus simulations altogether. This chapter's primary function was therefore diagnostic; it empirically established and precisely defined the theoretical and methodological gap that the remainder of the thesis would seek to fill.

In direct response to the gap identified in Chapter 2, Chapter 3 was dedicated to developing the theoretical architecture of the FWS nexus. This chapter argued for a fundamental paradigm shift, one that repositions soil from the periphery to the very center of the analytical framework. I conceptualized soil not as a simple input but as the central mediating component—a capital asset whose health and functional state govern the system's overall performance and resilience. To analyze the complex interactions within this nexus, I employed the dual lenses of ES and EDS. This approach allowed the analysis to move beyond a simple accounting of material flows to capture the qualitative, functional outcomes of the garden system as they relate to human well-being. The chapter culminated in a proposed management hierarchy, adapting the work of Vaz et al. (2017) to provide a structured, step-by-step process for the adaptive governance of FWS interactions, addressing challenges from soil contamination to water scarcity.

To ensure the FWS framework was not merely a theoretical abstraction, Chapter 4 was dedicated to grounding its core concepts in empirical, social reality through an in-depth case study of 54 urban gardeners in Nantes Nord. This mixed-methods study systematically deconstructed the gardeners' worldview, using a multi-phase statistical analysis to model their perceptions of the benefits and drawbacks of their gardens. The EFA revealed that the gardeners' complex belief system could be synthesized into four fundamental and statistically coherent bundles: Nuisance (personal discomforts), Ecological Stewardship (an appreciation

for the garden's ecosystem functions), Pollution (a specific anxiety about chemical contamination), and Well-being (a holistic integration of physical, mental, social, and provisioning benefits). Critically, the analysis found that these perceptions were shaped more profoundly by social identity—most notably gender—than by the physical location of a specific garden. This chapter also illuminated the dynamic interplay between top-down municipal governance and bottom-up grassroots action in co-producing the lived experience of the gardens. In doing so, it provided the essential social data needed to ensure the final assessment tool would be not only technically robust but also socially relevant and legitimate.

The final step in the research dissertation, presented in Chapter 5, was to synthesize the preceding theoretical and empirical work into a tangible, applied output: the FWS Sustainability Index. This chapter detailed the operationalization of the FWS nexus by critically adapting the established, participatory MESMIS methodology. The key innovation of this adaptation was the direct integration of the empirical findings from Chapter 4 into the very design of the tool. The final weights assigned to the index's indicators and scopes were explicitly justified by the values, concerns, and perceptual bundles expressed by the Nantes gardener community. For instance, the high weight assigned to water conservation directly reflected the community's primary anxiety about water consumption. This process ensured that the resulting index was not a top-down, expert-driven imposition but

a socially validated instrument, fulfilling the participatory ethos of the MESMIS philosophy and the dissertation's overall aim.

6.3 STRENGTHS OF THE RESEARCH

Upon reflection, I identify three primary strengths that underpin the contribution of this dissertation. First, the core strength is the theoretical innovation of the FWS nexus itself. By replacing the 'Energy' component of the dominant WEF nexus with 'Soil', the framework offers a more accurate and physiologically relevant paradigm for a vast and growing class of low-energy, soil-based urban ecosystems, including not only UFGs but also parks, bioswales, and other forms of green infrastructure. This is not a minor adaptation but a fundamental re-grounding of nexus thinking. It positions a living ecosystem, soil, as the central, dynamic capital asset that mediates resource flows, rather than treating it as an external factor or a passive substrate. This shift provides a more robust conceptual foundation for understanding and managing these systems.

Second, a unique strength lies in the thesis's methodological coherence and integration. The research was designed to follow a complete and logical scientific arc, progressing systematically from one stage to the next. It began by empirically identifying a gap in the existing literature (Chapter 2), proceeded to propose a novel theoretical solution to fill that gap (Chapter 3), then grounded that theory in rich, empirical social data from a specific community (Chapter 4), and finally, synthesized all preceding work to operationalize the framework into a practical, validated tool (Chapter 5). This integrated, multi-stage design creates a powerful, self-reinforcing argument where each chapter validates and informs the next. This

structure is more robust and comprehensive than a simple collection of disparate studies, demonstrating a complete cycle of scientific inquiry from problem to solution.

Third, and perhaps most significantly, the research is distinguished by its participatory and socially-grounded assessment approach. The decision to base the indicator weights in the FWS Sustainability Index (Chapter 5) directly on the empirically derived perceptual bundles from the Nantes gardeners (Chapter 4) is a direct answer to the well-documented limitations of top-down, purely technocratic assessment models that often fail to capture local values and priorities. This approach is a practical embodiment of the participatory ethos of the MESMIS framework and the principles of knowledge co-production, which call for the active engagement of stakeholders in the creation of knowledge and tools that affect them. This integration of stakeholder values enhances the index's legitimacy, its social relevance, and, ultimately, its potential for genuine uptake and use by the communities it is designed to serve.

6.4 LIMITATIONS

While this research makes several important contributions, it is also bounded by specific methodological choices and conceptual frameworks that create limitations and open avenues for critical reflection. From a methodological standpoint, the systematic review in Chapter 2, while robust in its PRISMA-guided

process, was constrained by its reliance on the Web of Science core collection and its focus on English-language articles. Furthermore, its deliberate exclusion of commercial urban agriculture models, while necessary for analytical clarity, means its findings on assessment tools may not fully capture the state of the art in that rapidly evolving sector.

Similarly, the empirical study in Chapter 4, which provides the crucial social data for the thesis, is based on a relatively small sample size of 54 gardeners. The demographic profile of this sample is also heavily skewed towards older adults, with 85% of respondents aged 55 or older.¹ As the chapter itself discusses, this group represents a "self-selected community of practice," which implies that the specific findings on gardener perceptions in Nantes Nord may not be generalizable to the broader urban population of Nantes or to gardening communities in different cultural contexts.

Beyond these practical limitations, there is a more profound conceptual limitation that I term the political ecology blind spot. My thesis, in its focus on the internal socio-ecological functioning of UFGs, largely frames them as positive catalysts for urban renewal and resilience. However, in doing so, it does not adequately engage with the large and critical body of literature in urban political ecology on the phenomenon of "green gentrification". This research argues that the creation of desirable green amenities, such as well-maintained and productive community gardens, can paradoxically increase neighbourhood attractiveness, drive up

property values and rents, and ultimately contribute to the displacement of the very long-standing, often low-income, residents they were intended to benefit. This creates a potential contradiction at the heart of my research. A UFG that scores highly on the FWS Sustainability Index, one that is productive, biodiverse, and fosters strong social cohesion, is precisely the kind of amenity that could accelerate gentrification pressures. Therefore, a key limitation of my framework is that by focusing on the internal sustainability of the garden system, it may overlook the external political-economic processes of displacement that can undermine the project's ultimate social justice outcomes. The FWS index can effectively measure the health of the garden, but it does not, in its current form, measure the stability and affordability of the community surrounding it. This is a crucial area for future research.

6.5 OUTLOOK: IMPLICATIONS FOR RESEARCH, POLICY, AND PRACTICE

6.5.1 Implications for Future Research

This dissertation lays the groundwork for several promising avenues of future research that can build upon and extend its contributions. First, the immediate and most logical next step is the longitudinal and comparative application of the FWS Sustainability Index. Applying the index in a longitudinal study across the

gardens of Nantes would allow for the tracking of sustainability performance over time, providing invaluable data on the long-term impacts of different management practices and policy interventions. Furthermore, conducting comparative studies that apply the index in different socio-political and climatic contexts, such as in Thu Dau Mot city in Vietnam, a site initially envisioned in my research plan, is essential for testing its transferability, adaptability, and cultural relevance. Second, the conceptual power of the FWS nexus should be tested by expanding its application to other soil-based NbS. Urban parks, bioswales, green roofs, and constructed wetlands are all ecosystems where soil health is a critical, yet often overlooked, determinant of performance. Applying the FWS logic to these systems could help to develop a more unified theory for assessing the role of soil in urban ecosystem functioning, providing a common language for planners and ecologists. Third, and most importantly, future research must directly address the limitation of the political ecology blind spot by integrating a justice lens. This would involve augmenting the existing framework with indicators that explicitly measure social equity outcomes, such as housing affordability trends, land tenure security for gardeners, and the risk of residential displacement in the surrounding neighbourhood. Such a tool would create a more holistic assessment that evaluates not only the ecological sustainability of green spaces but also their contribution to a just and equitable city.

6.5.2 Implications for Policy and Practice

One of the core aims of this research was to produce actionable knowledge. The empirical findings from Chapter 4, in particular, provide a direct, evidence-based roadmap for municipal action in Nantes. The policy implications derived from this work are not generic; they are tailored to the specific, expressed needs and values of the city's gardening community.

The analysis in Chapter 4 revealed that "Water Consumption" is the single greatest and most uniformly shared concern among gardeners.¹ At the same time, "Education" and "Social Cohesion" were among the most highly valued ecosystem services, and the data showed clear gendered differences in the perception of risks and care-related services. These findings point to a set of targeted interventions:

- **Target Water Scarcity:** The Nantes Métropole should prioritize investments and policies that directly address the community's primary anxiety. This could include subsidizing the installation of larger rainwater harvesting systems in all community gardens and co-developing a series of practical, hands-on educational workshops with expert local groups like ECOS on topics such as drought-tolerant crop selection, mulching techniques, and efficient irrigation practices.
- **Amplify Social and Educational Roles:** The city should recognize and increase its support for the non-profit organizations (such as MAAD and ECOS) that are instrumental in turning gardens into vibrant hubs for social

cohesion and learning. This leverages the positive interplay between governance and grassroots action identified in Chapter 4 and strategically invests in the benefits most valued by the community. This support could take the form of increased funding, streamlined administrative processes, or formal integration into the city's educational outreach programs.

- **Adopt Gender-Responsive Management:** Garden associations and municipal planners should acknowledge that different members of the community experience the garden in different ways. The finding that women expressed significantly higher concern for pests and personal safety should inform practical management decisions.¹ This could translate into simple but meaningful measures like ensuring garden layouts are well-lit and pathways are clear to enhance feelings of security, or offering targeted workshops on organic pest management and the identification of beneficial insects to build confidence and address specific risk perceptions.

More broadly, for urban governance beyond Nantes, the FWS Sustainability Index offers a transformative tool for evidence-based decision-making.¹ As detailed in the Impact Statement, it allows city planners and policymakers to move from anecdotal success stories to a quantitative, diagnostic understanding of their green infrastructure. The index can be used to **benchmark performance** across a city-wide network of gardens, to **diagnose specific weaknesses** (such as systemic soil degradation or inefficient water use) and deploy targeted support where it is

most needed, and to **foster integrated policy** by providing a common language and dataset that helps break down the administrative silos that often exist between municipal departments of parks, water, and waste management.

6.6 CONCLUDING REMARKS

This dissertation began with a simple premise: that to understand the sustainability of urban food gardens, we must first be willing to get our hands dirty. The dominant, high-level nexus frameworks, with their technocratic focus on large-scale resource flows, have consistently overlooked the most fundamental component of these vital urban ecosystems: the living soil beneath our feet. This theoretical oversight could result in practical consequences, leaving urban planners and communities without the tools they need to properly assess and manage these complex socio-ecological systems.

In response, this research has delivered two primary, interwoven contributions. First, it has proposed a new conceptual model, the Food-Water-Soil nexus, that re-grounds sustainability assessment in the material reality and physiological dynamics of the urban agroecosystem. By placing soil at its rightful center, the FWS nexus offers a more accurate and relevant paradigm for a whole class of nature-based solutions. Second, it has translated this theory into a validated, practical tool—the FWS Sustainability Index. This index is more than just a list of indicators; it is an instrument for adaptive management and evidence-based governance that is uniquely and powerfully informed by the voices, values, and concerns of the gardeners themselves.

Ultimately, this work argues for a more humble, grounded, and participatory science of urban sustainability. It demonstrates that by listening carefully to communities and by paying close attention to the foundational ecological processes that support them, we can develop more effective, resilient, and equitable strategies for greening our cities. The path to a sustainable urban future is not built solely on grand plans and abstract models; it is also cultivated, plot by plot, through the nurturing of the soil and, in doing so, the cultivation of the human and ecological capital upon which true urban resilience depends.

7

REFERENCES

- Al-Hamaiedeh, H., Bino, M., 2010. Effect of treated grey water reuse in irrigation on soil and plants. *Desalination* 256, 115–119. <https://doi.org/10.1016/j.desal.2010.02.004>
- Al-Ismaili, A.M., Ahmed, M., Al-Busaidi, A., Al-Adawi, S., Tandlich, R., Al-Amri, M., 2017. Extended use of grey water for irrigating home gardens in an arid environment. *Environ. Sci. Pollut. Res.* 24, 13650–13658. <https://doi.org/10.1007/s11356-017-8963-z>
- Alessa, L., Earnhart, C., 2000. Effects of Soil Compaction on Root and Root Hair Morphology: Implications for Campsite Rehabilitation. *Wilderness Sci. a Time Chang. Conf.* 5.
- Allouche, J., 2024. Nexus Framing of Sustainability Issues: Feasibility, Synergies, and Trade-Offs in Terms of Water-Energy-Food. *Annu. Rev. Environ. Resour.* 49, 501–518. <https://doi.org/10.1146/annurev-environ-112321-112445>
- Artmann, M., Sartison, K., 2018. The role of urban agriculture as a nature-based solution: A review for developing a systemic assessment framework. *Sustain.* 10. <https://doi.org/10.3390/su10061937>
- Ashrafzadeh, S., Lehto, N.J., Oddy, G., McLaren, R.G., Kang, L., Dickinson, N.M., Welsch, J., Robinson, B.H., 2018. Heavy metals in suburban gardens and the implications of land-use change following a major earthquake. *Appl. Geochemistry* 88, 10–16. <https://doi.org/10.1016/j.apgeochem.2017.04.009>
- Astier, M., García-Barrios, L., Galván-Miyoshi, Y., González-Esquivel, C.E., Masera, O.R., 2012. Assessing the Sustainability of Small Farmer Natural Resource Management Systems. A Critical Analysis of the MESMIS Program (1995-2010). *Ecol. Soc.* 17, art25. <https://doi.org/10.5751/ES-04910-170325>
- Babaei, L., Niksokhan, M.H., Torabian, A., Negahban-Azar, M., 2024. Sustainability indicators for evaluation of the Water-Energy-Food Nexus in urban agriculture. *Int. J. Sustain. Dev. WORLD Ecol.* 31, 466–480. <https://doi.org/10.1080/13504509.2023.2296505>
- Barrett, P., Kline, P., 1981. The Observation to Variable Ratio in Factor Analysis. *Personal. Study Gr. Behav.* 1, 23–33.
- Barthel, S., Folke, C., Colding, J., 2010. Social–ecological memory in urban gardens—Retaining the capacity for management of ecosystem services. *Glob. Environ. Chang.* 20, 255–265. <https://doi.org/10.1016/J.GLOENVCHA.2010.01.001>
- Bechet, B., Joimel, S., Jean-Soro, L., Hursthouse, A., Agboola, A., Leitão, T.E., Costa, H., do Rosário Cameira, M., Le Guern, C., Schwartz, C., Lebeau, T., 2018. Spatial variability of trace elements in allotment gardens

- of four European cities: assessments at city, garden, and plot scale. *J. Soils Sediments* 18, 391–406.
<https://doi.org/10.1007/s11368-016-1515-1>
- Bernaards, C.A., Jennrich, R.I., 2005. Gradient Projection Algorithms and Software for Arbitrary Rotation Criteria in Factor Analysis. *Educ. Psychol. Meas.* 65, 676–696.
<https://doi.org/10.1177/0013164404272507>
- Bessette, G., 2004. Involving the community : a guide to participatory development communication. Southbound ; International Development Research Centre.
- Bodéan, P., Béchet, B., 2023. Des « solutions fondées sur la nature » dans un quartier de grands ensembles ? Métropolitiques. <https://doi.org/10.56698/metropolitiques.1890>
- Brauman, K.A., Garibaldi, L.A., Polasky, S., Aumeeruddy-Thomas, Y., Brancalion, P.H.S., DeClerck, F., Jacob, U., Mastrangelo, M.E., Nkongolo, N. V., Palang, H., Pérez-Méndez, N., Shannon, L.J., Shrestha, U.B., Strombom, E., Verma, M., 2020. Global trends in nature’s contributions to people. *Proc. Natl. Acad. Sci.* 117, 32799–32805. <https://doi.org/10.1073/pnas.2010473117>
- Bronick, C.J., Lal, R., 2005. Soil structure and management: a review. *Geoderma* 124, 3–22.
<https://doi.org/10.1016/j.geoderma.2004.03.005>
- Bulkeley, H., Marvin, S., Palgan, Y.V., McCormick, K., Breitfuss-Loidl, M., Mai, L., von Wirth, T., Frantzeskaki, N., 2019. Urban living laboratories: Conducting the experimental city? *Eur. Urban Reg. Stud.* 26, 317–335.
<https://doi.org/10.1177/0969776418787222>
- Bush, J., Doyon, A., 2019. Building urban resilience with nature-based solutions: How can urban planning contribute? *Cities* 95, 102483. <https://doi.org/10.1016/J.CITIES.2019.102483>
- Cabral, I., Costa, S., Weiland, U., Bonn, A., 2017a. Urban Gardens as Multifunctional Nature-Based Solutions for Societal Goals in a Changing Climate. pp. 237–253. https://doi.org/10.1007/978-3-319-56091-5_14
- Cabral, I., Keim, J., Engelmann, R., Kraemer, R., Siebert, J., Bonn, A., 2017b. Ecosystem services of allotment and community gardens: A Leipzig, Germany case study. *Urban For. Urban Green.* 23, 44–53.
<https://doi.org/10.1016/j.ufug.2017.02.008>
- Camacho-Caballero, D., Langemeyer, J., Segura-Barrero, R., Ventura, S., Beltran, A.M., Villalba, G., 2024. Assessing Nature-based solutions in the face of urban vulnerabilities: A multi-criteria decision approach. *Sustain. Cities Soc.* 103. <https://doi.org/10.1016/j.scs.2024.105257>
- Camps-Calvet, M., Langemeyer, J., Calvet-Mir, L., Gómez-Baggethun, E., 2016. Ecosystem services provided by urban gardens in Barcelona, Spain: Insights for policy and planning. *Environ. Sci. Policy* 62, 14–23.
<https://doi.org/10.1016/j.envsci.2016.01.007>
- Caputo, S., Schoen, V., Blythe, C., 2023. Productivity and Efficiency of Community Gardens: Case Studies from the UK. *Land* 12, 238. <https://doi.org/10.3390/land12010238>
- Caputo, S., Schoen, V., Specht, K., Grard, B., Blythe, C., Cohen, N., Fox-Kämper, R., Hawes, J., Newell, J., Ponížý, L., 2021. Applying the food-energy-water nexus approach to urban agriculture: From FEW to FEWP (Food-Energy-Water-People). *Urban For. Urban Green.* 58, 126934.
<https://doi.org/10.1016/j.ufug.2020.126934>

- Castellar, J.A.C., Popartan, L.A., Pueyo-Ros, J., Atanasova, N., Langergraber, G., Säumel, I., Corominas, L., Comas, J., Acuña, V., 2021. Nature-based solutions in the urban context: terminology, classification and scoring for urban challenges and ecosystem services. *Sci. Total Environ.* 779, 146237. <https://doi.org/10.1016/j.scitotenv.2021.146237>
- Chen, S., Chen, H., Yang, R., Ye, Y., 2023. Linking social-ecological management and ecosystem service bundles: Lessons from a peri-urban agriculture landscape. *Land use policy* 131. <https://doi.org/10.1016/j.landusepol.2023.106697>
- Cogger, C.G., 2005. Potential Compost Benefits for Restoration Of Soils Disturbed by Urban Development. *Compost Sci. Util.* 13, 243–251. <https://doi.org/10.1080/1065657X.2005.10702248>
- Colding, J., Barthel, S., 2013. The potential of ‘Urban Green Commons’ in the resilience building of cities. *Ecol. Econ.* 86, 156–166. <https://doi.org/10.1016/j.ecolecon.2012.10.016>
- Cortinovis, C., Alzetta, C., Geneletti, D., 2021. Mapping Ecosystem Services, Disservices, and Ecological Requirements to Enhance Urban Forest Planning and Management in Padova. pp. 167–179. https://doi.org/10.1007/978-3-030-54345-7_13
- Costa, O.Y.A., Raaijmakers, J.M., Kuramae, E.E., 2018. Microbial Extracellular Polymeric Substances: Ecological Function and Impact on Soil Aggregation. *Front. Microbiol.* 9. <https://doi.org/10.3389/fmicb.2018.01636>
- Costanza, R., de Groot, R., Sutton, P., van der Ploeg, S., Anderson, S.J., Kubiszewski, I., Farber, S., Turner, R.K., 2014. Changes in the global value of ecosystem services. *Glob. Environ. Chang.* 26, 152–158. <https://doi.org/10.1016/j.gloenvcha.2014.04.002>
- Cox, M., 2015. A basic guide for empirical environmental social science. *Ecol. Soc.* 20, art63. <https://doi.org/10.5751/ES-07400-200163>
- Cumo, F., Pennacchia, E., Maurelli, P., Rosa, F., Zylka, C., 2025. Spaces, Energy and Shared Resources: New Technologies for Promoting More Inclusive and Sustainable Urban Communities. *Energies* 18, 1–25. <https://doi.org/10.3390/en18164410>
- D’Amore, C., Cohen, L., Chen, J., Owen, P., Ball, C., 2026. Community gardens and the cultivation of social capital. *Curr. Opin. Psychol.* 67, 102149. <https://doi.org/10.1016/j.copsyc.2025.102149>
- Daily, G.C., Postel, S., Bawa, K., Kaufman, L., Peterson, C.H., Carpenter, S., Tillman, D., Dayton, P., Alexander, S., Lagerquist, K., others, Daily, G.C., Postel, S., Bawa, K., Kaufman, L., Peterson, C.H., Carpenter, S., Tillman, D., Dayton, P., Alexander, S., Lagerquist, K., others, 1997. *Nature’s Services: Societal Dependence On Natural Ecosystems, Nature’s Services: Societal Dependence On Natural Ecosystems, Nature / science.* Island Press.
- De la Sota, C., Ruffato-Ferreira, V.J., Ruiz-García, L., Alvarez, S., la Sota, C., Ruffato-Ferreira, V.J., Ruiz-Garcia, L., Alvarez, S., 2019. Urban green infrastructure as a strategy of climate change mitigation. A case study in northern Spain. *Urban For. Urban Green.* 40, 145–151. <https://doi.org/10.1016/j.ufug.2018.09.004>
- Delshad, A.B., 2022. Community gardens: An investment in social cohesion, public health, economic sustainability, and the urban environment. *Urban For. Urban Green.* 70, 127549. <https://doi.org/10.1016/j.ufug.2022.127549>

- Dennis, M., James, P., 2017. Ecosystem services of collectively managed urban gardens: Exploring factors affecting synergies and trade-offs at the site level. *Ecosyst. Serv.* 26, 17–26. <https://doi.org/10.1016/J.ECOSER.2017.05.009>
- Dennis, M., James, P., 2016. Site-specific factors in the production of local urban ecosystem services: A case study of community-managed green space. *Ecosyst. Serv.* 17, 208–216. <https://doi.org/10.1016/j.ecoser.2016.01.003>
- Depietri, Y., 2022. Planning for urban green infrastructure: addressing tradeoffs and synergies. *Curr. Opin. Environ. Sustain.* 54, 101148. <https://doi.org/10.1016/J.COSUST.2021.12.001>
- Dorr, E., Hawes, J.K., Goldstein, B., Fargue-Lelievre, A., Fox-Kaemper, R., Specht, K., Fedenczak, K., Caputo, S., Cohen, N., Ponizy, L., Schoen, V., Gorecki, T., Newell, J.P., Jean-Soro, L., Grard, B., Fargue-Lelièvre, A., Fox-Kämper, R., Specht, K., Fedęńczak, K., Caputo, S., Cohen, N., Ponizy, L., Schoen, V., Górecki, T., Newell, J.P., Jean-Soro, L., Grard, B., 2023. Food production and resource use of urban farms and gardens: a five-country study. *Agron. Sustain. Dev.* 43. <https://doi.org/10.1007/s13593-022-00859-4>
- Dorr, E., Sanyé-Mengual, E., Gabrielle, B.B., Grard, B.J.-P.J.P., Aubry, C., Sanyé-Mengual, E., Gabrielle, B.B., Grard, B.J.-P.J.P., Aubry, C., 2017. Proper selection of substrates and crops enhances the sustainability of Paris rooftop garden. *Agron. Sustain. Dev.* 37. <https://doi.org/10.1007/s13593-017-0459-1>
- Dorst, H., van der Jagt, A., Raven, R., Runhaar, H., 2019. Urban greening through nature-based solutions – Key characteristics of an emerging concept. *Sustain. Cities Soc.* 49, 101620. <https://doi.org/10.1016/j.scs.2019.101620>
- Douglas, I., Anderson, P.M.L., Goode, D., Houck, M.C., Maddox, D., Nagendra, H., Yok, T.P. (Eds.), 2020. *The Routledge Handbook of Urban Ecology*. Routledge, Other titles: *Handbook of urban ecology Description: Second Edition.* | New York: Routledge, 2020. <https://doi.org/10.4324/9780429506758>
- Egerer, M.H., Lin, B.B., Threlfall, C.G., Kendal, D., 2019. Temperature variability influences urban garden plant richness and gardener water use behavior, but not planting decisions. *Sci. Total Environ.* 646, 111–120. <https://doi.org/10.1016/j.scitotenv.2018.07.270>
- Ehrlich, D., T, K., JH, U., I, M.R., P, F., M, P., M, M., P, P., L, M., M, S., P, T., A, C., K, K., 2024. *Atlas of the Human Planet 2024*. Publications Office of the European Union, Luxembourg (Luxembourg). <https://doi.org/10.2760/808886> (online), [10.2760/855738](https://doi.org/10.2760/855738) (print)
- Eisen, M.B., Spellman, P.T., Brown, P.O., Botstein, D., 1998. Cluster analysis and display of genome-wide expression patterns. *Proc. Natl. Acad. Sci.* 95, 14863–14868. <https://doi.org/10.1073/pnas.95.25.14863>
- Eisenhauer, N., Frank, K., Weigelt, A., Bartkowski, B., Beugnon, R., Liebal, K., Mahecha, M., Quaas, M., Al-Halbouni, D., Bastos, A., Bohn, F.J., de Brito, M.M., Denzler, J., Feilhauer, H., Fischer, R., Fritsche, I., Guimaraes-Steinicke, C., Hänsel, M., Haun, D.B.M., Herrmann, H., Huth, A., Kalesse-Los, H., Koetter, M., Kolleck, N., Krause, M., Kretschmer, M., Leitão, P.J., Masson, T., Mora, K., Müller, B., Peng, J., Pöhlker, M.L., Ratzke, L., Reichstein, M., Richter, S., Rüger, N., Sánchez-Parra, B., Shadaydeh, M., Sippel, S., Tegen, I., Thrän, D., Umlauf, J., Wendisch, M., Wolf, K., Wirth, C., Zacher, H., Zaehle, S., Quaas, J., 2024. A belowground perspective on the nexus between biodiversity change, climate change, and human well-being. *J. Sustain. Agric. Environ.* 3, e212108. <https://doi.org/10.1002/SAE2.12108>

- Elkamel, M., Valencia, A., Zhang, W., Zheng, Q.P., Chang, N.-B., 2023. Multi-agent modeling for linking a green transportation system with an urban agriculture network in a food-energy-water nexus. *Sustain. CITIES Soc.* 89. <https://doi.org/10.1016/j.scs.2022.104354>
- Evans, D.L., Falagán, N., Hardman, C.A., Kourmpetli, S., Liu, L., Mead, B.R., Davies, J.A.C., 2022. Ecosystem service delivery by urban agriculture and green infrastructure – a systematic review. *Ecosyst. Serv.* 54, 101405. <https://doi.org/10.1016/j.ecoser.2022.101405>
- Fabrigar, L.R., Wegener, D.T., MacCallum, R.C., Strahan, E.J., 1999. Evaluating the use of exploratory factor analysis in psychological research. *Psychol. Methods* 4, 272–299. <https://doi.org/10.1037/1082-989X.4.3.272>
- Fanfani, D., Duzi, B., Mancino, M., Rovai, M., 2022. Multiple evaluation of urban and peri-urban agriculture and its relation to spatial planning: The case of Prato territory (Italy). *Sustain. CITIES Soc.* 79. <https://doi.org/10.1016/j.scs.2021.103636>
- Faruqui, N., Al-Jayyousi, O., 2002. Greywater Reuse in Urban Agriculture for Poverty Alleviation. *Water Int.* 27, 387–394. <https://doi.org/10.1080/02508060208687018>
- Foley, J.A., DeFries, R., Asner, G.P., Barford, C., Bonan, G., Carpenter, S.R., Chapin, F.S., Coe, M.T., Daily, G.C., Gibbs, H.K., Helkowski, J.H., Holloway, T., Howard, E.A., Kucharik, C.J., Monfreda, C., Patz, J.A., Prentice, I.C., Ramankutty, N., Snyder, P.K., 2005. Global consequences of land use. *Science* (80-.). 309, 570–574. <https://doi.org/10.1126/science.1111772>
- Fox-Kämper, R., Wesener, A., Münderlein, D., Sondermann, M., McWilliam, W., Kirk, N., 2018. Urban community gardens: An evaluation of governance approaches and related enablers and barriers at different development stages. *Landsc. Urban Plan.* 170, 59–68. <https://doi.org/10.1016/j.landurbplan.2017.06.023>
- Frouz, J., 2024. Plant-soil feedback across spatiotemporal scales from immediate effects to legacy. *Soil Biol. Biochem.* 189, 109289. <https://doi.org/10.1016/j.soilbio.2023.109289>
- Fürst, C., Luque, S., Geneletti, D., 2017. Nexus thinking – how ecosystem services can contribute to enhancing the cross-scale and cross-sectoral coherence between land use, spatial planning and policy-making. *Int. J. Biodivers. Sci. Ecosyst. Serv. Manag.* 13, 412–421. <https://doi.org/10.1080/21513732.2017.1396257>
- Galdámez, E., Kauark-Fontes, B., Vásquez, A., Salbitano, F., Mercado, G., Wild, T., Pauleit, S., 2025. Beyond experimentation: how are urban nature-based solutions living labs in Europe and Latin America designed and implemented for transformative change? *Sustain. Sci. Pract. Policy* 21. <https://doi.org/10.1080/15487733.2025.2545644>
- Garcia, D.J., Lovett, B.M., You, F., 2019. Considering agricultural wastes and ecosystem services in Food-Energy-Water-Waste Nexus system design. *J. Clean. Prod.* 228, 941–955. <https://doi.org/10.1016/j.jclepro.2019.04.314>
- Genter, C., Roberts, A., Richardson, J., Sheaff, M., 2015. The contribution of allotment gardening to health and wellbeing: A systematic review of the literature. *Br. J. Occup. Ther.* 78, 593–605. <https://doi.org/10.1177/0308022615599408>
- Ghodsvai, M., Krishnamurthy, S., de Vries, B., 2019. Review of transdisciplinary approaches to food-water-

- energy nexus: A guide towards sustainable development. *Environ. Sci. Policy* 101, 266–278.
<https://doi.org/10.1016/j.envsci.2019.09.003>
- Gómez-Baggethun, E., Barton, D.N., 2013. Classifying and valuing ecosystem services for urban planning. *Ecol. Econ.* 86, 235–245. <https://doi.org/10.1016/j.ecolecon.2012.08.019>
- Gondhalekar, D., Ramsauer, T., 2017. Nexus City: Operationalizing the urban Water-Energy-Food Nexus for climate change adaptation in Munich, Germany. *URBAN Clim.* 19, 28–40.
<https://doi.org/10.1016/j.uclim.2016.11.004>
- Goodarzi, M.R., Mohtar, R.H., Piryaci, R., Fatchifar, A., Niazkar, M., 2022. Urban WEF Nexus: An Approach for the Use of Internal Resources under Climate Change. *HYDROLOGY* 9.
<https://doi.org/10.3390/hydrology9100176>
- Grard, B.J.-P.-P., Chenu, C., Manouchehri, N., Houot, S., Frascaria-Lacoste, N., Aubry, C., 2018. Rooftop farming on urban waste provides many ecosystem services. *Agron. Sustain. Dev.* 38.
<https://doi.org/10.1007/s13593-017-0474-2>
- Grebitus, C., 2021. Small-scale urban agriculture: Drivers of growing produce at home and in community gardens in Detroit. *PLoS One* 16, e0256913. <https://doi.org/10.1371/journal.pone.0256913>
- Gregory, J.H., Dukes, M.D., Jones, P.H., Miller, G.L., 2006. Effect of urban soil compaction on infiltration rate. *J. Soil Water Conserv.* 61, 117–124. <https://doi.org/10.1080/00224561.2006.12435870>
- Gregory, M.M., Leslie, T.W., Drinkwater, L.E., 2016. Agroecological and social characteristics of New York city community gardens: contributions to urban food security, ecosystem services, and environmental education, *Urban Ecosystems*. *Urban Ecosystems*. <https://doi.org/10.1007/s11252-015-0505-1>
- Guo, C., Li, J., Li, H., Li, Y., 2020. Influences of stormwater concentration infiltration on the heavy metal contents of soil in rain gardens. *Water Sci. Technol.* 81, 1039–1051. <https://doi.org/10.2166/wst.2020.189>
- Haase, D., Gaeva, D., 2023. Allotments for all? Social–environmental values of urban gardens for gardeners and the public in cities: The example of Berlin, Germany. *People Nat.* <https://doi.org/10.1002/pan3.10488>
- Haase, D., Larondelle, N., Andersson, E., Artmann, M., Borgström, S., Breuste, J., Gomez-Baggethun, E., Gren, Å., Hamstead, Z., Hansen, R., Kabisch, N., Kremer, P., Langemeyer, J., Rall, E.L., McPhearson, T., Pauleit, S., Qureshi, S., Schwarz, N., Voigt, A., Wurster, D., Elmqvist, T., 2014. A quantitative review of urban ecosystem service assessments: Concepts, models, and implementation. *Ambio*.
<https://doi.org/10.1007/s13280-014-0504-0>
- Hanon, I., 2015. Agriculture urbaine et autogestion à Cuba. *Rev. Int. l'économie Soc. Recma* 84.
<https://doi.org/10.7202/1032525ar>
- Hatfield, J.L., Sauer, T.J., Cruse, R.M., 2017. Soil: The Forgotten Piece of the Water, Food, Energy Nexus. *Adv. Agron.* 143, 1–46. <https://doi.org/10.1016/BS.AGRON.2017.02.001>
- Herd-Hoare, S., Shackleton, C.M., 2020. Ecosystem disservices matter when valuing ecosystem benefits from small-scale arable agriculture. *Ecosyst. Serv.* 46, 101201. <https://doi.org/10.1016/j.ecoser.2020.101201>
- Hernández-Blanco, M., Costanza, R., 2018. Natural capital and ecosystem services, in: *The Routledge Handbook of Agricultural Economics*. Routledge, New York, NY : Routledge, 2018., pp. 254–268.

- <https://doi.org/10.4324/9781315623351-15>
- Hoff, H., 2011. Understanding the nexus. background paper for the bonn2011 nexus conference: the water, energy and food security nexus.
- Holgado-Tello, F.P., Chacón-Moscoso, S., Barbero-García, I., Vila-Abad, E., 2010. Polychoric versus Pearson correlations in exploratory and confirmatory factor analysis of ordinal variables. *Qual. Quant.* 44, 153–166. <https://doi.org/10.1007/s11135-008-9190-y>
- Holt, N., Shukla, S., Hochmuth, G., Munoz-Carpena, R., Ozores-Hampton, M., 2017. Transforming the food-water-energy-land-economic nexus of plasticulture production through compact bed geometries. *Adv. Water Resour.* 110, 515–527. <https://doi.org/10.1016/j.advwatres.2017.04.023>
- Horn, J.L., 1965. A Rationale and Test for the Number of Factors in Factor Analysis. *Psychometrika* 30, 179–185. <https://doi.org/10.1007/BF02289447>
- Howard, J.L., Dubay, B.R., Daniels, W.L., 2013. Artifact weathering, anthropogenic microparticles and lead contamination in urban soils at former demolition sites, Detroit, Michigan. *Environ. Pollut.* 179, 1–12. <https://doi.org/10.1016/j.envpol.2013.03.053>
- Huang, A., Chang, F.-J., 2021. Prospects for Rooftop Farming System Dynamics: An Action to Stimulate Water-Energy-Food Nexus Synergies toward Green Cities of Tomorrow. *SUSTAINABILITY* 13. <https://doi.org/10.3390/su13169042>
- Huntington, T.G., 2020. Soil: Organic Matter and Available Water Capacity, in: *Landscape and Land Capacity*. CRC Press, Second edition. | Boca Raton: CRC Press, [2020] | Revised edition of: *Encyclopedia of natural resources*. [2014]., [2020] | Revised edition of: *Encyclopedia of natural resources*. [2014]., pp. 273–281. <https://doi.org/10.1201/9780429445552-36>
- Ignaciuk, A., D'Croz, D., Islam, S., 2015. Better Drip than Flood: Reaping the Benefits of Efficient Irrigation. *EuroChoices* 14, 26–32. <https://doi.org/10.1111/1746-692X.12088>
- Imam, M., Glaros, A., Chen, C., Dsouza, A., Brander, A., Ferguson, R., 2025. A carbon-centric evaluation framework for building-integrated agriculture: a comparison of three farm types and building standards. *Front. Sustain. FOOD Syst.* 9. <https://doi.org/10.3389/fsufs.2025.1533433>
- Insee, 2025. Dossier complet – Commune de Nantes (44109) | Insee [WWW Document]. URL <https://www.insee.fr/fr/statistiques/2011101?geo=COM-44109> (accessed 10.19.25).
- IPBES, 2019. Global assessment report of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. Island Press, Bonn, Germany.
- Jahrl, I., Moschitz, H., Cavin, J.S., 2021. The role of food gardening in addressing urban sustainability – A new framework for analysing policy approaches. *Land use policy* 108, 105564. <https://doi.org/10.1016/j.landusepol.2021.105564>
- Jans-Singh, M., Ward, R., Choudhary, R., 2021. Co-simulating a greenhouse in a building to quantify co-benefits of different coupled configurations. *J. Build. Perform. Simul.* 14, 247–276. <https://doi.org/10.1080/19401493.2021.1908426>
- Jean-Soro, L., Le Guern, C.C., Bechet, B.B., Lebeau, T., Ringear, M.-F.F., 2015. Origin of trace elements in an

- urban garden in Nantes, France. *J. Soils Sediments* 15, 1802–1812. <https://doi.org/10.1007/s11368-014-0952-y>
- Jha, S., Egerer, M., Bichier, P., Cohen, H., Liere, H., Lin, B., Lucatero, A., Philpott, S.M., 2023. Multiple ecosystem service synergies and landscape mediation of biodiversity within urban agroecosystems. *Ecol. Lett.* 26, 369–383. <https://doi.org/10.1111/ele.14146>
- Jin, N., Zhang, D., Jin, L., Wang, S., Yang, X., Lei, Y., Meng, X., Xu, Z., Sun, J., Lyu, J., Yu, J., 2023. Controlling water deficiency as an abiotic stress factor to improve tomato nutritional and flavour quality. *Food Chem. X* 19, 100756. <https://doi.org/10.1016/j.fochx.2023.100756>
- Kafle, A., Hopeward, J., Myers, B., 2023. Exploring Trade-Offs between Potential Economic, Social and Environmental Outcomes of Urban Agriculture in Adelaide, Australia and the Kathmandu Valley, Nepal. *SUSTAINABILITY* 15. <https://doi.org/10.3390/su151411251>
- Kamali, M.I., Nazari, R., Karimi, M., Nikoo, M.R., 2025. Enhancing urban food production: A framework for optimal site selection and policy development. *Sustain. CITIES Soc.* 126, 106375. <https://doi.org/10.1016/j.scs.2025.106375>
- Keller, A.B., Handler, S.D., 2024. Soil organic carbon in temperate managed ecosystems. <https://doi.org/10.32747/2024.8633528.ch>
- Kessler, R., 2013. Urban gardening: Managing the risks of contaminated soil. *Environ. Health Perspect.* 121. <https://doi.org/10.1289/ehp.121-A326>
- Kirby, C.K., Specht, K., Fox-Kämper, R., Hawes, J.K., Cohen, N., Caputo, S., Ilieva, R.T., Lelièvre, A., Ponizy, L., Schoen, V., Blythe, C., 2021. Differences in motivations and social impacts across urban agriculture types: Case studies in Europe and the US. *Landsc. Urban Plan.* 212, 104110. <https://doi.org/10.1016/j.landurbplan.2021.104110>
- Kong, J., Gao, M., Alofaysan, H., Fayziyeva, D., Liu, Z., 2025. Enhancing urban agriculture networks: A clustering and multicriteria decision-making approach to sustainability indicators and governance. *Ecol. Indic.* 170. <https://doi.org/10.1016/j.ecolind.2024.112997>
- Krasny, M.E., Russ, A., Tidball, K.G., Elmqvist, T., 2014. Civic ecology practices: Participatory approaches to generating and measuring ecosystem services in cities. *Ecosyst. Serv.* 7, 177–186. <https://doi.org/10.1016/j.ecoser.2013.11.002>
- Krasny, M.E., Tidball, K.G., 2012. Civic ecology: a pathway for Earth Stewardship in cities. *Front. Ecol. Environ.* 10, 267–273. <https://doi.org/10.1890/110230>
- Kruskal, W.H., Wallis, W.A., 1952. Use of Ranks in One-Criterion Variance Analysis. *J. Am. Stat. Assoc.* 47, 583. <https://doi.org/10.2307/2280779>
- Lal, R., Mohtar, R.H., Assi, A.T., Ray, R., Baybil, H., Jahn, M., 2017. Soil as a Basic Nexus Tool: Soils at the Center of the Food–Energy–Water Nexus, Current Sustainable/Renewable Energy Reports. Springer Nature. <https://doi.org/10.1007/s40518-017-0082-4>
- Langemeyer, J., Latkowska, M.J., Gómez-Baggethun, E.N., Voigt, A., Calvet-Mir, L., Pourias, J., Camps-Calvet, M., Orsini, F., Breuste, J., Artmann, M., Jokinen, A., Béchet, B., da Luz, P.B., Hursthouse, A., Stępień,

- M.P., Baležentienė, L., 2016. Ecosystem services from urban gardens, in: *Urban Allotment Gardens in Europe*. Routledge, New York : Routledge, 2016., pp. 115–141. <https://doi.org/10.4324/9781315686608-6>
- Le Guern, C., Jean-Soro, L., Béchet, B., Lebeau, T., Bouquet, D., 2018. Management initiatives in support of the soil quality of urban allotment gardens: Examples from Nantes (France). *L. Degrad. Dev.* 29, 3681–3692. <https://doi.org/10.1002/LDR.3123>
- Lehmann, S., 2018. Implementing the Urban Nexus approach for improved resource-efficiency of developing cities in Southeast-Asia. *City, Cult. Soc.* 13, 46–56. <https://doi.org/10.1016/j.ccs.2017.10.003>
- Leitão, T.E., Cameira, M.R., Costa, H.D., Pacheco, J.M., Henriques, M.J., Martins, L.L., Mourato, M.P., 2018. Environmental Quality in Urban Allotment Gardens: Atmospheric Deposition, Soil, Water and Vegetable Assessment at LISBON City. *Water. Air. Soil Pollut.* 229, 31. <https://doi.org/10.1007/s11270-017-3681-1>
- Li, H., Geng, J., Liu, Z., Ao, H., Wang, Z., Xue, Q., 2025. Mulching Improves the Yield and Water Use Efficiency of Millet in Northern China: A Meta-Analysis. *Agriculture* 15, 397. <https://doi.org/10.3390/agriculture15040397>
- Lin, B.B., 2010. The role of agroforestry in reducing water loss through soil evaporation and crop transpiration in coffee agroecosystems. *Agric. For. Meteorol.* 150, 510–518. <https://doi.org/10.1016/j.agrformet.2009.11.010>
- Lin, B.B., Egerer, M.H., Liere, H., Jha, S., Bichier, P., Philpott, S.M., 2018a. Local- and landscape-scale land cover affects microclimate and water use in urban gardens. *Sci. Total Environ.* 610–611, 570–575. <https://doi.org/10.1016/j.scitotenv.2017.08.091>
- Lin, B.B., Egerer, M.H., Liere, H., Jha, S., Philpott, S.M., 2018b. Soil management is key to maintaining soil moisture in urban gardens facing changing climatic conditions. *Sci. Rep.* 8, 17565. <https://doi.org/10.1038/s41598-018-35731-7>
- Lin, Z.-E., Lin, Y.-C., Lee, M., Chiueh, P.-T., Wu, C.-W., Chang, Y.-S., 2025. An integrated quantitative framework for exploring sustainable scenarios in urban agriculture planning. *Sustain. Environ. Res.* 35. <https://doi.org/10.1186/s42834-025-00241-5>
- Liu, S.-K.K., Lin, Z.-E.E., Chiueh, P.-T. Te, 2022. Improving urban sustainability and resilience with the optimal arrangement of water-energy-food related practices. *Sci. Total Environ.* 812, 152559. <https://doi.org/10.1016/j.scitotenv.2021.152559>
- López-Ridaura, S., 2002. Evaluating the sustainability of complex socio-environmental systems. the MESMIS framework. *Ecol. Indic.* 2, 135–148. [https://doi.org/10.1016/S1470-160X\(02\)00043-2](https://doi.org/10.1016/S1470-160X(02)00043-2)
- Lourdes, K.T., Gibbins, C.N., Hamel, P., Sanusi, R., Azhar, B., Lechner, A.M., 2021. A Review of Urban Ecosystem Services Research in Southeast Asia. *Land*. <https://doi.org/10.3390/land10010040>
- Lowan-Trudeau, M., Keough, N., Wong, J., Haidey, S., 2020. The affordable housing, transportation, and food nexus: Community gardens and healthy affordable living in Calgary. *Can. Geogr.* 64, 502–515. <https://doi.org/10.1111/cag.12606>
- Lucertini, G., Di Giustino, G., 2021. Urban and Peri-Urban Agriculture as a Tool for Food Security and Climate

- Change Mitigation and Adaptation: The Case of Mestre. *SUSTAINABILITY* 13. <https://doi.org/10.3390/su13115999>
- Lupp, G., Zingraff-Hamed, A., Huang, J.J., Oen, A., Pauleit, S., 2020. Living Labs—A Concept for Co-Designing Nature-Based Solutions. *Sustainability* 13, 188. <https://doi.org/10.3390/su13010188>
- Lyytimäki, J., 2015. Prospects for Environmental Communication Based on 25 Years of Newspaper Coverage of Climate Change and Eutrophication in Finland. *Appl. Environ. Educ. Commun.* 14, 246–255. <https://doi.org/10.1080/1533015X.2015.1109486>
- Lyytimäki, J., Petersen, L.K., Normander, B., Bezák, P., 2008. Nature as a nuisance? Ecosystem services and disservices to urban lifestyle. *Environ. Sci.* 5, 161–172. <https://doi.org/10.1080/15693430802055524>
- Lyytimäki, J., Sipilä, M., 2009. Hopping on one leg – The challenge of ecosystem disservices for urban green management. *Urban For. Urban Green.* 8, 309–315. <https://doi.org/10.1016/j.ufug.2009.09.003>
- M. Susana, O.O., Davide Geneletti, Susana, O.O.M., Davide, G., 2023. Prioritizing urban nature-based solutions to support scaling-out strategies: A case study in Las Palmas de Gran Canaria. *Environ. Impact Assess. Rev.* 102, 107158. <https://doi.org/10.1016/j.eiar.2023.107158>
- Marche, G., 2015. What Can Urban Gardening Really Do About Gentrification? A Case-Study of Three San Francisco Community Gardens. *Eur. J. Am. Stud.* 10. <https://doi.org/10.4000/ejas.11316>
- Martinez-Alier, J., 2002. *The Environmentalism of the Poor*. Edward Elgar Publishing. <https://doi.org/10.4337/9781843765486>
- Materechera, S.A., Dexter, A.R., Alston, A.M., 1992. Formation of aggregates by plant roots in homogenised soils. *Plant Soil* 142, 69–79. <https://doi.org/10.1007/BF00010176>
- McDougall, R., Kristiansen, P., Rader, R., 2019. Small-scale urban agriculture results in high yields but requires judicious management of inputs to achieve sustainability. *Proc. Natl. Acad. Sci. U. S. A.* 116, 129–134. <https://doi.org/10.1073/pnas.1809707115>
- McKnight, P.E., Najab, J., 2010. Mann-Whitney U Test, in: *The Corsini Encyclopedia of Psychology*. Wiley, pp. 1–1. <https://doi.org/10.1002/9780470479216.corpsy0524>
- Meadows, D.H., Wright, D., 2015. *Thinking in systems : a primer*. Chelsea Green Publishing.
- Meng, F., Yuan, Q., Bellezoni, R.A., de Oliveira, J.P.A., Hu, Y., Jing, R., Liu, G., Yang, Z., Seto, K.C., 2023. The food-water-energy nexus and green roofs in Sao Jose dos Campos, Brazil, and Johannesburg, South Africa. *NPJ URBAN Sustain.* 3. <https://doi.org/10.1038/s42949-023-00091-3>
- Mensah, J.K., 2023. Urban agriculture, local economic development and climate change: conceptual linkages. *Int. J. Urban Sustain. Dev.* 15, 141–151. <https://doi.org/10.1080/19463138.2023.2207523>
- Millennium Ecosystem Assessment, 2005. *Ecosystems and human well-being : synthesis*. Island Press, Washington, DC.
- Mueller, C.W., Baumert, V., Carminati, A., Germon, A., Holz, M., Kögel-Knabner, I., Peth, S., Schlüter, S., Uteau, D., Vetterlein, D., Teixeira, P., Vidal, A., 2024. From rhizosphere to detritosphere – Soil structure formation driven by plant roots and the interactions with soil biota. *Soil Biol. Biochem.* 193, 109396.

- <https://doi.org/10.1016/j.soilbio.2024.109396>
- Mulier, M.C.G.H., van de Ven, F.H.M., Kirshen, P., 2022. Circularity in the Urban Water-Energy-Nutrients-Food nexus. *ENERGY NEXUS* 7. <https://doi.org/10.1016/j.nexus.2022.100081>
- Mumaw, L., Mata, L., 2022. Wildlife gardening: an urban nexus of social and ecological relationships. *Front. Ecol. Environ.* 20, 379–385. <https://doi.org/10.1002/fee.2484>
- Munda, G., 2004. Social multi-criteria evaluation: Methodological foundations and operational consequences. *Eur. J. Oper. Res.* 158, 662–677. [https://doi.org/10.1016/S0377-2217\(03\)00369-2](https://doi.org/10.1016/S0377-2217(03)00369-2)
- Muthee, K., Duguma, L., Nzyoka, J., Minang, P., 2021. Ecosystem-Based Adaptation Practices as a Nature-Based Solution to Promote Water-Energy-Food Nexus Balance. *Sustainability* 13, 1142. <https://doi.org/10.3390/su13031142>
- Nacem, M.Y., 2025. The Impact of Drought Stress on the Nutritional Quality of Vegetables, in: *Drought Stress*. Springer Nature Switzerland, Cham, pp. 143–158. https://doi.org/10.1007/978-3-031-80610-0_5
- Nantes Métropole, 2025. Jardins familiaux de la ville de Nantes — Open Data Nantes Métropole [WWW Document]. *Jard. Fam. la V. Nantes*. URL https://data.nantesmetropole.fr/explore/dataset/244400404_jardins-familiaux-nantes/table/?disjunctive.association (accessed 10.19.25).
- Nassary, E.K., 2025. Exploring the role of nature-based solutions and emerging technologies in advancing circular and sustainable agriculture: An opinionated review for environmental resilience. *Clean. Circ. Bioeconomy* 10, 100142. <https://doi.org/10.1016/j.clcb.2025.100142>
- Nemeskéri, E., Helyes, L., 2019. Physiological Responses of Selected Vegetable Crop Species to Water Stress. *Agronomy* 9, 447. <https://doi.org/10.3390/agronomy9080447>
- Neurath, R.A., Pett-Ridge, J., Chu-Jacoby, I., Herman, D., Whitman, T., Nico, P.S., Lipton, A.S., Kyle, J., Tfaily, M.M., Thompson, A., Firestone, M.K., 2021. Root Carbon Interaction with Soil Minerals Is Dynamic, Leaving a Legacy of Microbially Derived Residues. *Environ. Sci. Technol.* *acs.est.1c00300*. <https://doi.org/10.1021/acs.est.1c00300>
- Nie, J., Kiminami, A., Yagi, H., 2024. Assessing the Sustainability of Urban Agriculture in Shanghai’s Nine Agriculture Districts: A Decadal Analysis (2010–2020). *AGRICULTURE-BASEL* 14. <https://doi.org/10.3390/agriculture14040631>
- Nunes, N., Björner, E., Hilding-Hamann, K.E., 2021. Guidelines for Citizen Engagement and the Co-Creation of Nature-Based Solutions: Living Knowledge in the URBiNAT Project. *Sustainability* 13, 13378. <https://doi.org/10.3390/su132313378>
- Olsson, U., 1979. Maximum Likelihood Estimation of the Polychoric Correlation Coefficient. *Psychometrika* 44, 443–460. <https://doi.org/10.1007/BF02296207>
- Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for

- reporting systematic reviews. *BMJ* n71. <https://doi.org/10.1136/bmj.n71>
- Palazzo, A., Kahil, T., Willaarts, B.A., Burek, P., van Dijk, M., Tang, T., Magnuszewski, P., Havlík, P., Langan, S., Wada, Y., 2024. Assessing sustainable development pathways for water, food, and energy security in a transboundary river basin. *Environ. Dev.* 51, 101030. <https://doi.org/10.1016/j.envdev.2024.101030>
- Paltseva, A.A., Cheng, Z., McBride, M., Deeb, M., Egendorf, S.P., Groffman, P.M., 2022. Legacy Lead in Urban Garden Soils: Communicating Risk and Limiting Exposure. *Front. Ecol. Evol.* 10. <https://doi.org/10.3389/fevo.2022.873542>
- Panagos, P., Borrelli, P., Jones, A., Robinson, D.A., 2024. A 1 billion euro mission: A Soil Deal for Europe. *Eur. J. Soil Sci.* 75. <https://doi.org/10.1111/ejss.13466>
- Pedro, J., Reis, A., Silva, C., Pinheiro, M.D., 2021. Evaluating the economic benefits of moving from a single building to a community approach for sustainable urban redevelopment: Lisbon neighborhood case study. *J. Clean. Prod.* 304, 126810. <https://doi.org/10.1016/j.jclepro.2021.126810>
- Pereira, P., Yin, C., Hua, T., 2023. Nature-based solutions, ecosystem services, disservices, and impacts on well-being in urban environments. *Curr. Opin. Environ. Sci. Heal.* 33, 100465. <https://doi.org/10.1016/j.coesh.2023.100465>
- Pirasteh-Anosheh, H., Saed-Moucheshi, A., Pakniyat, H., Pessarakli, M., 2016. Stomatal responses to drought stress, in: *Water Stress and Crop Plants*. Wiley, pp. 24–40. <https://doi.org/10.1002/9781119054450.ch3>
- Pollard, G., Ward, J., Roetman, P., 2018. Water Use Efficiency in Urban Food Gardens: Insights from a Systematic Review and Case Study. *Horticulturae* 4, 27. <https://doi.org/10.3390/horticulturae4030027>
- Ponge, J.-F., 2003. Humus forms in terrestrial ecosystems: a framework to biodiversity. *Soil Biol. Biochem.* 35, 935–945. [https://doi.org/10.1016/S0038-0717\(03\)00149-4](https://doi.org/10.1016/S0038-0717(03)00149-4)
- Pueyo-Ros, J., Škerjanec, M., Castellar, J.A.C.C., Atanasova, N.N.N., Comas, J., Corominas, L.L.L., Škerjanec, M., Castellar, J.A.C.C., Atanasova, N.N.N., Comas, J., Corominas, L.L.L., 2024. Beyond food: A stochastic model to estimate the contributions of urban agriculture to sustainability. *Landsc. Urban Plan.* 241. <https://doi.org/10.1016/j.landurbplan.2023.104930>
- Puppim de Oliveira, J.A., Bellezoni, R.A., Shih, W. yu, Bayulken, B., 2022. Innovations in Urban Green and Blue Infrastructure: Tackling local and global challenges in cities. *J. Clean. Prod.* 362. <https://doi.org/10.1016/j.jclepro.2022.132355>
- Quinton, J., Nesbitt, L., Sax, D., 2022. How well do we know green gentrification? A systematic review of the methods. *Prog. Hum. Geogr.* 46, 960–987. <https://doi.org/10.1177/03091325221104478>
- R Core Team, 2025. R: A Language and Environment for Statistical Computing Team RDCVienna, Austria2006, in: *R Foundation for Statistical Computing*. Vienna, Austria.
- Rahman, M.M., Shahriyar, A.A., Hagare, D., Maheshwari, B., 2022. Impact of Recycled Water Irrigation on Soil Salinity and Its Remediation. *Soil Syst.* 6, 13. <https://doi.org/10.3390/soilsystems6010013>
- Raisson-Victor, V., Charlot, A., Bechet, B., Bertrand, J.-L., Chancibault, K., Devisme, L., Fleury-Bahi, G., Habets, F., Robin, M., Rospars, C., others, 2022. GIEC DES PAYS DE LA LOIRE. 1er rapport—Juin 2022.

- Rao, N., Patil, S., Singh, C., Roy, P., Pryor, C., Poonacha, P., Genes, M., 2022. Cultivating sustainable and healthy cities: A systematic literature review of the outcomes of urban and peri-urban agriculture. *Sustain. Cities Soc.* 85, 104063. <https://doi.org/10.1016/j.scs.2022.104063>
- Ravar, Z., Zahraie, B., Sharifinejad, A., Gozini, H., Jafari, S., 2020. System dynamics modeling for assessment of water–food–energy resources security and nexus in Gavkhuni basin in Iran. *Ecol. Indic.* 108, 105682. <https://doi.org/10.1016/j.ecolind.2019.105682>
- Redmile-Gordon, M.A., Brookes, P.C., Evershed, R.P., Goulding, K.W.T., Hirsch, P.R., 2014. Measuring the soil-microbial interface: Extraction of extracellular polymeric substances (EPS) from soil biofilms. *Soil Biol. Biochem.* 72, 163–171. <https://doi.org/10.1016/j.soilbio.2014.01.025>
- Robinson, D.A., Lebron, I., Vereecken, H., 2009. On the Definition of the Natural Capital of Soils: A Framework for Description, Evaluation, and Monitoring. *Soil Sci. Soc. Am. J.* 73, 1904–1911. <https://doi.org/10.2136/sssaj2008.0332>
- Rodríguez González, M.I., Scott, C.K., Marquina, T., Mewa, D.B., Polo, J.G.G., Peng, B., Gonzalez, M.I.R., Scott, C.K., Marquina, T., Mewa, D.B., Polo, J.G.G., Peng, B., 2023. Integration of the Livelihood and Ecosystem Services Frameworks-A Case Study on Urban Agriculture in Mexico City. *EARTH Interact.* 27. <https://doi.org/10.1175/EI-D-22-0010.1>
- Russo, A., Cirella, G.T., 2019. Edible urbanism 5.0. *Palgrave Commun.* 5, 163. <https://doi.org/10.1057/s41599-019-0377-8>
- Russo, A., Escobedo, F.J., Cirella, G.T., Zerbe, S., 2017. Edible green infrastructure: An approach and review of provisioning ecosystem services and disservices in urban environments. *Agric. Ecosyst. Environ.* 242, 53–66. <https://doi.org/10.1016/j.agee.2017.03.026>
- Salomon, M.J., Watts-Williams, S.J., McLaughlin, M.J., Cavagnaro, T.R., 2020. Urban soil health: A city-wide survey of chemical and biological properties of urban agriculture soils. *J. Clean. Prod.* 275. <https://doi.org/10.1016/j.jclepro.2020.122900>
- Sambo, B., Sperotto, A., Torresan, S., Pittore, M., Zebisch, M., Critto, A., 2024. Looking at the Water-Energy-Food nexus through the lens of Ecosystem Services: a new perspective. *Environ. Res. Lett.* 19, 121003. <https://doi.org/10.1088/1748-9326/ad96cf>
- Sanjuan-Delmás, D., Llorach-Massana, P., Nadal, A., Ercilla-Montserrat, M., Muñoz, P., Montero, J.I., Josa, A., Gabarrell, X., Rieradevall, J., Sanjuan-Delmas, D., Llorach-Massana, P., Nadal, A., Ercilla-Montserrat, M., Munoz, P., Ignacio Montero, J., Josa, A., Gabarrell, X., Rieradevall, J., 2018. Environmental assessment of an integrated rooftop greenhouse for food production in cities. *J. Clean. Prod.* 177, 326–337. <https://doi.org/10.1016/j.jclepro.2017.12.147>
- Sanyé-Mengual, E., Specht, K., Krikser, T., Vanni, C., Pennisi, G., Orsini, F., Gianquinto, G.P., 2018. Social acceptance and perceived ecosystem services of urban agriculture in Southern Europe: The case of Bologna, Italy. *PLoS One* 13, e0200993. <https://doi.org/10.1371/journal.pone.0200993>
- Satz, D., Gould, R.K., Chan, K.M.A., Guerry, A., Norton, B., Satterfield, T., Halpern, B.S., Levine, J., Woodside, U., Hannahs, N., Basurto, X., Klain, S., 2013. The Challenges of Incorporating Cultural Ecosystem Services into Environmental Assessment. *Ambio* 42, 675–684. <https://doi.org/10.1007/s13280-013-0386->

- Schaubroeck, T., 2017. A need for equal consideration of ecosystem disservices and services when valuing nature; countering arguments against disservices. *Ecosyst. Serv.* 26, 95–97.
<https://doi.org/10.1016/j.ecoser.2017.06.009>
- Schreiber, J.B., 2021. Issues and recommendations for exploratory factor analysis and principal component analysis. *Res. Soc. Adm. Pharm.* 17, 1004–1011. <https://doi.org/10.1016/j.sapharm.2020.07.027>
- Séré, G., Le Guern, C., Bispo, A., Layet, C., Ducommun, C., Clesse, M., Schwartz, C., Vidal-Beaudet, L., 2024. Selection of soil health indicators for modelling soil functions to promote smart urban planning. *Sci. Total Environ.* 924, 171347. <https://doi.org/10.1016/j.scitotenv.2024.171347>
- Shqerat, N., Al-Tabbal, J., 2025. Potential reuse of greywater for irrigation of tomato (*Solanum lycopersicum*) plants and its effect on plants growth and soil. *Int. J. Phytoremediation* 27, 561–582.
<https://doi.org/10.1080/15226514.2024.2427926>
- Siegner, A.B., Acey, C., Sowerwine, J., 2020. Producing urban agroecology in the East Bay: from soil health to community empowerment. *Agroecol. Sustain. Food Syst.* 44, 566–593.
<https://doi.org/10.1080/21683565.2019.1690615>
- Six, J., Bossuyt, H., Degryze, S., Denef, K., 2004. A history of research on the link between (micro)aggregates, soil biota, and soil organic matter dynamics. *Soil Tillage Res.* 79, 7–31.
<https://doi.org/10.1016/j.still.2004.03.008>
- Skiba, M.A., Abramiuk, I., 2025. Women’s Preferences and Perspectives on the Use of Parks and Urban Forests: A Case Study. *Land* 14, 1345. <https://doi.org/10.3390/land14071345>
- Small, G.E., Martensson, N., Janke, B.D., Metson, G.S.G.S.G.S., 2023. Potential for high contribution of urban gardens to nutrient export in urban watersheds. *Landsc. Urban Plan.* 229.
<https://doi.org/10.1016/j.landurbplan.2022.104602>
- Smith, J.P., Meerow, S., Turner, B.L., 2021. Planning urban community gardens strategically through multicriteria decision analysis. *Urban For. Urban Green.* 58, 126897. <https://doi.org/10.1016/j.ufug.2020.126897>
- Snyder, H., 2019. Literature review as a research methodology: An overview and guidelines. *J. Bus. Res.* 104, 333–339. <https://doi.org/10.1016/j.jbusres.2019.07.039>
- South, L., Saffo, D., Vitek, O., Dunne, C., Borkin, M.A., 2022. Effective Use of Likert Scales in Visualization Evaluations: A Systematic Review. *Comput. Graph. Forum* 41, 43–55. <https://doi.org/10.1111/cgf.14521>
- Speak, A.F., Mizgajski, A., Borysiak, J., 2015. Allotment gardens and parks: Provision of ecosystem services with an emphasis on biodiversity. *Urban For. Urban Green.* 14, 772–781.
<https://doi.org/10.1016/j.ufug.2015.07.007>
- Speelman, E.N., López-Ridaura, S., Colomer, N.A., Astier, M., Masera, O.R., 2007. Ten years of sustainability evaluation using the MESMIS framework: Lessons learned from its application in 28 Latin American case studies. *Int. J. Sustain. Dev. World Ecol.* 14, 345–361. <https://doi.org/10.1080/13504500709469735>
- Stone, T.F., Thompson, J.R., Rosentrater, K.A., Nair, A., 2021. A Life Cycle Assessment Approach for Vegetables in Large-, Mid-, and Small-Scale Food Systems in the Midwest US. *SUSTAINABILITY* 13.

- <https://doi.org/10.3390/su132011368>
- Sylla, M., Hagemann, N., Szewranski, S., 2020. Mapping trade-offs and synergies among peri-urban ecosystem services to address spatial policy. *Environ. Sci. & POLICY* 112, 79–90.
<https://doi.org/10.1016/j.envsci.2020.06.002>
- Taylor, J.R., 2020. Modeling the Potential Productivity of Urban Agriculture and Its Impacts on Soil Quality Through Experimental Research on Scale-Appropriate Systems. *Front. Sustain. Food Syst.* 4.
<https://doi.org/10.3389/fsufs.2020.00089>
- Ternell, A., Lagerqvist, B., Nilsson, A.M., Sagastuy Klie, M., Berg, M., Pedersen, M.A.-B., Lasocki, M., Kreidlmayer, M., Oláh, C., Szekér, Z., Némethy, S., 2024. Multifunctional approach in ecosystem-centred urban planning and urban living labs in landscape conservation. *Ecocycles* 10, 114–127.
<https://doi.org/10.19040/ecocycles.v10i2.503>
- Tidball, K., Krasny, M., 2007. From Risk to Resilience: What Role for Community Greening and Civic Ecology in Cities?, in: *Social Learning Towards a More Sustainable World*. pp. 149–164.
https://doi.org/10.3920/9789086865949_011
- Toboso-Chavero, S., Madrid-López, C., Durany, X.G., Villalba, G., 2021a. Incorporating user preferences in rooftop food-energy-water production through integrated sustainability assessment*. *Environ. Res. Commun.* 3. <https://doi.org/10.1088/2515-7620/abffa5>
- Toboso-Chavero, S., Montealegre, A.L., Garcia-Perez, S., Sierra-Perez, J., Munoz-Liesa, J., Villalba, G., Durany, X.G., Madrid-Lopez, C., 2023. The potential of local food, energy, and water production systems on urban rooftops considering consumption patterns and urban morphology. *Sustain. CITIES Soc.* 95.
<https://doi.org/10.1016/j.scs.2023.104599>
- Toboso-Chavero, S., Nadal, A., Petit-Boix, A., Pons, O., Villalba, G., Gabarrell, X., Josa, A., Rieradevall, J., 2019. Towards Productive Cities: Environmental Assessment of the Food-Energy-Water Nexus of the Urban Roof Mosaic. *J. Ind. Ecol.* 23, 767–780. <https://doi.org/10.1111/jiec.12829>
- Toboso-Chavero, S., Villalba, G., Gabarrell Durany, X., Madrid-Lopez, C., 2021b. More than the sum of the parts: System analysis of the usability of roofs in housing estates. *J. Ind. Ecol.* 25, 1284–1299.
<https://doi.org/10.1111/jiec.13114>
- Tom, M., Fletcher, T.D., McCarthy, D.T., 2014. Heavy Metal Contamination of Vegetables Irrigated by Urban Stormwater: A Matter of Time? *PLoS One* 9, e112441. <https://doi.org/10.1371/journal.pone.0112441>
- UC Agriculture and Natural Resources, n.d. Your Food Garden in Drought [WWW Document]. URL <https://ucanr.edu/site/uc-master-gardener-program-alameda-county/your-food-garden-drought> (accessed 10.12.25).
- Ustaoglu, E., Sisman, S., Aydinoglu, A.C., 2021. Determining agricultural suitable land in peri-urban geography using GIS and Multi Criteria Decision Analysis (MCDA) techniques. *Ecol. Modell.* 455.
<https://doi.org/10.1016/j.ecolmodel.2021.109610>
- Valencia, A., Hossain, M.U., Chang, N.-B., 2022a. Building energy retrofit simulation for exploring decarbonization pathways in a community-scale food-energy-water-waste nexus. *Sustain. CITIES Soc.* 87.

- <https://doi.org/10.1016/j.scs.2022.104173>
- Valencia, A., Qiu, J., Chang, N.-B., 2022b. Integrating sustainability indicators and governance structures via clustering analysis and multicriteria decision making for an urban agriculture network. *Ecol. Indic.* 142. <https://doi.org/10.1016/j.ecolind.2022.109237>
- Valencia, A., Zhang, W., Chang, N.-B. Bin, 2022c. Sustainability transitions of urban food-energy-water-waste infrastructure: A living laboratory approach for circular economy. *Resour. Conserv. Recycl.* 177, 105991. <https://doi.org/10.1016/j.resconrec.2021.105991>
- van der Jagt, A.P.N., Smith, M., Ambrose-Oji, B., Konijnendijk, C.C., Giannico, V., Haase, D., Laforteza, R., Nastran, M., Pintar, M., Železnikar, Š., Cvejić, R., 2019. Co-creating urban green infrastructure connecting people and nature: A guiding framework and approach. *J. Environ. Manage.* 233, 757–767. <https://doi.org/10.1016/j.jenvman.2018.09.083>
- van der Jagt, A.P.N., Szaraz, L.R., Delshammar, T., Cvejić, R., Santos, A., Goodness, J., Buijs, A., 2017. Cultivating nature-based solutions: The governance of communal urban gardens in the European Union. *Environ. Res.* 159, 264–275. <https://doi.org/10.1016/J.ENVRES.2017.08.013>
- van Es, H., Magdoff, F., 2009. *Building Soils for Better Crops*.
- Vaz, A.S., Kueffer, C., Kull, C.A., Richardson, D.M., Vicente, J.R., Kühn, I., Schröter, M., Hauck, J., Bonn, A., Honrado, J.P., 2017. Integrating ecosystem services and disservices: insights from plant invasions. *Ecosyst. Serv.* 23, 94–107. <https://doi.org/10.1016/j.ecoser.2016.11.017>
- Viana, J.G.A., de Barros, C.A.P., Ribeiro, C.G., Minella, J.P.G., Fleck dos Santos, C., Ribeiro, C.M., Silveira, V.C.P., 2023. Sustainability attributes from the water-energy-food nexus: An application to livestock systems in the Brazilian Pampa biome. *Energy Nexus* 12, 100248. <https://doi.org/10.1016/j.nexus.2023.100248>
- Viretto, A., Gontard, N., Angellier-Coussy, H., 2021. Urban parks and gardens green waste: A valuable resource for the production of fillers for biocomposites applications. *Waste Manag.* 120, 538–548. <https://doi.org/10.1016/j.wasman.2020.10.018>
- von Döhren, P., Haase, D., 2015. Ecosystem disservices research: A review of the state of the art with a focus on cities. *Ecol. Indic.* 52, 490–497. <https://doi.org/10.1016/j.ecolind.2014.12.027>
- Wang, J., Zhang, S., Sainju, U.M., Ghimire, R., Zhao, F., 2021. A meta-analysis on cover crop impact on soil water storage, succeeding crop yield, and water-use efficiency. *Agric. Water Manag.* 256, 107085. <https://doi.org/10.1016/j.agwat.2021.107085>
- Watkins, M.W., 2018. *Exploratory Factor Analysis: A Guide to Best Practice*. J. Black Psychol. 44, 219–246. <https://doi.org/10.1177/0095798418771807>
- Webb, R., Bai, X., Smith, M.S., Costanza, R., Griggs, D., Moglia, M., Neuman, M., Newman, P., Newton, P., Norman, B., Ryan, C., Schandl, H., Steffen, W., Tapper, N., Thomson, G., 2018. Sustainable urban systems: Co-design and framing for transformation. *Ambio* 47, 57–77. <https://doi.org/10.1007/s13280-017-0934-6>
- Weidner, T., Yang, A., 2020. The potential of urban agriculture in combination with organic waste valorization:

- Assessment of resource flows and emissions for two european cities. *J. Clean. Prod.* 244. <https://doi.org/10.1016/j.jclepro.2019.118490>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemond, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., Yutani, H., 2019. Welcome to the Tidyverse. *J. Open Source Softw.* 4, 1686. <https://doi.org/10.21105/joss.01686>
- Wielemaker, R., Oenema, O., Zeeman, G., Weijma, J., 2019. Fertile cities: Nutrient management practices in urban agriculture. *Sci. Total Environ.* 668, 1277–1288. <https://doi.org/10.1016/j.scitotenv.2019.02.424>
- Wolch, J.R., Byrne, J., Newell, J.P., 2014. Urban green space, public health, and environmental justice: The challenge of making cities “just green enough.” *Landsc. Urban Plan.* 125, 234–244. <https://doi.org/10.1016/j.landurbplan.2014.01.017>
- Wright, J., Lytle, J., Santillo, D., Marcos, L., Mai, K.V., 2021. Addressing the Water-Energy-Food Nexus through Enhanced Green Roof Performance. *SUSTAINABILITY* 13. <https://doi.org/10.3390/su13041972>
- Wut, T.M., Ng, P., Kan, Hing-Ki, M., Fong, Chiu, S., 2021. Does gender matter? Attitude towards waste charging policy and pro-environmental behaviours. *Soc. Responsib. J.* 17, 1100–1115. <https://doi.org/10.1108/SRJ-03-2020-0102>
- Yan, D., Liu, L., Liu, X., Zhang, M., 2022. Global Trends in Urban Agriculture Research: A Pathway toward Urban Resilience and Sustainability. *LAND* 11. <https://doi.org/10.3390/land11010117>
- Zambrano-Prado, P., Muñoz-Liesa, J., Josa, A., Rieradevall, J., Alamús, R., Gasso-Domingo, S., Gabarrell, X., 2021. Assessment of the food-water-energy nexus suitability of rooftops. A methodological remote sensing approach in an urban Mediterranean area. *Sustain. Cities Soc.* 75. <https://doi.org/10.1016/j.scs.2021.103287>
- Zhang, B., MacKenzie, A., 2024. Trade-offs and synergies in urban green infrastructure: A systematic review. *Urban For. Urban Green.* 94, 128262. <https://doi.org/10.1016/J.UFUG.2024.128262>
- Zhang, Y., Smith, J.P., Tong, D., Turner, B.L., 2022. Optimizing the co-benefits of food desert and urban heat mitigation through community garden planning. *Landsc. Urban Plan.* 226, 104488. <https://doi.org/10.1016/J.LANDURBPLAN.2022.104488>
- Zhen, H., Gao, W., Jia, L., Qiao, Y., Ju, X., 2020. Environmental and economic life cycle assessment of alternative greenhouse vegetable production farms in peri-urban Beijing, China. *J. Clean. Prod.* 269. <https://doi.org/10.1016/j.jclepro.2020.122380>
- Zheng, J., Li, H., Sun, L., Li, M., Zhang, Y., 2025. Optimizing the Food-Energy-Water Nexus: A Multi-Objective Spatial Configuration Framework for High-Density Communities. *BUILDINGS* 15. <https://doi.org/10.3390/buildings15132196>
- Zipper, S.C., Schatz, J., Kucharik, C.J., Loheide, S.P., 2017. Urban heat island-induced increases in evapotranspirative demand. *Geophys. Res. Lett.* 44, 873–881. <https://doi.org/10.1002/2016GL072190>

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APPENDIX

8.1 APPENDIX 1 – SUPPLEMENT MATERIALS FOR CHAPTER 2

Table A.1.1 Scales of analysis of reviewed papers

Scale of Analysis	Methodological Cluster	Number of Studies	References (Author, Year)
1. Plot & Building	Quantitative Sustainability Assessment	4	Dorr et al., 2017; Meng et al., 2023; Sanjuan-Delmás et al., 2018; Carrasco-Astudillo et al., 2024
	Systems & Nexus Modeling	5	Huang & Chang, 2021; Jans-Singh et al., 2021; Johnson et al., 2015; Piezer et al., 2019; Holt et al., 2017
	Empirical & Field-Based Studies	3	Grard et al., 2018; Taylor, 2020; Wright et al., 2021
2. Community & Neighborhood	Spatial Planning & Optimization	5	Cumo et al., 2025; Pedro et al., 2021; Smith et al., 2021; Zhang et al., 2022; Zheng et al., 2025
	Systems & Nexus Modeling	1	Valencia, Hossain, & Chang, 2022
	Socio-Ecological & Community Assessment	3	Kirby et al., 2021; Lowan-Trudeau et al., 2020; Siegner, Acey, & Sowerwine, 2020

	Empirical & Field-Based Studies	2	Caputo et al., 2023; Jha et al., 2023
3. City & Regional	Spatial Planning & Optimization	8	Fanfani et al., 2022; Lin et al., 2025; Liu, Lin, & Chiueh, 2022; Lucertini & Di Giustino, 2021; M. Susana & Geneletti, 2023; Ustaoglu et al., 2021; Zambrano-Prado et al., 2021; Small et al., 2023
	Systems & Nexus Modeling	6	Elkamel et al., 2023; Gondhalekar & Ramsauer, 2017; Goodarzi et al., 2022; Pueyo-Ros et al., 2024; Valencia, Zhang, & Chang, 2022; Weidner & Yang, 2020
	Socio-Ecological & Community Assessment	2	Nie, Kiminami, & Yagi, 2024; Valencia, Qiu, & Chang, 2022
	Empirical & Field-Based Studies	4	Chen et al., 2023; Dennis & James, 2017; Mulier et al., 2022; Sylla et al., 2020
4. Conceptual / Multi-Scale	Quantitative Sustainability Assessment	5	Imam et al., 2025; McDougall et al., 2019; Stone et al., 2021; Toboso-Chavero et al., 2019; Zhen et al., 2020
	Spatial Planning & Optimization	3	Camacho-Caballero et al., 2024; Kafle et al., 2023; Toboso-Chavero et al., 2023
	Socio-Ecological & Community Assessment	5	Babaei et al., 2024; Caputo et al., 2021; Kong et al., 2025; Rodríguez González et al., 2023; Toboso-Chavero et al., 2021
	Empirical & Field-Based Studies	1	Dorr et al., 2023

Total	58
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Table A.1.2 Common indicators used in integrated assessment for UFGs

Methodological Cluster	Indicator Category	Specific Input Parameter / Indicator	Frequency (Count of Papers)
Quantitative Sustainability Assessment	Resource Inputs (Material & Energy)	Infrastructure Materials (steel, plastic, wood, concrete)	10
		Substrate & Grow Media (compost, potting soil, etc.)	8
		Fertilizers & Soil Amendments (synthetic, organic, NPK)	8
		Water (tap water, irrigation)	7
		Energy (electricity, fuel, diesel)	7
		Pesticides & Herbicides	5
		Packaging Materials	2
		Other Inputs (seeds, earthworms)	2
	Economic Inputs	Operational Costs (labor, water, energy)	4
		Capital Costs (infrastructure, machinery)	3
		Land Rent / Value	2
	System & Production Parameters	Crop Type & Yield Data	9
		Life Cycle Stages (production, transport, waste)	8
		Food Waste Ratios	2
Spatial Planning & Optimization	Spatial & Geographic Data	Land Use / Land Cover (vacant land, green areas, etc.)	18

		Building Data (footprints, rooftop area, height)	14
		Infrastructure Networks (roads, public transit)	11
		Environmental Data (slope, elevation, aspect, hydrology)	9
		Climate Data (solar radiation, temperature, precipitation)	8
		Proximity to Features (markets, community spaces)	8
		Soil Properties (depth, drainage, capability class)	5
	Socio-Economic & Demographic Data	Population Data (density, age structure)	12
		Economic Indicators (income, unemployment)	10
		Health & Vulnerability Indicators (food deserts, heat vulnerability)	9
		Social Indicators (minority population, education level)	7
	Policy & Planning Scenarios	Land-Use Scenarios	7
		Siting Criteria (from stakeholders/literature)	6
		Development Goals & Constraints	4
		Budget / Resource Constraints	2
Systems & Nexus Modeling	Resource Flow & Consumption Data	Water Demand & Supply Data	11
		Energy Demand & Supply Data	11
		Food Demand & Supply Data	10
		Organic Waste Generation Rates	5
		Technology Specifications (PV efficiency, tank capacity)	8

	System Design & Operational Parameters	UA System Parameters (farm area, greenhouse setpoints)	7
		Agent Attributes (for ABM: income, preferences)	2
	External Drivers & Scenarios	Climate Change Projections (RCPs)	6
		Policy Scenarios	5
		Economic Scenarios (price inflation)	2
	Biophysical & Ecological Models	Plant Growth Models	3
		Evapotranspiration (ET) Models	2
		Hydrological Models (runoff, infiltration)	2
	Socio-Ecological & Community Assessment	Participant & Community Demographics	Socio-Economic Data (age, income, education)
Gardener Experience & Skill Level			3
Social & Institutional Factors		Participant Motivations	5
		Stakeholder Perceptions & Preferences (AHP/Delphi)	5
		Governance Structures	3
		Social Networks & Cohesion	2
Livelihood Capitals		Human Capital (knowledge, skills)	2
		Social Capital (networks, trust)	2
		Financial Capital (income, savings)	2
		Physical & Natural Capital (tools, land access)	2
UA Site Characteristics		Site Typology (community garden, allotment, etc.)	8
		Food Distribution Model (self-consumption, sales)	4

		Management Practices (organic, conventional)	3
Empirical & Field-Based Studies	Resource Inputs (Measured)	Water Consumption (irrigation volume)	7
		Soil Amendments (compost, fertilizer application rates)	6
		Energy Use (fuel for machinery)	2
	Biophysical Site Characteristics	Site & Cultivated Area (m ²)	9
		Soil Quality Metrics (%OM, pH, CEC, nutrients)	5
		Floral & Vegetative Diversity (richness, abundance)	4
		Landscape Context (% impervious surface)	2
	Management Practices	Tillage System	2
		Crop Diversity & Polyculture Design	2
	Human Activity (Measured)	Labor Hours	4
		Volunteer Journeys (transport)	2

8.2 APPENDIX 2 – SUPPLEMENT MATERIALS FOR CHAPTER 4

Table A.2.1. Summary of the gardener demographics and gardening intensity

Category	Age	Daily_time	Day_week
25-34	1	0	0
35-54	7	0	0
55-64	21	0	0
65+	25	0	0
1-2hr	0	8	0
2-3hr	0	21	0
<1hr	0	3	0
>3hr	0	22	0
1-2 days per week	0	0	3
3-5days perweek	0	0	31
Everyday	0	0	20

Table A.2.2. Summary of waiting time to acquire an urban garden plot (year)

Garden	Mean	Median	SD	IQR
Overall	1,87	1,5	1,34	2
AC	1,5	1,5		
AM	1,18	1		
BR	2,17	2,5		
CR	1,86	1,5		
EG	2,2	2		

Table A.2.3. Results of Inferential Tests for Differences in Gardening Behavior by Gender and Age

Comparison	statistic	p.value	estimate	method
Gender vs Days/Week	347			
Gender vs Hours/Visit	370			
Age vs Days/Week	28341,18141	0,563889615	-0,080281357	Spearman's rank correlation rho
Age vs Hours/Visit	23581,55797	0,466789825	0,101141301	Spearman's rank correlation rho

Table A.2.4. Percentage, Median, and Interquartile Distribution of Perceived Importance of Ecosystem Services

Ecosystem service	Not important (%)	Slightly important (%)	Moderately important (%)	Highly important (%)	Median	IQR
<i>Food Provision</i>	0,0	7,4	38,9	53,7	4	1
<i>Medicinal & Aromatic</i>	0,0	13,0	51,9	35,2	3	1
<i>Physical Health</i>	0,0	11,1	25,9	63,0	4	1
<i>Biodiversity</i>	1,9	11,1	35,2	51,9	4	1
<i>Soil Conservation</i>	5,6	9,3	20,4	64,8	4	1
<i>Education</i>	0,0	3,7	22,2	74,1	4	0,75
<i>Water Regulation</i>	0,0	5,6	24,1	70,4	4	1
<i>Social Cohesion</i>	1,9	11,1	31,5	55,6	4	1
<i>Air Quality</i>	16,7	7,4	35,2	40,7	3	1
<i>Mental Health</i>	3,7	3,7	25,9	66,7	4	1

Note: Responses were coded into a Likert scale system, with 1 = not important; 2 = slightly important, 3 = moderately important, 4 = highly important.

Table A.2.5. Percentage, Median, and Interquartile Distribution of Perceived Concern for Ecosystem Disservices

Ecosystem disservice	Not concerned (%)	Low concern (%)	Moderate concern (%)	High concern (%)	Median	IQR
<i>Food Contamination</i>	22,2	24,1	25,9	27,8	3	2
<i>Green Waste</i>	55,6	11,1	5,6	27,8	1	3
<i>Water Consumption</i>	3,7	9,3	25,9	61,1	4	1
<i>Water Pollution</i>	44,4	7,4	11,1	37,0	2	3
<i>Soil Pollution</i>	37,0	9,3	9,3	44,4	3	3
<i>Air Pollution</i>	53,7	20,4	11,1	14,8	1	1,75
<i>Allergies</i>	51,9	5,6	24,1	18,5	1	2
<i>Maintenance Cost</i>	46,3	22,2	25,9	5,6	2	2
<i>Pest/Insect Bites</i>	38,9	18,5	18,5	24,1	2	2
<i>Fear of Wilderness</i>	59,3	14,8	22,2	3,7	1	1,75
<i>Unpleasant Odor</i>	66,7	14,8	13,0	5,6	1	1

Note: Responses were coded into a Likert scale system, with 1 = not concerned; 2 = low concern, 3 = moderate concern, 4 = high concern.

Table A.2.6. Kruskal-Wallis H Test Results for Differences in Perceptions Across Garden Sites

Item	statistic	df	p	Effect Size (η^2)	magnitude
<i>Ecosystem disservices</i>					
<i>Air Pollution</i>	3,302134	4	0,509	-0,01424	small
<i>Allergies</i>	2,594788	4	0,628	-0,02868	small
<i>Pet/Insect Bites</i>	9,02966	4	0,0604	0,102646	moderate
<i>Food Contamination</i>	7,867545	4	0,0966	0,078929	moderate
<i>Maintenance Cost</i>	5,197139	4	0,268	0,024431	small
<i>Fear of Wilderness</i>	1,334525	4	0,855	-0,0544	small
<i>Unpleasant Odor</i>	0,960967	4	0,916	-0,06202	moderate
<i>Soil Pollution</i>	3,053896	4	0,549	-0,01931	small
<i>Water Consumption</i>	2,511272	4	0,643	-0,03038	small
<i>Water Pollution</i>	2,912591	4	0,573	-0,02219	small
<i>Green Waste</i>	0,391347	4	0,983	-0,07365	moderate
<i>Ecosystem services</i>					
<i>Air Quality</i>	0,828225	4	0,935	-0,06473	moderate
<i>Biodiversity</i>	4,100938	4	0,393	0,00206	small
<i>Education</i>	3,416873	4	0,491	-0,0119	small
<i>Food Provision</i>	2,433323	4	0,657	-0,03197	small
<i>Medicinal & Aromatic</i>	1,028371	4	0,905	-0,06065	moderate
<i>Mental Health</i>	2,025075	4	0,731	-0,0403	small
<i>Physical Health</i>	2,372224	4	0,668	-0,03322	small
<i>Social Cohesion</i>	2,441561	4	0,655	-0,0318	small
<i>Soil Conservation</i>	1,375358	4	0,848	-0,05356	small
<i>Water Regulation</i>	1,679924	4	0,794	-0,04735	small

Table A.2.7. Mann-Whitney U Test Results for Differences in Perceptions Between Genders

Item	statistic	p	Effect Size (η^2)	magnitude
<i>Ecosystem disservices</i>				
<i>Air Pollution</i>	405,5	0,437	0,107164	small
<i>Allergies</i>	452,5	0,0966	0,227429	small
<i>Pet/Insect Bites</i>	512,5	0,00739	0,365762	moderate
<i>Food Contamination</i>	408	0,436	0,107112	small
<i>Maint</i>	356,5	0,897	0,018916	small
<i>Fear of Wilderness</i>	500	0,00782	0,363267	moderate
<i>Unpleasant Odor</i>	438,5	0,125	0,210024	small
<i>Soil Pollution</i>	374,5	0,852	0,026672	small
<i>Water Consumption</i>	438	0,143	0,200799	small
<i>Water Pollution</i>	414	0,355	0,127025	small
<i>Green Waste</i>	352	0,824	0,031492	small
<i>Ecosystem services</i>				
<i>Air Quality</i>	398	0,537	0,085188	small
<i>Biodiversity</i>	504	0,0075	0,36513	moderate
<i>Education</i>	378	0,759	0,043206	small
<i>Food Provision</i>	376,5	0,815	0,033212	small
<i>Medicinal & Aromatic</i>	601,5	5,51E-06	0,619721	large
<i>Mental Health</i>	423	0,222	0,167767	small
<i>Physical Health</i>	354	0,848	0,027538	small
<i>Social Cohesion</i>	310	0,299	0,142568	small
<i>Soil Conservation</i>	360	0,943	0,011117	small
<i>Water Regulation</i>	537	0,000184	0,510401	large

Table A.2.8. Polychoric Correlation Matrix of Ecosystem Service and Disservice Perception

Item	Food Provision	Medicinal & Aromatic	Physical Health	Biodiversity	Soil Conservation	Education	Water Regulation	Social Cohesion	Air Quality	Mental Health	Food Contamination	Green Waste	Water Consumption	Water Pollution	Soil Pollution	Air Pollution	Allergies	Maintenance Cost	Pest/Insect Bites	Fear of Wilderness	Unpleasant Odor
Food Provision	1																				
Medicinal & Aromatic	-0,16	1																			
Physical Health	0,39	-0,10	1																		
Biodiversity	-0,06	0,41	0,15	1																	
Soil Conservation	-0,02	-0,06	0,19	0,74	1																
Education	-0,13	0,16	0,28	0,58	0,45	1															
Water Regulation	0,24	0,23	0,09	0,55	0,37	-0,04	1														
Social Cohesion	-0,07	-0,17	0,42	0,19	0,47	0,14	0,25	1													
Air Quality	-0,08	-0,06	0,28	0,38	0,51	0,50	0,45	0,49	1												
Mental Health	0,42	0,08	0,54	0,44	0,41	-0,03	0,53	0,51	0,17	1											
Food Contamination	0,03	0,06	0,11	0,27	0,11	0,03	0,20	-0,20	-0,08	0,30	1										
Green Waste	-0,33	0,04	-0,01	0,45	0,10	0,13	0,28	-0,10	0,02	0,06	0,34	1									
Water Consumption	-0,17	0,04	0,10	0,43	0,21	0,15	0,17	-0,05	0,16	0,07	0,32	0,53	1								
Water Pollution	-0,42	0,00	-0,20	0,44	0,34	0,32	-0,09	0,02	0,04	0,17	0,37	0,30	0,27	1							
Soil Pollution	-0,31	-0,12	-0,20	0,17	0,15	0,04	-0,10	-0,11	0,07	0,14	0,46	0,20	0,30	0,87	1						
Air Pollution	-0,24	0,11	-0,03	0,43	0,13	0,50	0,21	-0,25	0,27	0,07	0,27	0,46	0,08	0,57	0,51	1					
Allergies	-0,32	0,32	-0,17	0,29	-0,08	-0,20	0,28	-0,13	-0,13	0,24	0,25	0,66	0,44	0,42	0,46	0,44	1				
Maintenance Cost	0,05	-0,06	0,10	0,23	0,22	0,16	0,31	0,20	0,22	0,23	0,28	0,37	0,10	0,04	-0,01	0,25	0,37	1			
Pest/Insect Bites	0,03	0,20	0,21	0,45	0,15	0,19	0,42	0,04	0,23	0,41	0,33	0,24	0,32	0,34	0,30	0,46	0,46	0,40	1		
Fear of Wilderness	0,12	0,38	0,00	0,46	0,06	0,05	0,43	0,07	-0,12	0,39	0,43	0,36	0,47	0,22	0,12	0,11	0,56	0,52	0,48	1	

6 | CONCLUSION AND RECOMMENDATION

Unpleasant Odor	-0,02	0,39	-0,32	0,25	-0,12	-0,05	0,31	-0,30	-0,06	0,10	0,34	0,37	0,23	0,14	0,26	0,42	0,65	0,48	0,40	0,60	1
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Table A.2.9. Rotated Factor Loading Matrix (Pattern Matrix) from Exploratory Factor Analysis

ES and EDS	PA1	PA2	PA3	PA4
Ecosystem services				
<i>Food Provision</i>	-0,03132	-0,22458	-0,31712	0,513779
<i>Medicinal & Aromatic</i>	0,467272	0,163671	-0,23258	-0,18472
<i>Physical Health</i>	-0,2041	0,182587	-0,10358	0,534477
<i>Biodiversity</i>	0,365923	0,752903	0,035312	0,100842
<i>Soil Conservation</i>	-0,17505	0,661263	0,134565	0,282299
<i>Education</i>	-0,09584	0,794697	0,034759	-0,20373
<i>Water Regulation</i>	0,453009	0,300298	-0,26935	0,367909
<i>Social Cohesion</i>	-0,31196	0,321361	-0,04241	0,485238
<i>Air Quality</i>	-0,16139	0,684776	-0,0503	0,115309
<i>Mental Health</i>	0,109541	0,007436	0,122486	0,906891
Ecosystem disservices				
<i>Food Contamination</i>	0,337213	-0,08376	0,341731	0,233502
<i>Green Waste</i>	0,533951	0,210838	0,155838	-0,12513
<i>Water Consumption</i>	0,349586	0,177419	0,184422	0,026925
<i>Water Pollution</i>	0,005055	0,184046	0,906046	-0,02592
<i>Soil Pollution</i>	0,004813	-0,11005	0,979985	0,069452
<i>Air Pollution</i>	0,316397	0,377747	0,364794	-0,21528
<i>Allergies</i>	0,761998	-0,11557	0,282318	-0,01817
<i>Maintenance Cost</i>	0,443525	0,166818	-0,08113	0,194021
<i>Pet/Insect Bites</i>	0,449553	0,185841	0,161417	0,259838
<i>Fear of Wilderness</i>	0,761435	-0,01111	-0,05825	0,26811
<i>Unpleasant Odor</i>	0,847633	-0,07124	-0,02761	-0,11731

Table A.2.10. Variance Explained by the Extracted Factors

Statistic	SS loadings	Proportion Var	Cumulative Var	Proportion Explained	Cumulative Proportion
PA1	3,833806	0,182562	0,182562	0,326341	0,326341
PA2	2,931666	0,139603	0,322165	0,249549	0,575891
PA3	2,637462	0,125593	0,447759	0,224506	0,800397
PA4	2,344908	0,111662	0,559421	0,199603	1

Table A.2.11. Inter-Factor Correlation Matrix (Phi Matrix)

Factor	PA1	PA2	PA3	PA4
PA1	1	0,12136	0,263191	0,123024
PA2	0,12136	1	0,179014	0,241945
PA3	0,263191	0,179014	1	-0,0491
PA4	0,123024	0,241945	-0,0491	1

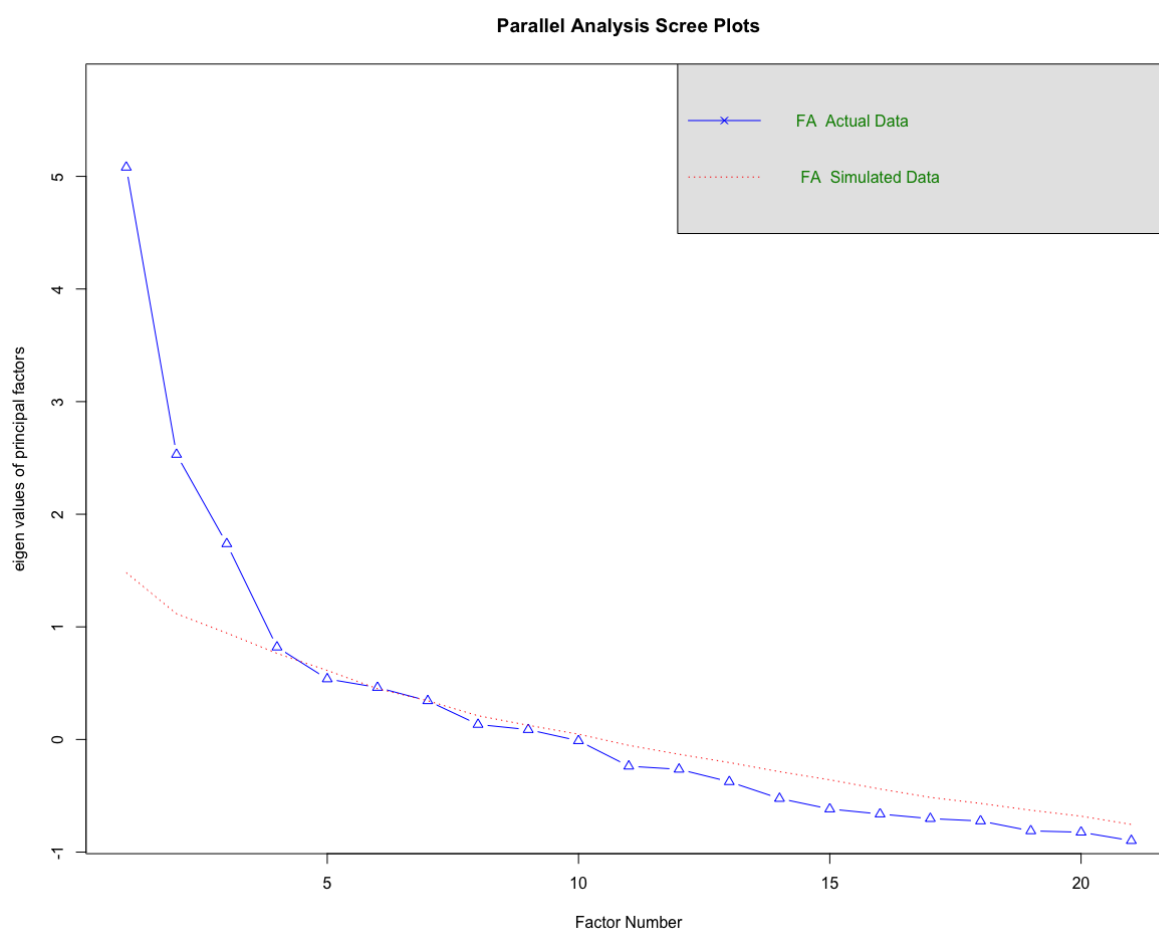


Figure A.2.12. Scree Plots of Result from Parallel Analysis

APPENDIX B - Questionnaire

Jardin: *Date de l'entretien:* *Taille de la parcelle:*
Nom du jardinier : *Sexe:*
Tranche d'âge: *Statut professionnel :*

Questions générales

1. Combien de temps vous a-t-il fallu pour obtenir l'attribution d'une parcelle ?

.....

3. Quel pourcentage de produits issus des jardins couvre votre consommation journalière?
(80-90-100% etc)

.....

4. Venez-vous du monde rural? Avez vous des compétences en jardinage, en culture avant de prendre le jardin?

☐ oui ☐ non

5. Combien d'heures par jour consacrez-vous au jardinage?

☐ moins d'une heure ☐ 1 - 2 heures ☐ 2 - 3 heures ☐ plus de 3 heures

6. Combien de fois allez-vous au jardin?

☐ tous les jours ☐ plusieurs fois par semaine ☐ une fois par semaine ☐ moins souvent

7. Quelle végétation (fruits et légumes) cultivez-vous dans le jardin ?

.....

.....

.....

.....

9. Qui consomme les produits du jardin ?

☐ famille exclusivement ☐ partager volontairement avec des amis/voisins ☐ autre

Si vous partagiez des légumes avec des amis ou des voisins, à quelle fréquence le faisiez-vous ?

☐ Non ☐ Occasionnellement ☐ Fréquemment

10. Quels types d'outils de gestion utilisez-vous pour le choix des légumes ou les dépenses du jardin ?

☐ Logiciel pour ordinateur/smartphone ☐ Livre/tableur pour la culture des légumes ☐ Je ne les utilise pas..

Services et disservices écosystémiques

11. Nous allons maintenant vous présenter une liste benefices des jardins urbains, et nous aimerions que vous les classiez en fonction de leur importance pour vous, allant de : "Ne me concerne pas", "Priorité basse", "Priorité moyenne" et "Priorité élevée".

	<i>Ne me concerne pas</i>	<i>Priorité basse</i>	<i>Priorité moyenne</i>	<i>Priorité élevée</i>
<i>Production alimentaire</i>				
<i>Herbes médicinales/aromatiques et fleurs</i>				
<i>Le jardinage dans le cadre de l'exercice physique pour la santé</i>				
<i>Protéger et améliorer la biodiversité et la nature</i>				
<i>Protéger les sols et réduire leur érosion</i>				
<i>Améliorer les connaissances sur le jardinage et la nature</i>				
<i>Améliorer l'efficacité de l'utilisation de l'eau et la conservation de l'eau</i>				
<i>Renforcer les relations entre les résidents de la communauté</i>				
<i>Renforcer les relations entre les personnes âgées et les jeunes</i>				
<i>Réduire la dépendance alimentaire au marché et aux grandes surfaces</i>				
<i>Améliorer les conditions environnementales atmosphériques et climatiques dans la zone environnante.</i>				
<i>Améliorer les conditions économiques de la famille</i>				
<i>Amélioration de l'humeur par l'embellissement et l'amélioration esthétique</i>				
<i>Améliorer vos connaissances sur les plantes/insectes/animaux/ la nature et les interactions entre eux</i>				

12. Nous allons vous proposer une liste d'éléments qui peuvent être considérées comme des nuisances liées aux jardins. Pour chaque élément, pourriez-vous indiquer si cela vous pose problème ou non, en utilisant cette échelle : Pas de préoccupation, Préoccupation légère, Préoccupation modérée, Forte préoccupation.

	<i>Pas de préoccupation</i>	<i>Préoccupation légère</i>	<i>Préoccupation modérée</i>	<i>Forte préoccupation</i>
<i>Végétaux contaminés (air, eau et sol contaminés)</i>				
<i>Déchets de jardinage</i>				
<i>Consommation d'eau</i>				
<i>Pollution de l'eau par les intrants chimiques</i>				
<i>Pollution des sols par les intrants chimiques</i>				
<i>Risque de pollution lié aux trafics routiers</i>				
<i>Allergies et maladies de la peau</i>				
<i>Coûts d'installation et de maintenance (frais de location, frais d'équipement, frais de graines pour les légumes et les herbes.)</i>				
<i>Animaux de compagnie et piqûres d'insectes</i>				
<i>Peur et risques liés à la nature sauvage</i>				
<i>Sons et odeurs désagréables</i>				

8.3 APPENDIX 3 – SUPPLEMENT MATERIALS FOR CHAPTER 5

Table A.3.1 Soil Axis Indicators - Condition

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Condition	40	Physical properties	12	Top-soil depth (cm)	Ordinal	3	>100
						2	50-100
						1	25-50
						0	<25 c
				Texture class	Ordinal	3	Balanced
						2	Loamy
						1	Clayey

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						0	Sandy
				Bulk density (g cm ⁻³)	Ordinal	3	0.8-1.2
						2	1.2-1.5
						1	1.5-1.7
						0	<0.8 or >1.7
				Slope (%)	Ordinal	3	0-2
						2	2-5
						1	5-15
						0	>15
		Chemical properties	18	Electrical conductivity (μS cm ⁻¹)	Ordinal	3	<500
						2	500-1000
						1	1000-2000

						0	>2000
				Total organic carbon (g/kg ⁻¹)	Ordinal	3	>5.8
						2	3.5-5.8
						1	1.2-3.5
						0	<1.5
				Total Nitrogen (g/kg ⁻¹)	Ordinal	3	10-20
						2	>20
						1	2-10
						0	<2
				Total phosphorus (g/kg ⁻¹)	Ordinal	3	0.06-0.1
						2	>0.1
						1	0.04-0.06

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						0	<0.04
				Trace elements (mg kg ⁻¹) (NPGB - natural pedo- geochemical background)	Ordinal	3	<NPGB
						2	NPGB-2xNPGB
						1	2xNPGB-10NPGB
						0	>10xNPGB
				Organic pollutants (PAH) (mg kg ⁻¹)	Ordinal	3	<Quantification limit < NPGB for persistent substances
						2	<Quantification limit or <A
						1	A-C
						0	>C (Pollution to manage)
			10	Earthworm		3	>300

		Biological property				2	150-300
						1	50-150
						0	<25

Table A.3.2 Soil Axis Indicators – Degradation

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Degradation	30	Evidence of the erosive process	10	Wind erosion	Nominal	5	No
						0	Yes
				Water erosion (diffuse/concentrated)	Nominal	5	No
						0	Yes
		Synthetic input	10	Integrated pest management	Ordinal	5	Prevention practices only

						3	Physical/Mechanical controls
						1	Biological/Biochemical
						0	Board-spectrum synthetic pesticides
		Extreme weather conditions	10	Heavy rain	Nominal	5	Rarely
						0	Every season
				Flooding	Nominal	5	No
						0	Yes

Table A.3.3 Soil Axis Indicators – Stewardship

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Stewardship	30		20		Ordinal	5	No tilling

		Preservation practices		Disturbance minimization		2.5	1-3 times
						0	4 or more
				Cover crop/ Munching	Nominal	5	Yes
						0	No
				Crop rotation	Nominal	5	Yes
						0	No
				Vegetation cover %	Ordinal	5	>75%
						2.5	50 – 75 %
						0	<50%
		Institutional supports	10	Land tenure policy	Nominal	5	Unlimited
						0	Limited
					Nominal	5	Yes

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				Public participation		0	No
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Table A.3.4 Water Axis Indicators – Human Consumption

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Human consumption	10	Water quantity	5	Source meets consumption	Scalar	5	total sources
						4	total sources)
						3	total sources
						2	total sources
						1	total
						0	No access
		Water quality	5	Quality	Ordinal	5	Good
						2.5	Average
						0	Bad

Table A.3.5 Water Axis Indicators – Production

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Production	50	Production Water	10	Source meets consumption	Scalar	10	total (scale 5)
						8	total (scale 4)
						6	total (scale 3)
						4	total (scale 2)
						2	total (scale 1)
						0	No network
		Water-use efficiency	30	Conventional system	Ordinal	20	>50%
						10	<50%
				Intensive system	Ordinal	30	>80%

						20	50-80%
						10	<50
						30	>80%
						20	50-80%
						10	<50%
		Susceptibility to drought	10	Occurrence	Nominal	5	No
						0	Yes
				Frequency	Ordinal	5	Low
						3	Average
						0	Discharge

Table A.3.6 Water Axis Indicators – Conservation

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Conservation	40	Plant management	6	Origin of vegetables	Ordinal	6	Native
						3	Mixed
						0	Non-native
		Water management	12	Irrigation system	Ordinal	12	Buried
						6	Drip
						3	Surface
		Soil management	12	Soil moisture protection methods	Ordinal	12	>2
						6	1-2
						0	0

		Water reuse/recycle	10	Collection pathways	Ordinal	10	>2
						5	1-2
						0	0

Table A.3.7 Food Axis Indicators – Consumption

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Consumption	40	Self-consumption	20	Portion of vegetables covered by products from the gardens	Ordinal	20	>75%
						15	50-75%
						10	25-50%
						5	10-25%
						0	<10%
			10		Scalar	10	Frequently

		Close network sharing		Frequency of product shared with neighbours/friends/colleagues		5	Sometime
						0	Never
		Social sharing	10	Frequency of product shared with public (school/food bank)	Scalar	10	Frequently
						5	Sometime
						0	Never

Table A.3.8 Food Axis Indicators – Organizational and Institutional Environment

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Organizational and Institutional Environment	30	Social and associative participation	10	Degree of participation in garden association and local community	Scalar	0	Never
						2.5	Low
						5	Average
						7.5	High

						10	Fully participated
		Support organizations	5	Degree of relationship with support organizations	Scalar	0	Never
						1	Rarely
						3	Occasionally
						5	Frequently
		Traditional and culture	5	Importance of culture and tradition in the activity	Scalar	0	Nothing important
						1	Little
						3	Moderate
						5	High
		Public policies	5	Supporting local policies and government	Ordinal	0	Non-existent/ Unclear
						5	Exist

		Logistics and infrastructure	5	Condition of equipment and infrastructures	Scalar	0	Non-existent
						1	Bad
						3	Regular
						5	Good

Table A.3.9 Food Axis Indicators – Productive and Technological Environment

Scope	Weight	Indicator	Weight	Variable	Measurement form	Weight	Measurement
Productive and Technological Environment	30	Garden management	10	Use of crop management tools	Ordinal	0	No
						5	Yes
				Use of economic management tools	Ordinal	0	No
						5	Yes
			5		Scalar	0	High

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		Dependence on economic input		Degree of economic concern		2.5	Moderate
						5	No concern
		Labour availability	5	Time availability	Scalar	0	Less than once a week
						2.5	Few times a week
						5	Every day
		Ownership	5	Garden acquisition time	Scalar	0	< 1 year
						2.5	1-2 years
						5	> 2 years
		Biodiversity enhancement	5	Biodiversity protection pathways	Scalar	0	Non-existent
						2.5	1-2
						5	More than 2

Titre : Renforcement de la capacité de résilience urbaine à travers les jardins alimentaires urbains multifonctionnels : approche du nexus Alimentation-Eau-Sol

Mots clés : jardins urbains, services et disservices écosystémiques, nexus Alimentation-Eau-Sol

Résumé : Les jardins alimentaires urbains (UFG) sont de plus en plus reconnus comme des Solutions fondées sur la Nature essentielles pour renforcer la résilience et la durabilité urbaines. Cependant, leur viabilité à long terme est mise au défi par un manque d'outils d'évaluation capables de saisir de manière holistique leurs dynamiques socio-écologiques complexes. Les paradigmes dominants, tels que le nexus Eau-Énergie-Alimentation, se révèlent souvent inadaptés à ces systèmes à faible consommation d'énergie et basés sur le sol, car ils tendent à marginaliser le rôle fondamental du sol, le traitant comme un substrat statique plutôt que comme un capital dynamique. Cette thèse comble cette lacune critique en formulant, développant et validant un nouveau cadre conceptuel, le nexus Alimentation-Eau-Sol, un

changement de paradigme qui positionne le sol comme l'élément médiateur central dans les agroécosystèmes urbains. La recherche emploie une approche mixte et multi-étapes, débutant par une revue systématique qui critique les méthodes d'évaluation existantes et établit la nécessité théorique d'un paradigme centré sur le sol. Ce cadre est testé empiriquement à travers une étude de cas approfondie de 54 jardins collectifs à Nantes Nord, en France. L'application de l'indice génère un profil de durabilité quantitatif et multidimensionnel des jardins, révélant les forces, faiblesses et compromis critiques entre les dimensions de l'alimentation, de l'eau et du sol. En fin de compte, cette thèse apporte à la fois un nouveau cadre théorique et un outil pratique validé pour l'évaluation intégrée des UFGs.

Title : Enhancing urban resilience capacity through multifunctional urban food gardens: Food-Water-Soil nexus approach .

Keywords : urban gardens, ecosystem services and disservices, Food-Water-Soil nexus

Abstract : Urban Food Gardens (UFGs) are increasingly recognized as vital Nature-Based Solutions for enhancing urban resilience and sustainability. However, their long-term viability is challenged by a lack of assessment tools that can holistically capture their complex socio-ecological dynamics. Dominant paradigms, such as the Water-Energy-Food nexus, often prove inadequate for these low-energy, soil-based systems, as they tend to marginalize the foundational role of soil, treating it as a static substrate rather than a dynamic capital asset. This dissertation addresses this critical gap by formulating, developing, and validating a novel Food-Water-Soil (FWS) nexus framework, a paradigm shift that positions soil as the central mediating component in urban agroecosystems.

The research employs a multi-stage, mixed-methods approach, beginning with a systematic review that critiques existing assessment methods and establishes the theoretical necessity of a soil-centric paradigm. This framework is empirically tested through an in-depth case study of 54 community gardens in Nantes Nord, France. The application of the index generates a quantitative, multi-dimensional sustainability profile of the gardens, revealing critical strengths, weaknesses, and trade-offs between the food, water, and soil dimensions. Ultimately, this thesis contributes both a new theoretical framework and a validated, practical tool for the integrated assessment of UFGs.